

Chapter 3

The Theory of the Kinetics of Elementary Gas Phase Reactions

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1. Introduction

By no means all experimental kineticists measure rates and rate coefficients with the intention that the data should be used to test the validity, or otherwise, of a theory of reaction rates. Indeed, to many the main value of theoretical reaction kinetics is predictive, although the predictions are often themselves of limited application. For this reason, it was felt that *Comprehensive Chemical Kinetics* should include a humble introduction to theoretical kinetics, written from the point of view of an experimentalist who from time to time needs to examine his findings in the light of theoretical predictions.

It is clear that this article cannot hope to give a description of recent advances in theoretical kinetics at the same time as setting out the basic tenets of the theories, nor is it necessarily profitable to do so. A great wealth of material is published in the field of theoretical kinetics which provides increasingly sophisticated approximations to a number of fundamental problems. Other chapters of *Comprehensive Chemical Kinetics* include appropriate discussions of the theoretical interpretations of specific experimental results, and it is the intention of the present author merely to summarise the *concepts* of theoretical gas-phase kinetics within the confines of a single chapter. Other articles will discuss the special considerations applicable to condensed phase systems. A Bibliography is given at the end of the chapter for the reader who wishes to delve more deeply into particular aspects of the theories, and references are given where it is felt that the reader may wish to acquaint himself with the details of a particular piece of research.

A theory of reaction kinetics should ideally predict the rates at which reactants possessing a given energy distributed about specific excitation modes pass over to products of equally well defined energy, and would presumably have as its basis the many-dimensioned potential energy hypersurface linking reactants and products. Such a theory is, however, unrealistic at the present time. In the first place a complete description of the potential surface is not yet possible, although the advent of high speed computers has enabled considerable advances to be made in this respect. Secondly, the experimentally determined rates of reaction are frequently not sufficiently precise, nor the energy distributions of reactants and products well enough known, for unequivocal tests of such theories to be possible.

Thus the theories currently in use are of necessity in the nature of a compromise, and it is hoped that their limitations will become apparent in the following discussion.

The simplest of these theories is the *collision* theory, which considers the collision rate between reactant species assumed to be rigid spheres in thermal equilibrium with their surroundings. It will be shown later that this collision theory can be regarded as a special case of the *transition state* theory, although in accordance with normal practice the theories are treated separately here. This approach is justified in so far as the collision formulation may be quite realistic for certain reaction systems, and the predictions in suitable cases are as good as those obtained from the more complex transition state theory. The transition state theory starts from a consideration of the system at the highest point of the potential surface—the “transition state” or “activated complex”—and its relation to reactants and products. The passage to products could be regarded as a truly unimolecular reaction (as distinct from a kinetically first order reaction) if the activated complex had a real *chemical* identity. Unimolecular reaction of activated chemical species is, of course, well established, and such processes are considered in some detail in a further section of this chapter. The actual activation itself may be derived in some way from a bimolecular process, and these activation steps are discussed, somewhat arbitrarily, together with the subsequent unimolecular reaction. Reactions of molecularity higher than two are relatively rare, and of some interest: they are considered separately.

2. The Arrhenius equation

The main features which a theory of kinetics must interpret are the effect on reaction velocity of variations of reactant concentration and temperature. For an elementary reaction step the molecularity and kinetic order will be numerically identical, and it remains to be seen whether the various theories can provide rate expressions containing the appropriate concentration terms. The experimentally determined rate of an elementary bimolecular reaction



can be represented at any temperature by the expression

$$\text{Rate} = k[A][B] \quad (2)$$

Temperature usually has a marked effect on the rate of a chemical reaction [and thus on the rate coefficient, k , in equation (2)], although the analytical expression for the relation between rate and temperature is less immediately apparent.

Arrhenius¹ suggested the relation which bears his name as long ago as 1889, and it is still universally regarded as the most satisfactory simple way of expressing the rate coefficient for a reaction as a function of temperature. In the form most frequently used it is

$$k = Ae^{-E/RT} \quad (3)$$

where A is a "pre-exponential" constant, and E is a characteristic energy, for the particular reaction. Arrhenius arrived at the relation by consideration of an equilibrium process



The equilibrium is established by equal rates of forward and reverse reaction, and

$$K = \frac{[C][D]}{[A][B]} = \frac{k_f}{k_r} \quad (5)$$

Since

$$\frac{d \ln K}{dT} = \frac{\Delta H}{RT^2} \quad (6)$$

where ΔH is the heat of reaction,

$$\frac{d \ln k_f}{dT} - \frac{d \ln k_r}{dT} = \frac{\Delta H}{RT^2} \quad (7)$$

Arrhenius suggested that ΔH could be split into two components, E_f and E_r , such that $\Delta H = E_f - E_r$, and argued that (7) could then be separated into two equations of the form

$$\frac{d \ln k}{dT} = \frac{E}{RT^2} \quad (8)$$

Integration of (8) then gives $\ln k = -E/RT + \text{constant}$, or

$$k = Ae^{-E/RT} \quad (3)$$

That (7) can indeed be separated into the two equations (8) is adequately justified by the vast amount of kinetic data which is roughly consistent with equation (3).

It is now realised, on both theoretical and experimental grounds, that neither A nor E is necessarily temperature independent, although the largest contribution to the temperature dependence of k almost always arises from the exponential term. Some discussion of these matters will be found in the following sections.

3. Collision theory

3.1 PRINCIPLES

It seems reasonable to suppose that before two species A and B can participate in a chemical reaction they must approach each other sufficiently closely for there to exist some interaction between them. The *collision theory*, in its simplest form, goes further than this, and suggests that the reactant molecules must have their centres separated by the sum of the gas-kinetic radii before reaction can occur: that is to say, a "collision" must take place. The rate of reaction will therefore be related to the gas-kinetic collision rate, Z_{AB} , between the two species A and B . The kinetic theory of gases can be used to calculate a value for the collision rate

$$Z_{AB} = n_A n_B \left\{ \frac{\sigma_A + \sigma_B}{2} \right\}^2 \left(\frac{8\pi kT}{\mu} \right)^{\frac{1}{2}} \quad (9)$$

where Z_{AB} is the number of collisions per cm^3 per second

n_A, n_B are the numbers of A and B molecules per cm^3

σ_A, σ_B are the molecular diameters of A and B

$\mu = \frac{m_A m_B}{m_A + m_B}$ is the "reduced mass" of A and B .

This expression predicts correctly the concentration dependence of the rate of reaction, although the magnitude of Z_{AB} is much greater than the experimentally determined rate for most reactions. Furthermore, the observed variation of reaction rate with temperature bears little relation to the $\sqrt{T}$ dependence suggested by (9). It is clear, therefore, that the rate coefficient for reaction is not to be identified directly with $Z_{AB}/n_A n_B$, and that some modification of the theory is needed to introduce the dependence of rate on temperature suggested by the Arrhenius equation. The required modification is to assume that not all collisions are reactive, but rather that reaction occurs only after those collisions which possess relative kinetic energy greater than a critical value, E_c , along the line of collision centres. It will be shown in the next section that the rate at which such energetic collisions

occur is equal to

$$Z_{AB}e^{-\epsilon_c/kT} = Z_{AB}e^{-E_c/RT} \quad (10)$$

The equation for the rate of "energetic" collisions is now of a form directly equivalent to the Arrhenius expression (ϵ_c is the excess energy per molecular collision while E_c is a *molar* quantity $E_c = N_0\epsilon_c$ where N_0 is Avogadro's number). That collisions need to possess energy in excess of ϵ_c suggests that there is some energy barrier to reaction which must be overcome. This barrier is related to the energy required to form the "activated complex" of the transition state theory (see section 4), and its height is associated with the potential energy of the highest point on the potential energy surface connecting reactants and products. For this reason, the energy is universally known as the *Activation Energy*, E_a ; E of the Arrhenius expression (3) and E_c of the rate of collision with energy greater than E_c (10) are usually referred to as activation energies, E_a , although according to the definitions of (3) and (10) they are not necessarily identical. We shall adopt the practice of referring to $E = E_a$ as the activation energy (per mole) since this is normally the experimentally determined quantity.

The simple collision theory clearly has a number of inadequacies. First, the theory treats the reactants as hard spheres; secondly it assumes that collision in all directions is equally favourable; and thirdly it makes no allowance for interactions at a longer range than so-called collision. It is, perhaps, surprising for how many simple reactions the observed pre-exponential factor A is of the same order of magnitude as Z_{AB} . The inadequacies of the theory do, however, lead to an even larger number of incorrect predictions, and it has become common practice to multiply the predicted rate $Z_{AB}e^{-E_c/RT}$ by a probability factor P which is chosen, more or less empirically, to fit the experimentally determined rates. It may then be convenient to define a reaction cross section, σ_r^2 , for the reaction such that

$$\sigma_r^2 = P \left\{ \frac{\sigma_A + \sigma_B}{2} \right\}^2 \quad (11)$$

This cross section then allows for the geometrical factors and for the unrealistic assumption of a hard collision. Various attempts have been made to modify the collision theory to allow for the factors described above. For example, Hinshelwood derived (see, for example, ref. 2) the fraction of effective collisions if the critical energy, E_c , could be accommodated in degrees of freedom other than the kinetic energy of collision. The derivation, which is discussed in section 3.6.3, gives the chance of a total energy exceeding E_c without restriction as to how the energy is shared among the separate modes. It does not, however, take any account of the possibility that species may react preferentially if the energy is in one mode rather than another. More recent suggestions concerning modifications to the collision theory are discussed in section 3.7.

3.2 RATE OF BIMOLECULAR COLLISIONS

The gas-collision rate Z_{AB} between two hard spheres of diameter σ_A , σ_B and masses m_A , m_B may be derived from the simple kinetic theory of gases, and the derivation is given in most textbooks. In this section, an expression is obtained for the rate of collisions possessing a relative kinetic energy along the line of centres greater than ϵ_c . The most straightforward way of obtaining the expression is to use the Maxwell distribution law applied to the relative velocity of two colliding species A and B to give the collision rate distribution as a function of velocity. The condition that the energy along the line of centres shall be greater than ϵ_c is then applied in the form of a velocity restriction.

The distribution law for the relative velocity, r , between any two species A and B may be given in the form

$$dn_{AB} = 4\pi n_A n_B \left(\frac{\mu}{2\pi kT} \right)^{\frac{3}{2}} e^{-\mu r^2/2kT} r^2 dr \quad (12)$$

where dn_{AB} is the number of pairs of A and B whose relative velocity lies between r and $r+dr$. [The other symbols have the meaning given in connection with equation (9)].

Now suppose that this relative velocity lies at an angle between θ and $\theta+d\theta$ to the line of centres. If σ is taken to be $\frac{1}{2}(\sigma_A + \sigma_B)$, then the possible orientations of B during collision are those in which the centre of B lies on a ring-shaped surface on a sphere of radius σ centred on A. The surface has an area $2\pi\sigma^2 \sin \theta d\theta$, so that as A moves with respect to B it sweeps out a volume

$$V = 2\pi\sigma^2 \sin \theta d\theta \cdot r \cos \theta \quad (13)$$

in unit time. The number of collisions occurring in unit volume and unit time, dZ , whose relative velocity lies between r and $r+dr$ at an angle between θ and $\theta+d\theta$ to the line of centres may now be calculated from equations (12) and (13)

$$dZ = 8\pi^2 n_A n_B \left(\frac{\mu}{2\pi kT} \right)^{\frac{3}{2}} e^{-\mu r^2/2kT} r^3 dr \sigma^2 \sin \theta \cos \theta d\theta \quad (14)$$

Now for the kinetic energy to be greater than ϵ_c along the line of centres of A and B it follows that $\frac{1}{2}\mu r^2 \cos^2 \theta > \epsilon_c$. Thus the total number of collisions of required energy is obtained by integrating (14) between the limits $\theta = 0$, $\theta = \cos^{-1}(2\epsilon_c/\mu r^2)^{\frac{1}{2}}$; $r = (2\epsilon_c/\mu)^{\frac{1}{2}}$, $r = \infty$

$$Z(\epsilon > \epsilon_c) = n_A n_B \sigma^2 \left(\frac{8\pi kT}{\mu} \right)^{\frac{3}{2}} e^{-\epsilon_c/kT} \quad (15)$$

The expression is identical with that given by equations (9) and (10) with $\sigma = \frac{1}{2}(\sigma_A + \sigma_B)$. Note that ε_c is here the critical energy per *molecule*; in practice it is more convenient to work in terms of the energy per *mole* so that k is replaced by R in the exponential term.

3.3 COLLISIONS BETWEEN LIKE SPECIES

So far only collisions between unlike species, A and B, have been considered. Bimolecular reaction between like species



is of chemical importance, and it is of interest to consider how the rate coefficient, in terms of the collision theory, is to be adjusted for this type of reaction.

The rate of collisions with energy greater than ε_c can be obtained by setting $m_A = \mu/2$ in equation (15) and dividing by two. The factor of two is required since Z_{AB} effectively counts the rate of collision between a given A and all B and then multiplies the result by the number of A. To do this for Z_{AA} would count every collision twice. Thus

$$Z_{AA}(\varepsilon > \varepsilon_c) = 2n_A^2 \left(\frac{\pi kT}{m_A} \right)^{\frac{1}{2}} \sigma_A^2 e^{-\varepsilon_c/kT} \quad (17)$$

There is, however, a problem in setting the rate coefficient for reaction (16) equal to $PZ_{AA} e^{-\varepsilon_c/kT}/n_A^2$. The rate equation for the reaction of unlike species in (1) is universally given in terms of the reactants, *i.e.*

$$\frac{-dn_A}{dt} = \frac{-dn_B}{dt} = k[A][B] \quad (18)$$

It is desirable, therefore, that the rate equation for (16) should be given in the same form

$$\frac{-dn_A}{dt} = k'[A]^2 \quad (19)$$

Every effective collision counted by Z_{AA} removes *two*, rather than one, A, so that the rate of removal of reactant is twice the rate of effective collision. Thus we should write

$$k' = 2PZ_{AA} e^{-\varepsilon_c/kT}/n_A^2 \quad (20)$$

The argument used above is by no means artificial. Consider a system consisting of equal concentrations of an alkyl radical C_nH_{2n+1} and a monodeuterated radical $C_nH_{2n}D$. The following combination reactions may occur



According to our definition for rate coefficients, the rate coefficients for (21), (22) and (23) will be approximately *identical* if the radicals are sufficiently massive. This result accords with the normal idea of a rate coefficient. On the other hand, the rate of formation of the product $C_{2n}H_{4n+1}D$ in (22) will be twice as great as the rate of formation of the other two products. Experimental determinations of rates of product formation in reaction systems involving even two alkyl radicals of *different* masses show that the cross-combination product appears approximately twice as fast as the other combination products.

3.4 COMPARISON WITH EXPERIMENT: THE "P" FACTOR

A complete comparison of the predicted rate coefficients with experimentally determined values would require a knowledge not only of the mass and collision diameter of the reactants, but also of the critical energy for reactive collisions. While m is in principle a known quantity and some estimate of σ may be obtained from, say, viscosity data, the calculation of E_c from data not involving rate determinations presents some difficulty. For the purposes of this section, E_c is set equal to the experimental activation energy, E_a , so that the comparisons to be made are essentially of the Arrhenius pre-exponential " A " factor and the calculated collision rate at unit concentration. Some remarks are offered in section 3.4 about calculations of E_c without reference to rate data.

The calculation of A from the rate of a reaction is sensitive to the value of E_a , and any attempt to calculate A from the activation energy and the rate constant at a given temperature can give a grossly misleading value of A if E_a is in error. It is probably better to obtain A from the logarithmic form of the Arrhenius equation which is used to calculate E_a

$$\ln k = \ln A - E_a/RT \quad (24)$$

The value of A is then the extrapolated value of k at $1/T = 0$. This practice should give the most reliable value of A , although it still depends critically upon the extrapolation and thus on the value chosen for E_a . Since the rate determinations

may be over a relatively narrow temperature range, the extrapolation to $1/T = 0$ can introduce severe errors into the calculated value of A . Thus an agreement of an order of magnitude between experimental and theoretical pre-exponential factors is the best that can be expected.

For a typical elementary gas-phase reaction, the molecular or atomic weights will be about 40, so that $\mu = 20$, and the molecular diameter, σ , about 3×10^{-8} A. At 300° K this gives a value of $A_{\text{calculated}} = Z_{AB}/n_A n_B \approx 2 \times 10^{-10} \text{ cm}^3 \cdot \text{molec}^{-1} \cdot \text{sec}^{-1}$. Changes in either temperature or reduced mass have little effect on this value since they both appear under the square root sign. Even for the reaction



performed at 600° K, for which μ is but slightly greater than two, the value of $A_{\text{calculated}}$ is only $\sqrt{20}$ times (*i.e.* less than five times) faster than that calculated for $\mu = 20$, $T = 300^\circ$ K. Thus under the ordinary conditions of chemical reactions in the gas phase, the pre-exponential factor lies between about 10^{-10} and $10^{-9} \text{ cm}^3 \cdot \text{molec}^{-1} \cdot \text{sec}^{-1}$ or about 10^{11} and 10^{12} in $\text{l. mole}^{-1} \cdot \text{sec}^{-1}$ units.

Table 1 gives values of $\log A$ (for A in $\text{l. mole}^{-1} \cdot \text{sec}^{-1}$ units) and E_a (in kcal. mole^{-1}) for a number of elementary reactions, and it is arranged in order of

TABLE 1

RATE DATA FOR SOME BIMOLECULAR REACTIONS ARRANGED IN ORDER OF DECREASING PRE-EXPONENTIAL FACTOR (A) [$\text{l. mole}^{-1} \cdot \text{sec}^{-1}$ units]

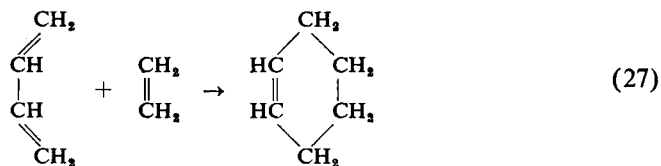
Reaction	$\log A$	E_a (kcal. mole^{-1})	Reference
$\text{H} + \text{O}_2 \rightarrow \text{OH} + \text{O}$	11.1	18.8	6
$\dagger \text{H}_2 + \text{I}_2 \rightarrow 2\text{HI}$	11.1	39.0	3
$2\text{HI} \rightarrow \text{H}_2 + \text{I}_2$	10.9	44.0	3
$\text{OH} + \text{H}_2 \rightarrow \text{H}_2\text{O} + \text{H}$	10.9	10.0	7
$\text{Cl} + \text{H}_2 \rightarrow \text{HCl} + \text{H}$	10.9	5.5	8
$\text{H} + \text{H}_2 \rightarrow \text{H}_2 + \text{H}$	10.7	7.5	9
$\text{CH}_3 + \text{CH}_3 \rightarrow \text{C}_2\text{H}_6$	10.3	0	10
$\text{NO}_2 + \text{O}_3 \rightarrow \text{NO}_3 + \text{O}_2$	9.8	7.0	11
$\text{NO} + \text{Cl}_2 \rightarrow \text{NOCl} + \text{Cl}$	9.6	20.3	12
$\text{H} + \text{C}_2\text{H}_6 \rightarrow \text{H}_2 + \text{C}_2\text{H}_5$	9.5	6.8	13
$\text{NO} + \text{O}_3 \rightarrow \text{NO}_2 + \text{O}_2$	8.8	2.5	14
$\text{CH}_3 + \text{C}_2\text{H}_4 \rightarrow \text{C}_3\text{H}_7$	8.4	~ 7	15
$\text{CH}_3 + \text{C}_2\text{H}_6 \rightarrow \text{CH}_4 + \text{C}_2\text{H}_5$	8.3	10.4	16
$\text{C}_4\text{H}_6 + \text{C}_4\text{H}_6 \rightarrow \text{cyclo-C}_8\text{H}_{12}$	8.1	26.8	17
$\text{CH}_3 + \text{C}_3\text{H}_6 \rightarrow \text{C}_4\text{H}_9$	8.0	~ 6	15
$\text{C}_4\text{H}_6 + \text{C}_2\text{H}_4 \rightarrow \text{cyclo-C}_6\text{H}_{10}$	7.5	27.5	17
$\text{O}_3 + \text{C}_3\text{H}_8 \rightarrow \text{C}_3\text{H}_7\text{O} + \text{HO}_2$	6.5	12.1	18

$\dagger$ The H_2/I_2 reaction has been included because of its importance in the historical development of the theory of gas-phase kinetics, although recent work⁵ suggests that it is not in fact a simple bimolecular process.

decreasing A . It is seen that the pre-exponential factor observed for the first few of these reactions is quite close to that calculated above. The hydrogen-iodine reaction (25) rate data of Bodenstein³ was in fact used by McC. Lewis⁴ in his earliest comparisons (1918) of the collision theory with experiment. The empirical probability factor, P , is defined by

$$P = A_{\text{observed}}/A_{\text{calculated}} \quad (26)$$

and for the hydrogen-iodine reaction is about 0.33. P lies in the range 0.1 to 1 for the simplest reactions given in Table 1, but becomes progressively smaller with an increase in complexity of the reaction. Two factors contribute to this result. First is the so-called "steric factor". The collision theory does not consider the necessity for reaction of any specific geometric orientation of the reactants. It is, however, quite apparent that the reactive portions of reactants must be brought geometrically close to each other before chemical change can take place. To some extent, therefore, the probability factor P may be taken as representing the fraction of collisions with the correct orientation. Of the reactions in Table 1, the Diels-Alder cyclizations of butadiene on its own, or with ethylene are good examples of reactions with low steric P factors. The reaction of butadiene with ethylene to form cyclohexene



clearly requires rather specifically oriented collisions, and, in agreement with this expectation, the P factor for this reaction (27) is of the order of 10^{-4} . The second factor which contributes towards the overall magnitude of P concerns the distribution of the kinetic energy of collision. There is no reason to suppose that all the kinetic energy of a collision will appear where or when it is needed to overcome the barrier to reaction, and the probability of enough energy being at the right place will presumably decrease as the total number of bonds increases (see, however, section 3.6.3). Once again, this general conclusion is borne out for the reactions shown in Table 1: for example, the P factor of the reaction



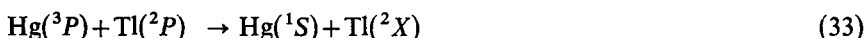
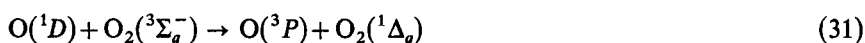
is about one tenth that of



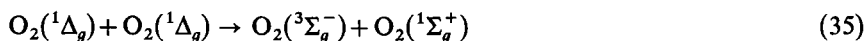
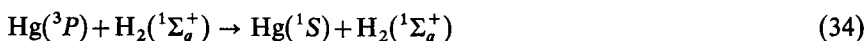
Certain further, rather special, cases of reactions with low P factors must be discussed. The first of these concerns processes in which there is a change of electronic state during reaction: perhaps the most important members of this class are those reactions in which there is a change of multiplicity. Such reactions are usually known as "non-adiabatic" processes, although this description is rather misleading. An adiabatic process is certainly one in which there is no change in electronic state, and it can be shown that the probability of a non-adiabatic transition between two potential energy surfaces decreases rapidly as the smallest energy separation increases. The perturbation energy of spin-spin or spin-orbital interactions is small, so that for processes involving a change in resultant spin of a system the probability of the process occurring adiabatically but with the change in multiplicity is low. Thus the probability of the adiabatic reaction



is low. Emphasis is placed on this point since reactions such as (30) are frequently said to be non-adiabatic and therefore of low probability. In practice confusion does not arise, since it is relatively easy to decide for which reactions *as written* the probability is low. In particular, processes for which there is a change of multiplicity may be expected to be slow: this postulate is known as the Wigner Spin Conservation rule¹⁹. For example, the reactions



are "spin-allowed" since it is possible to find a total spin quantum number which is common to both reactants and product systems. On the other hand, processes such as



are "spin-forbidden", and, other things being equal, will be slow. Adiabaticity is discussed further in section 4.5.7.

Reactions involving the exchange of energy in modes other than electronic may also proceed with a reduced probability as a result of restrictions on the exchange of energy. The transfer of vibrational energy to translation is inefficient if the vibrational energy levels are widely spaced (*i.e.* if $h\nu \gg kT$) while vibration-vibration transfer proceeds slowly if the energy discrepancy between vibrational levels

of the exchanging species, which must be taken up by translation, is large (see, for example, the review by Callear²⁰). Thus in quenching or de-activation reactions the P factor is likely to be small unless certain closely defined conditions are fulfilled.

A reaction in which the product molecules can spontaneously and rapidly revert to reactants will have an overall rate considerably less than that predicted on the basis of the collision rate of reactants. The most important members of this class are the bimolecular association reactions. For example, the hydrogen molecule formed in the process



possesses as vibration in the newly-formed H-H bond all the heat of reaction. Since (36) proceeds with little or no activation energy, it follows that H_2^* will dissociate in the course of a single vibration unless it can somehow be stabilised in that time. Collision stabilization means that the reaction is effectively termolecular



and will be treated further in a later section. The true bimolecular rate is therefore very much smaller than the collision frequency. With increasing complexity of the reacting species a greater number of additional bonds will exist in the product which can accommodate the heat of reaction, and bimolecular association may then proceed at a normal rate. For the association of atoms, however, no such vibrational degrees of freedom exist, and the reactions proceed only *via* a termolecular mechanism. It might be noted at this point that the possibility of alternative reaction paths will alter the probability of any one path being followed. For example, in the association of two radicals (doublet species) there is one singlet and a (triply degenerate) triplet product which may not be an attractive state. The chance of a reaction following the path leading to the stable singlet is thus one in four, and a P factor of $\frac{1}{4}$ might be expected for the reaction. The question of statistical weights for reactions is explored further in section 4.3.3.

We have seen that, except for the very simplest of reactions, the empirical P factor tends to be considerably less than unity. A number of reasons for the low probability of reactions has been discussed, although no attempt has been made to make numerical estimates of the value of P . Such estimates are, in fact, not possible on the basis of the simple collision theory, although, of course, it may be possible to guess from experience with similar reactions whether P for a given reaction will be large or small. The transition state theory (section 4) can make allowance for certain of the factors treated empirically by the collision theory, and, in particular, it identifies the "geometric" parts of P with an "entropy of activation".

3.5 CALCULATION OF ACTIVATION ENERGIES

The activation energy of a chemical reaction may be looked upon as the energy of the highest point of the reaction path over a potential surface connecting reactants and products. Prediction of an activation energy requires in principle a method for the determination of the form of the potential energy surface. The calculation of potential energy surfaces is discussed in some detail in section 4.5, and consideration of this topic is deferred. Such calculations are naturally of the greatest interest, since a complete knowledge of the potential surface would enable equally complete description of the reaction. Unfortunately, the methods at present available for the prediction even of just the maximum energy on the reaction path, let alone of the entire potential surface, give results which in many cases differ considerably from experimental data. In view of the great influence that the value chosen for the activation energy has on the calculated rate coefficient, it must be concluded that absolute calculations of activation energies are not sufficiently accurate for rate predictions in most instances. For the purposes of this section, therefore, some *empirical* methods for the calculation of activation energies are presented. The methods have proved successful for a number of reactions, and some examples of predicted and measured activation energies will be discussed.

Hirschfelder²¹ has stated rules for the estimation of activation energies from bond energy data. For metathetical reactions of the type



a rule was formulated to give agreement with activation energy and bond energy data for the hydrogen-iodine reaction. The relation used is

$$E_a \sim 0.28(Q_1 + Q_2) \quad (39)$$

where Q_1 , Q_2 are the dissociation energies of the bonds being broken. On the basis of semi-empirical calculations Hirschfelder suggested that for an exothermic displacement reaction involving atoms or simple radicals



the activation energy could be expressed by the relation

$$E_a \sim 0.055 Q \quad (41)$$

where, in this case, Q is the B-C bond dissociation energy. An endothermic reaction may be treated by the same methods.

Fig. 1 illustrates diagrammatically the relationship between heat of reaction and

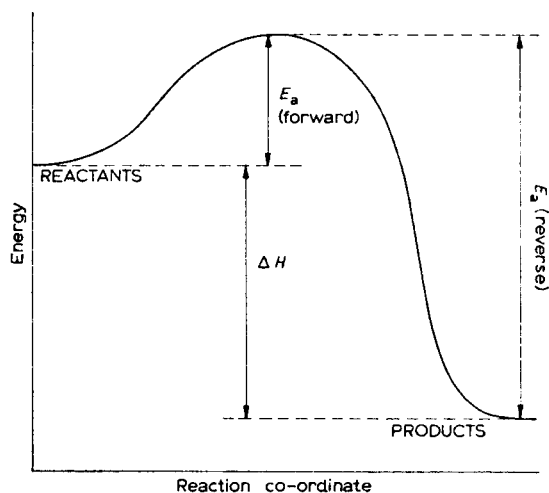


Fig. 1. Relationship between heat of reaction and activation energy for forward and reverse reactions.

activation energy for forward and reverse reactions. ΔH is the enthalpy change of reaction (and is taken to be negative for an exothermic reaction). It is apparent that

$$E_{a(\text{reverse})} = E_{a(\text{forward})} - \Delta H_{(\text{forward})} \quad (42)$$

The Hirschfelder rules may now be applied to endothermic reactions by calculating the activation energy for the reverse (exothermic) reaction and applying (42) to obtain the activation energy for the forward reaction.

Polanyi *et al.*^{22,23} investigated the series of reactions



where M = alkali metal and RX = alkyl halide. They suggested that the activation barrier was a result of the energy required to expand the X-R bond so that the postulated transition states MX^-R^+ and $\text{M}^+\text{X}^-\text{R}$ became degenerate. They then obtained the height of the activation barrier from the intersection point of the X-R attractive and repulsive curves, and proposed that

$$E_a = \text{constant} - \alpha \Delta H \quad (44)$$

where α is a constant lying in the range 0 to 1. For the series of reactions



the experimental results were consistent with (44) with a value of $\alpha = 0.27$. Semenov²⁴ subsequently empirically extended Polanyi's equation (44) for a number of exothermic atom and radical reactions by using the general relation

$$E_a \sim 0.25\Delta H + 11.5 \quad (46)$$

The appropriate value for endothermic reactions (ΔH positive) can once again be obtained by considering first the reverse exothermic reaction: the result is

$$E_a \sim 0.75\Delta H + 11.5 \quad (47)$$

Both Hirschfelder's and Semenov's empirical estimates of activation energies can only give a rough idea of the activation energy to be expected. In particular, the rules tend to give the same activation energies for a series of related reactions in which similar bonds are broken. Table 2 shows the experimental activation energies for a number of abstraction reactions together with the estimates made using the Hirschfelder and Semenov rules. The calculated activation energies do reflect the trends of the experimental values, although their absolute magnitudes frequently differ by a factor of two or more. For example, for the reaction



the activation energy calculated by the Semenov method is 10 kcal. mole⁻¹, while the measured value is about 3 kcal. mole⁻¹. This difference corresponds to a difference of about 10⁵ in the rate of reaction at room temperature, and it is clear

TABLE 2
EXPERIMENTAL AND CALCULATED ACTIVATION ENERGIES FOR SOME ABSTRACTION REACTIONS

Reaction	$E_a(\text{experimental})$ (kcal. mole ⁻¹)	ΔH (kcal. mole ⁻¹)	Calculated activation energies (kcal. mole ⁻¹)		Reference
			Hirschfelder	Semenov	
$\text{H} + \text{H}_2 \rightarrow \text{H}_2 + \text{H}$	7.5	0	6	12	9
$\text{Cl} + \text{H}_2 \rightarrow \text{HCl} + \text{H}$	5.5	+1	7	12	8
$\text{Br} + \text{H}_2 \rightarrow \text{HBr} + \text{H}$	17.6	+17	22	24	25
$\text{H} + \text{CH}_4 \rightarrow \text{H}_2 + \text{CH}_3$	6.6	-2	6	11	26
$\text{Cl} + \text{CH}_4 \rightarrow \text{HCl} + \text{CH}_3$	3.9	-1	6	11	27
$\text{Br} + \text{CH}_4 \rightarrow \text{HBr} + \text{CH}_3$	18.3	+15	20	23	28
$\text{K} + \text{HBr} \rightarrow \text{KBr} + \text{H}$	3.4	-5	5	10	29
$\text{OH} + \text{H}_2 \rightarrow \text{H}_2\text{O} + \text{H}$	10.0	-15	6	7	7
$\text{CH}_3 + \text{C}_2\text{H}_6 \rightarrow \text{CH}_4 + \text{C}_2\text{H}_5$	13.0	-3	5	11	16
$\text{NO} + \text{O}_3 \rightarrow \text{NO}_2 + \text{O}_2$	2.5	-47	1	~0	14
$\text{NO}_2 + \text{O}_3 \rightarrow \text{NO}_3 + \text{O}_2$	7.0	-25	1	5	11

References pp. 298-301

that no great reliance can be placed on rates calculated from estimated activation energies.

3.6 TEMPERATURE DEPENDENCE OF A AND E

The simple form of the Arrhenius equation

$$k = Ae^{-E_a/RT} \quad (3)$$

implies that neither A nor E_a depends upon the temperature. That the implication is unfounded is demonstrated by experimental results which demand that either or both of the "constants" should, in fact, exhibit a temperature dependence. This dependence can arise from a number of causes, operating singly or in combination, and some of these causes are considered here.

3.6.1 Variation of collision frequency with temperature

The basic assumption of the collision theory is that the rate coefficient, k , for a bimolecular reaction is given by the equation

$$k = \frac{Z_{AB}}{n_A n_B} e^{-\varepsilon_c/kT} \quad (49)$$

Since

$$Z_{AB} = n_A n_B \left(\frac{\sigma_A + \sigma_B}{2} \right)^2 \left(\frac{8\pi kT}{\mu} \right)^{\frac{1}{2}} \quad (9)$$

it follows that

$$\frac{d \ln k}{dT} = \frac{1}{2T} + \frac{\varepsilon_c}{kT^2} = \frac{1}{2T} + \frac{E_c}{RT^2} \quad (50)$$

However, according to the Arrhenius expression (3)

$$\frac{d \ln k}{dT} = \frac{E_a}{RT^2} \quad (51)$$

equations (49) and (50) can be reconciled only if the experimental activation energy is temperature dependent in such a way as to accommodate the variation of Z_{AB}

with temperature. That is

$$E_a = E_c + \frac{1}{2}RT \quad (52)$$

Most determinations of activation energies employ results obtained over relatively narrow temperature ranges, and deviations from linearity of a $\ln k-1/T$ curve may well be masked by scatter of the experimental points. It should, however, be noted that the absolute value of activation energy measured from the curve will be in error by the amount $\frac{1}{2}RT$, which in many cases may not be negligible in comparison with E_c .

The temperature dependence of the collision rate might more properly be accommodated by allowing A of the Arrhenius equation to be temperature dependent. That is to say, the Arrhenius equation might be modified to read

$$k = A'\sqrt{T}e^{-E_a'/RT} \quad (53)$$

where A' is a temperature invariant pre-exponential factor. The value of E_a' obtained from (53) is *not* the same as E_a of the ordinary equation, and, in fact, quoted activation energies normally apply to E_a of equation (3).

3.6.2 Variation of barrier height with temperature

The relationship between enthalpy change of reaction, ΔH , and the height of the activation barrier, E_c , for forward and reverse reactions has been discussed already in terms of Fig. 1

$$\Delta H_{(\text{forward})} = E_{c(\text{forward})} - E_{c(\text{reverse})} \quad (42a)$$

Kirchoff's well-known law relates the change of ΔH to the specific heats of reactants and products for a chemical reaction

$$\frac{\partial(\Delta H)}{\partial T} = \sum_i C_{p_i} \quad (54)$$

and makes it clear that in general ΔH will vary with temperature. In principle, therefore, activation energies for both forward and reverse reactions will exhibit a temperature dependence, although in practice the variation from this cause of E_c with temperature is small. [Equation (54) is itself a simplification, since C_{p_i} may be a function of temperature].

3.6.3 Activation in many degrees of freedom

According to the simple collision theory, an activation barrier is overcome by the kinetic energy of collision between two "hard sphere" reaction partners, and takes no account of the energy of excitation which may be possessed by them in internal modes of excitation (in particular, by vibrations). The methods of classical statistical mechanics allow calculation of the number of species possessing a total energy greater than any given amount *in many degrees of freedom, and without reference to the way in which it is shared between them*. The result is

$$\frac{N(E > E_c)}{N} = \frac{1}{\Gamma(\frac{1}{2}S)(RT)^{\frac{1}{2}S}} \int_{E_c}^{\infty} e^{-E/RT} E^{\frac{1}{2}S-1} dE \quad (55)$$

where S is the number of *squared terms* which describe the energy; that is, each translation or rotation corresponds to one squared term, while each vibration corresponds to two squared terms. If $\frac{1}{2}S$ is an integer, we may write:

$$\Gamma(\frac{1}{2}S) = (\frac{1}{2}S - 1)! \quad (56)$$

and, therefore

$$\frac{N(E > E_c)}{N} = e^{-E_c/RT} \left[\frac{1}{(\frac{1}{2}S - 1)!} \left(\frac{E_c}{RT} \right)^{\frac{1}{2}S-1} + \frac{1}{(\frac{1}{2}S - 2)!} \left(\frac{E_c}{RT} \right)^{\frac{1}{2}S-2} \dots + 1 \right] \quad (57)$$

If E_c/RT is large, then (57) simplifies to

$$\frac{N(E > E_c)}{N} = \frac{e^{-E_c/RT} \left(\frac{E_c}{RT} \right)^{\frac{1}{2}S-1}}{(\frac{1}{2}S - 1)!} \quad (58)$$

Thus, if activation in many degrees of freedom can, in fact, overcome an energy barrier to reaction, then the reaction rate is equal to

$$Z_{AB} e^{-E_c/RT} \cdot \frac{\left(\frac{E_c}{RT} \right)^{\frac{1}{2}S-1}}{(\frac{1}{2}S - 1)!} \quad (59)$$

It follows that the experimental activation energy differs from E_c , since

$$E_a = RT^2 \frac{d \ln k}{dT} = E_c - (\frac{1}{2}S - 1)RT \quad (60)$$

and E_a will exhibit a linear temperature dependence.

It is of interest to derive a value for the experimental activation energy from the expression (55) for $N(E > E_c)/N$ which does not employ the simplifications introduced. The rate coefficient, k , may be written

$$k = B \cdot \frac{N(E > E_c)}{N} = B \cdot \frac{1}{\Gamma(\frac{1}{2}S)(RT)^{\frac{1}{2}S}} \int_{E_c}^{\infty} e^{-E/RT} \cdot E^{(\frac{1}{2}S-1)} dE \quad (61)$$

where B is a constant.

Thus

$$\begin{aligned} \frac{dk}{dT} = & \left\{ \frac{B}{\Gamma(\frac{1}{2}S)} \cdot (RT)^{-\frac{1}{2}S} \int_{E_c}^{\infty} \frac{E}{RT^2} e^{-E/RT} \cdot E^{(\frac{1}{2}S-1)} dE \right\} \\ & - \left\{ \frac{S}{2} \cdot \frac{B}{\Gamma(\frac{1}{2}S)} \cdot R^{-\frac{1}{2}S} \cdot T^{(-\frac{1}{2}S-1)} \int_{E_c}^{\infty} e^{-E/RT} \cdot E^{(\frac{1}{2}S-1)} dE \right\} \end{aligned} \quad (62)$$

and

$$E_a = RT^2 \frac{d \ln k}{dT} = \frac{\int_{E_c}^{\infty} E \cdot e^{-E/RT} \cdot E^{(\frac{1}{2}S-1)} dE}{\int_{E_c}^{\infty} e^{-E/RT} \cdot E^{(\frac{1}{2}S-1)} dE} - \frac{1}{2} S RT \quad (63)$$

The quotient of the two integrals is the *average* energy of all species with energy greater than E_c , while $\frac{1}{2}SRT$ is the *average* energy of all molecules.

Another important consequence of activation in many degrees of freedom is that the rate coefficient predicted from (59) on the basis of an experimental activation energy may be considerably greater than that predicted by the simple collision theory. Using (60) to give a value for E_c in terms of E_a , the ratio of the two rate constants is:

$$e^{-(\frac{1}{2}S-1)} \cdot \frac{\left\{ \frac{E_a}{RT} + (\frac{1}{2}S-1) \right\}^{\frac{1}{2}S-1}}{(\frac{1}{2}S-1)!} \quad (64)$$

In a typical case, S might be 10 and E_a about 60,000 cal. mole⁻¹; RT at room temperature is about 600 cal. mole⁻¹, so that the rate coefficient calculated from the simple collision theory is between 10⁵ and 10⁶ times slower than that expected if energy in other degrees of freedom can help to overcome the activation barrier. This result can be looked upon as an explanation of probability factors greater than unity in the simple theory.

If energy possessed by various degrees of freedom is to overcome the activation barrier to a bimolecular reaction it must be able to appear in the place and form

that it is needed *during the lifetime of a "collision"*. It might be expected, therefore, that activation in many degrees of freedom would be of greater importance in unimolecular reactions, since the species undergoing reaction retains its identity until it *does* react (although it can lose energy by quenching processes). Such experimental evidence as there is for the importance of activation in many degrees of freedom does, indeed, tend to concern unimolecular reactions, and discussion of the evidence will be deferred until section 5, where unimolecular reactions are considered in some detail.

3.6.4 Quantum mechanical tunnelling

Quantum mechanical tunnelling cannot properly be treated without reference to the forms of potential energy surfaces for reaction, and the subject is therefore discussed in section 4.5.6. For the present purposes, it is intended to investigate qualitatively any effect that tunnelling might have on the temperature dependence of A or E_a .

The picture of the activation barrier adopted so far is of a classical barrier through which no leakage may occur: the energy possessed by species is either sufficient to overcome the barrier or it is not. However, according to quantum mechanical reasoning there is a certain finite probability of a particle "tunnelling" through a barrier, and the probability depends, amongst other things, on the height and shape of the barrier, and is most important for particles of low mass such as electrons, and to a lesser extent protons and deuterons. Further, the probability of tunnelling increases with a decrease in temperature.

If tunnelling can occur, then reaction will proceed more rapidly than would be expected from the classical barrier height, and, in particular, the increase in reaction rate over that expected will become larger as the temperature is decreased. Fig. 2 shows the type of behaviour that might be expected for a $\log k - 1/T$ plot with and without tunnelling.

It is seen that the experimental activation energies will be temperature dependent, and will decrease with a decrease in temperature. Values of A calculated from $\log k - 1/T$ curves will also be affected by the deviation from linearity, since the extrapolation of experimental results obtained at finite temperatures to $1/T = 0$ will give a lower A than the "true" value of k at $1/T = 0$.

There appear to be a number of reactions in solution for which there are deviations from linearity of Arrhenius plots which can only reasonably be explained in terms of quantum mechanical tunnelling. These reactions will be discussed in the chapter on the Theory of Reactions in Solution. The evidence for the importance of tunnelling in gas-phase reactions is generally less convincing. The higher temperatures at which gas-phase reactions are usually performed tend to reduce the effect of tunnelling, while the accuracy of kinetic measurements in the gas phase is generally lower than that of solution phase measurements, and it is dif-

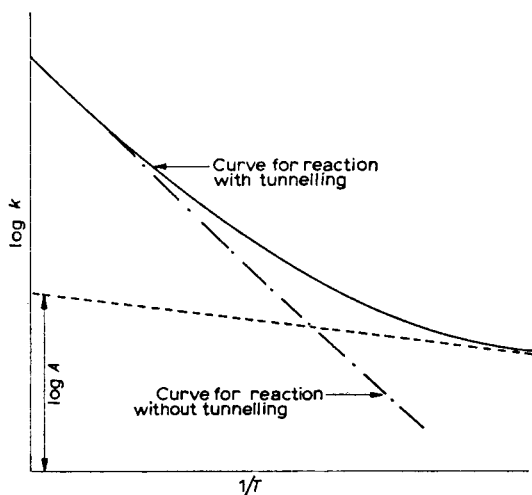


Fig. 2. Deviation from linearity of a $\log k$ versus $1/T$ plot for a reaction with quantum mechanical tunnelling.

difficult to demonstrate significant deviations of Arrhenius plots from linearity. Johnston *et al.*³⁰⁻³² have made a number of searches for experimental evidence of tunnelling, and have obtained some support for the view that tunnelling does occur. The most recent studies³³ of the $\text{H} + \text{H}_2$ reaction also show that the temperature dependence of reaction rate can only be explained reasonably on the basis of tunnelling.

3.7 SOPHISTICATED COLLISION THEORIES

One of the most obvious inadequacies of the simple collision theory is its treatment of collision partners as hard spheres. Attempts such as those described in section 3.6.3 to allow for activation in internal modes of excitation are also unrealistic in the sense that they do not differentiate between the effectiveness of energy in the various modes at overcoming activation barriers. To some extent, the Transition State Theory (to be developed in the next section) is less open to these criticisms, since it does allow for the internal motions of molecular species. However, the Transition State Theory assumes some kind of equilibrium between the reactants and an activated complex. In certain circumstances this assumption of equilibrium is not necessarily valid. Experiments using shock tubes or molecular beams, for example, do not even involve a Maxwell-Boltzmann distribution of velocities initially, while very fast reactions can cause depletion of the more energy rich species, and a consequent loss of thermal equilibrium. It is obvious that the

postulation of equilibrium between reactants and activated complex loses its meaning for systems in which the various degrees of freedom of the reactants are not in thermal equilibrium. The most useful treatment for systems such as these may well be an accurate treatment using a collisional approach. Although it is not the primary purpose of this chapter to discuss specialised treatments of theoretical kinetics, it may nevertheless be of some interest to outline some of the methods which have been adopted to modify the collision theory.

The process of energy transfer is to be regarded as a special case of chemical reaction, and Widom and Bauer³⁴ have developed a collision theory for the energy transfer between vibrationally excited carbon dioxide and water which uses a Lennard-Jones "6-12" potential for the forces between "colliding" molecules. They calculate the probability of transition between various vibrational states on collision and allow for the possibility of the conversion of some energy into translation. Thus this theory is a first move away from the hard sphere model of collision.

Other approaches to modification of the collision theory seem to be divided into two categories. There are those treatments in which the potential energy surface for reaction plays no direct part in the formulation, while on the other hand there are the *kinematic* models in which the behaviour is examined of collision partners on a hypothetical reaction surface. A number of treatments of the first kind exist, amongst them those by Kramers³⁵, by Montroll and Schuler³⁶ and by Eliason and Hirschfelder³⁷. Eliason and Hirschfelder divide a bimolecular reaction into a whole series of reactions



where the subscripts refer to different internal quantum states of the reactants and products. The overall rate equation is then given by

$$-\frac{dn_a}{dt} = \sum_{ij} \sum_{kl} k_{ij}^{kl} n_{ai} n_{bj} \quad (66)$$

A formal kinetic theory for polyatomic molecules which takes into account transitions between the internal quantum states of the molecules has been developed by Wang-Chang and Uhlenbeck³⁸. This theory forms the basis of Eliason and Hirschfelder's treatment of chemical reaction, since certain transitions may be taken to correspond to reaction. Eliason and Hirschfelder show that if a Boltzmann distribution among the internal energy states of the reactant molecules is assumed, then a rate equation of the usual form is obtained. However, perhaps the most interesting feature of Eliason and Hirschfelder's work is that they are able to show how the Eyring form of the rate expression, usually derived from the Transition State Theory (*cf.* section 4.3.2), may be derived from their collision

theory if a number of rather drastic approximations are used. The forms of these approximations are naturally of interest, since they may throw light on the essential nature of the Eyring approach to the Transition State Theory. A rather similar approach has been used recently by Levine³⁹ in a discussion of *non-reactive* molecular encounters which might well be extended to include chemical reaction.

Kinematic calculations may be either classical or quantum mechanical in their treatment of the motions on a potential surface. Classical approximations may be valid for systems of the type where tunnelling is of only minor importance, although for small, low energy, species approaching a narrow energy barrier the use of quantum mechanical methods becomes essential if meaningful results are to be obtained. This limitation quite possibly applies to the $\text{H} + \text{H}_2$ reaction on which, in fact, many of the *classical* theories have been tested. Indeed, the problems of quantum mechanical treatment are only beginning to be tackled, and offer considerable difficulties.

Many of the kinematic calculations give potentially useful information about energy distributions in the products of reaction, and these predictions can be tested against experimental data to afford confirmation of the potential surfaces adopted. Research of this kind is of the greatest interest, since in the final event chemical kinetics turns on the potential surfaces for reaction. For the present purposes, however, we are concerned only with the rate at which reactants move over to products, and the topic of energy distribution in products is discussed in connection with potential energy surfaces in section 4.5.5.

The first kinematic calculations were those of Hirschfelder *et al.*⁴⁰ for the $\text{H} + \text{H}_2$ system, and much of this early work is described by Glasstone *et al.* (see Bibliography). The treatment employed a point by point calculation of the motion of a mass sliding on a potential surface (the surface was skewed in order that the motion should represent the relative motion of three co-linear hydrogen atoms: *cf.* section 4.5.1). However, the manual calculations did not allow analysis of a sufficient number of initial approach conditions to enable meaningful predictions to be made. The first workers to employ computers in kinematic calculations appear to have been Wall *et al.*^{41,42}, and many computer calculations have appeared subsequently. One of the most interesting of these is that of Karplus *et al.*⁴³, who used an LEP surface (see section 4.5.3) in a three-dimensional treatment of the $\text{H} + \text{H}_2$ reaction. The relative velocity of approach of the reactants and the vibrational (v) and rotational (J) quantum levels of the hydrogen molecule were varied systematically to obtain values for the total reaction cross-section. The reaction cross-section was found to be a monotonically increasing function of the relative velocity after a threshold value, and it also increased with increasing v and J . The threshold velocity was about $0.9 \times 10^6 \text{ cm. sec}^{-1}$ (corresponding to an energy of $5.69 \text{ kcal. mole}^{-1}$), and the reaction cross-section increased to a plateau at about twice this velocity. It was shown that the zero point energy of vibration

contributes to the energy required for the threshold, while the rotational energy does not. The barrier height of the surface used by Karplus *et al.* was 9.13 kcal. mole⁻¹, while the zero point energy of hydrogen is about 6.20 kcal. mole⁻¹. It is immediately apparent that the calculated translational energy required for the threshold of reaction (5.7 kcal. mole⁻¹) is considerably in excess of the difference between the zero point energy and the barrier height. The exact computation in three dimensions of the amount of vibrational energy available was complicated, and this problem was discussed first in terms of the colinear model. The threshold energy for the colinear system *without zero point energy* is 9.4 kcal. mole⁻¹: that is, all but 0.3 kcal. mole⁻¹ of the translational energy is available for reaction. With vibration, the threshold translational energy is 5.3 kcal. mole⁻¹, so that apparently about 4 kcal. mole⁻¹ of the 6.20 kcal. mole⁻¹ zero point energy can contribute to reaction.

Integration of the reaction cross-section over the velocity, vibrational and rotational distributions corresponding to temperatures between 300° K and 1000° K yielded rate coefficients which fitted an Arrhenius expression fairly well. The best fit for the points gave

$$k = 4.334 \times 10^{10} \exp(-7,435/RT) \quad (67)$$

in l. mole⁻¹. sec⁻¹ units; the values of the pre-exponential factor and activation energy were compared with the best experimental estimates then available of 5.4×10^{10} l. mole⁻¹. sec⁻¹ and 7.5 ± 1 kcal. mole⁻¹ (see below). Detailed examination of the $\ln k$ versus $1/T$ plot led Karplus *et al.* to the conclusion that there was some slight deviation of their calculated rate coefficients from a simple Arrhenius relation, and the results were best fitted by an expression of the form

$$k = 7.867 \times 10^6 T^{1.1762} \exp(-6,234/RT) \quad (68)$$

Karplus *et al.* now performed the interesting exercise of calculating the expressions for the rate coefficients on the basis of the Transition State Theory *using the same potential energy surface* as they had employed for their collision calculations. They found that the rate coefficients (again in l. mole⁻¹. sec⁻¹ units) could be expressed in the Arrhenius form

$$k = 7.413 \times 10^{10} \exp(-8,812/RT) \quad (69)$$

or in the temperature-dependent pre-exponential form

$$k = 2.163 \times 10^8 T^{0.797} \exp(-7,998/RT) \quad (70)$$

Thus the results of the two approaches give similar, but by no means identical,

TABLE 3

COMPARISON OF COLLISION THEORY AND TRANSITION STATE THEORY PREDICTED RATES⁴³
FOR THE REACTION $\text{H} + \text{H}_2$ WITH EXPERIMENTAL RESULTS³³

Temperature (°K)	$10^{-8}k$	$10^{-8}k$ (l. mole ⁻¹ , sec ⁻¹)	$10^{-8}k$
	Collision	Transition State	Experimental
300	0.001845	0.000301	0.00118
400	0.03557	0.01112	0.0131
500	0.2213	0.0984	0.101
600	0.7795	0.4299	0.498
700	1.9742	1.260	1.18
800	4.0506	2.881	4.84
900	7.1965	5.584	11.1
1000	11.5280	9.640	22.0

Note: The experimental results were obtained over the approximate temperature range 300° K to 450° K, and should properly be compared with the predicted values over that range only.

values for the actual rate coefficients over the temperature range considered. The latest *experimental* results for the $\text{H} + \text{H}_2$ reaction are those of Schulz and LeRoy³³ who express their rate coefficients in the form

$$\log_{10} k = 12.45 - 3.49 \times \frac{10^3}{T} + 3.83 \times \frac{10^5}{T^2} \quad (71)$$

Table 3 lists the values of the rate coefficients calculated at various temperatures by Karplus *et al.* together with the experimental rate coefficients obtained by the use of relation (71). It is seen that the rate constants given by the kinematic calculations are consistently greater than those obtained by application of the Transition State Theory, the divergence between the two sets of figures being least at high temperatures. For the lowest temperatures the kinematic values appear to agree best with the experimental rate coefficient; less reliance can be placed on the "experimental" figures for temperatures above 450° K since they rely on large extrapolations.

Representative of the attempts to apply quantum mechanical principles to the problems of kinetics is the work of Golden *et al.*^{44, 45}. In view of the great difficulty of solution of the wave equation for any realistic potential surface, a highly simplified surface (based on the LEF surface) was used for calculations on the $\text{Br} + \text{H}_2$ reaction. The formulation derived for the rate coefficient predicts a linear $\ln k$ versus $1/T$ relationship, and the calculated activation energy of 17.0 kcal. mole⁻¹ is in fair agreement with the experimental value⁴⁶ of 18.1 ± 0.5 kcal. mole⁻¹. Bauer and Wu⁴⁷ have also performed simplified quantum mechanical calculations on the $\text{Br} + \text{H}_2$ reaction; their predicted rate coefficients are about

8×10^{-3} times lower than those predicted from the simple collision theory. The actual P factor seems to be rather less small than this result would indicate (*cf.* Table 1). A number of recent papers, of which one of the first is by Mortensen and Pitzer⁴⁸, describe attempts to use more realistic potential energy surfaces in quantum mechanical treatments, and the next few years should see the production of most useful results from this approach.

4. Transition state theory

4.1 INTRODUCTION

The collision theory, in its simpler forms, takes no account of the nature of the reacting species once they have collided except to consider whether or not they can carry sufficient energy to overcome an energy barrier to reaction. The transition state theory, on the other hand, concentrates on the configuration of the reactants just as they are about to pass over into products. This configuration is thought of as lying at the highest energy point of the potential energy surface describing the reaction, and is referred to as the *transition state* or *activated complex*.

The concept of the transition state has been used to develop theories of reaction kinetics in two equivalent ways. Both approaches start from an assumption that the reactants and activated complexes are in some kind of equilibrium, and they diverge according to whether they treat the equilibrium by the methods of classical thermodynamics or of statistical thermodynamics. The classical thermodynamic formulation is the more straightforward, and is discussed in section 4.2. The statistical thermodynamic formulation allows, in principle, the calculation of the reactant-activated complex equilibrium constant from a knowledge of partition functions for the reactants and activated complex. The partition functions are readily calculated from molecular parameters, although of course, in any real case the latter can be only rather roughly estimated for the activated complex. It will appear in section 4.3 that a knowledge of the potential surface for reaction should thus enable absolute calculation of the rate coefficient, and for this reason the theory is frequently rather hopefully known as *Absolute Rate Theory*. The major obstacle to the realisation of the hopes is the difficulty in the construction of sufficiently precise potential surfaces.

It seems appropriate, at this stage, to present a highly simplified derivation of the basic equation of the transition state theory. The formation of activated complexes, $AB^\ddagger$, is written as though reversible to reactants



and an “equilibrium” constant $K^\ddagger$ assigned to the process. Thus the concentration of $AB^\ddagger$ is given by

$$[AB^\ddagger] = K^\ddagger[A][B] \quad (73)$$

The rate of reaction is the rate at which the activated complexes pass over to products, and may be set equal to $\nu[AB^\ddagger]$ where ν is the frequency of a critical “vibration” which results in the formation of products. Since a vibration of the kind envisaged will be totally excited, its frequency, ν , will be given by

$$\nu = kT/h \quad (74)$$

(The derivation of equation (74) is given in section 4.3.2). Thus, using (73) and (74), the rate of reaction may be written

$$\text{Rate} = \frac{kT}{h} K^\ddagger[A][B] \quad (75)$$

Equation (75) is the basic form of the rate expression used in transition state calculations. A more rigorous justification for the introduction of the frequency, ν , will be given in section 4.3.2, and the underlying assumption of equilibrium between reactants and activated complex will be examined.

It is usual to modify equation (75) to read

$$\text{Rate} = \kappa \frac{kT}{h} K^\ddagger[A][B] \quad (76)$$

where κ is known as the *transmission coefficient*. κ is introduced to allow for the possibility that not all activated complexes move over to the specific products under consideration. For most reactions κ is, in fact, near unity, although in certain circumstances it can be very much less. The factors influencing the magnitude of κ are discussed in section 4.4.7; in the general treatments κ will be taken to be unity for the sake of simplicity.

4.2 THERMODYNAMIC FORMULATIONS

The concept of “equilibrium” of the ordinary kind for process (72) is clearly not justified since one is dealing with a situation of maximum rather than minimum potential energy (and, presumably, free energy—see section 4.4.1). Nevertheless, even under these conditions it is possible to use the Van’t Hoff relation

$$\Delta G_0^\ddagger = -RT \ln K_p^\ddagger \quad (77)$$

where $\Delta G_0^\ddagger$ is a "standard free energy of activation". Equation (75) is written in terms of concentration of reactants, and $K^\ddagger$ is the concentration equilibrium constant, $K_c^\ddagger$. For a gas-phase bimolecular reaction, as written in (72)

$$K_c^\ddagger = RT \cdot K_p^\ddagger \quad (78)$$

(and, in general, for a gas-phase reaction of molecularity n which produces a single activated complex)

$$K_c^\ddagger = (RT)^{n-1} K_p^\ddagger \quad (78a)$$

Equation (75) represents accurately the concentration dependence of rate of reaction, so that a rate coefficient for reaction, k , may be written in terms of $\Delta G_0^\ddagger$

$$k = \frac{kT}{h} \cdot e^{-\Delta G_0^\ddagger / RT} \quad (79)$$

$\Delta G_0^\ddagger$ may be expressed in terms of standard[†] enthalpy and entropy changes in the usual way

$$\Delta G_0^\ddagger = \Delta H_0^\ddagger - T\Delta S_0^\ddagger \quad (80)$$

and the rate coefficient is then given by

$$k = \frac{kT}{h} e^{\Delta S_0^\ddagger / R} e^{-\Delta H_0^\ddagger / RT} \cdot RT \quad (81)$$

Equation (81) clearly bears some resemblance to the Arrhenius expression, and $\Delta H_0^\ddagger$ can be related to the experimental activation energy, *viz.*

$$E_a = RT^2 \frac{d \ln k}{dT} = \Delta H_0^\ddagger + 2RT \text{ (in general } = \Delta H_0^\ddagger + nRT) \quad (82)$$

It should be noted that the free energy of activation is sometimes defined by the relation

$$\Delta G_0^* = -RT \ln K^\ddagger \quad (83)$$

[†] The standard conditions chosen for ΔG , ΔH and ΔS depend upon the concentration units adopted for [A] and [B]. In the present chapter rates are usually quoted in $\text{l.mole}^{-1} \cdot \text{sec}^{-1}$, so that the standard state is 1 mole.l^{-1} .

(we write here ΔG^* to distinguish this free energy change from $\Delta G^\ddagger$), and to define $\Delta H_0^* = \Delta H_0^\ddagger$ by the relation

$$\Delta H_0^* = RT^2 \cdot \frac{d \ln K_p^\ddagger}{dT} = RT^2 \cdot \frac{d \ln K^\ddagger}{dT} - RT \quad (84)$$

The entropy of activation, ΔS_0^* , is now defined by

$$\Delta S_0^* = \frac{\Delta H_0^* - \Delta G_0^*}{T} \quad (85)$$

and the rate coefficient may be written

$$k = \frac{kT}{h} e^{\Delta S_0^*/R} e^{-\Delta H_0^*/RT} \quad (86)$$

The relationship between experimental activation energy and heat of activation is still correctly represented by (82), and the difference between (81) and (86) is a result of the difference between $\Delta S_0^\ddagger$ and ΔS_0^*

$$\Delta S_0^* = \Delta S_0^\ddagger + R \ln RT \quad (87)$$

It is important to be sure which definition of $\Delta S_0^\ddagger$ is used since $R \ln RT$ (about 13 cal. degree⁻¹ at room temperature) may be of the same order of magnitude as $\Delta S_0^\ddagger$ itself. The slightly greater simplicity of (86) compared with (81) commends this form of the expression, although the thermodynamic quantities have a slightly artificial flavour. The artificiality arises, of course, from using a Gibbs free energy in (83): perhaps it would be more justifiable to use a Helmholtz free energy, $\Delta A^\ddagger$, although the real problem is the nature of the equilibrium.

Substitution of the value of $\Delta H_0^\ddagger$ from (82) into (86) yields

$$k = \frac{kT}{h} e^{\Delta S_0^*/R} \cdot e^2 \cdot e^{-E_a/RT} \quad (88)$$

and the pre-exponential A factor in the Arrhenius equation is represented in (88) by $(kT/h) e^{\Delta S_0^*/R} \cdot e^2$. Furthermore, the collision theory equates A to $PZ/[A][B]$, and since at normal temperatures kT/h is about 10^{13} sec⁻¹ and $Z/[A][B]$ is 10^{14} to 10^{15} cm³. mole⁻¹. sec⁻¹, the probability factor is about unity for a zero entropy of activation (for the standard state of 1 mole. l⁻¹). The actual dependence of $(kT/h) e^{\Delta S_0^*/R}$ on temperature will be related to the way in which ΔS_0^* itself varies with temperature.

In our discussion of the collision theory, it was pointed out that the "steric"

component of the probability factor was equivalent to the operation of an entropy of activation. That this is so can be seen from the preceding discussion. In any process involving specialised geometrical orientations for reaction, the formation of the activated complex will involve a decrease in entropy, and the term $e^{\Delta S^\ddagger/R}$ will become smaller than unity. A positive entropy of activation suggests that, since the entropy of the activated complex is greater than that of the reactants, the complex must be very loosely bound: a decrease in entropy is, in fact, more usual than an increase. For some reactions $\Delta S_0^\ddagger$ is not very different from ΔS_0 for the overall reaction, a result which indicates that the entropy of the complex is similar to that of the products. This kind of behaviour is seen in reactions of the type



Entropies may, in principle, be calculated from calorimetric data. However, direct calorimetric calculation for the activated complex is clearly not possible, and it is more usual to obtain entropies of reactants and activated complex from statistical mechanical calculations using known or estimated partition functions for the various species. Prediction of entropies of activation in this way is tantamount to using the statistical mechanical formulation of the Transition State Theory, and is more conveniently developed as part of the statistical treatment.

4.3 STATISTICAL THERMODYNAMIC FORMULATION

4.3.1 *The transition state*

Before proceeding to derivations of the rate equation, it is perhaps of interest to examine the nature of the transition state. Fig. 3 is a representation of the potential energy of a system of reactants and products as a function of "reaction co-ordinate".

The reaction co-ordinate is looked upon as the lowest energy path between reactants and products over the potential energy surface linking them. The highest point on this path corresponds to a saddle-point on the surface, and species with the configuration described by the potential surface at that point are said to be activated complexes. The question arises whether the activated complex should more properly be defined as lying at a *free-energy* maximum: that is to say, would calculation of the free energy of activation indicate a different configuration of the activated complex from that given by the potential energy surface? Szwarc⁴⁹ has discussed the problems involved in considering free-energy maxima for reacting species. Free energy and entropy are in essence statistical concepts, while potential energy is a purely mechanical concept. Potential energy is therefore

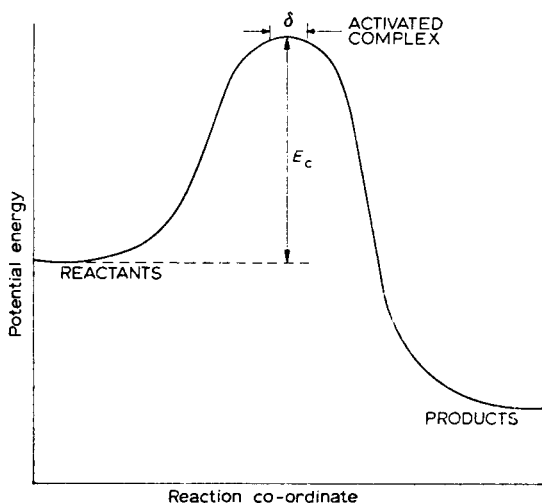


Fig. 3. Potential energy of a reaction system as a function of "reaction co-ordinate".

perfectly meaningful for a single set of reacting species while entropy and free energy are not. The potential energy of the system can be defined as a function of internal co-ordinates of the system, while the free energy cannot be defined until the potential energy is given as a function of those co-ordinates. Eyring⁵⁰ has suggested that even if the transition state were not at all well defined, the reaction rate could still be calculated by dividing into successful and unsuccessful paths all possible trajectories through the next most definitive gateway between reactants and products, and subsequent summation over the successful paths. Eyring goes on to say that for a well defined transition state the transmission coefficient could be accurately computed, so that the only problem is that of the difficulty of computation. There do, however, appear to be some cases where the identification of the transition state with a configuration of maximum free energy is necessary. Steel and Laidler⁵¹ have pointed out that in the dissociation of species to produce two free radicals, although the transition state is not defined by a potential energy surface since the latter reaches a plateau rather than a maximum, the free energy passes through a maximum. The free energy decreases beyond its highest point as a result of the increase in entropy accompanying the loosening of vibrations and the development of internal rotations as dissociation sets in. The height of the maximum must in principle be lower than the potential energy plateau, and it should be possible for the activation energy of a dissociative process to be less than the endothermicity. Unfortunately there is little experimental evidence to confirm this conclusion; there are, however, some liquid phase experiments involving large increases in entropy for which the activation energy appears to be less than the endothermicity.

The actual choice of the reaction co-ordinate is of some moment in the accurate

evaluation of rate coefficients. Precise determination of the lowest energy path requires a complete knowledge of the potential energy surface for reaction, and the problem is therefore discussed at greater length in connection with potential energy surfaces (section 4.5.8).

Finally, in this brief discussion of the nature of the transition state, some remarks should be offered about the stability of the activated complex. Certain theoretical potential surfaces possess a small potential well or "basin" at the configuration corresponding to the transition state. If the actual potential energy for the system does possess this basin then the activated complex could have an appreciable lifetime. On the other hand, if the potential basin arises only because of the approximations made in the calculations, then the activated complex has an existence only during the period of about one vibration at the top of the potential barrier. Experimental evidence about the stability of the activated complex tends to indicate that the complex is short-lived. Herschbach and collaborators (see, for example, references 52 and 53) have studied the angular scattering of alkali iodides formed in molecular beam experiments in the reaction of alkali metals with alkyl iodides. Most of the product molecules appear to recoil backwards along the direction of the relative velocity vector of the reactants. This result is only consistent with a short-lived complex. Again, Datz and Taylor⁵⁴ have shown that crossed molecular beams of $D + H_2$ show little angular scattering, and they attribute their results to the activated complex being short-lived (and linear).

4.3.2 The rate equation

The statistical mechanical formulation of the Transition State Theory treats the equilibrium of reaction (72) as a normal case, and derives a value for the equilibrium constant $K^\ddagger$ in terms of the partition functions of the reactants and activated complex and the height of the energy barrier (E_c of Fig. 3)

$$K^\ddagger = \frac{Q_{AB}^\ddagger}{Q_A^0 Q_B^0} e^{-E_c/RT} \quad (90)$$

where Q^0 is a volume independent total partition function referred to a standard state. Reaction is then said to correspond to one vibration of the activated complex becoming, in the limit, a translation along the reaction co-ordinate. The partition function for this special vibration is therefore factorised out from $Q_{AB}^\ddagger$ either directly in the form of a vibrational partition function of near-zero frequency, or as a translational partition function. The frequency of the vibration which leads to reaction now needs only to be multiplied by the concentration of activated complex to give the reaction rate.

In this section we shall give two alternative derivations of the rate equation, and *allow the assumption of equilibrium* for the time being. We shall then examine

the nature of the equilibrium assumption. It should be said, in anticipation of this examination, that the assumption of equilibrium appears to lead to no great errors. Non-equilibrium statistical treatments (for example, that of Present⁵⁵) indicate that the rate given by an equilibrium treatment is only significantly in error for very fast reactions in which the more highly energetic reactants are removed more rapidly than equilibrium within the reactants can be regained. Further confirmation that the equilibrium assumption is a reasonable approximation may be drawn from the apparent invariance of rate coefficients as equilibrium for an *overall reaction* is approached. For overall equilibrium, reactants, products and activated complexes are all in equilibrium with each other, and if measured rate coefficients are identical under these conditions with those obtained where reactants and products are *not* in equilibrium, then it would seem that the assumption of equilibrium between reactants and activated complexes is always justified.

The factor to be extracted from the partition function $Q_{AB^\ddagger}^0$ will be derived first as though it were a true vibrational factor. The partition function for a vibration of frequency ν is (see section 4.3.3)

$$f^{\text{vib}} = \frac{e^{-\frac{1}{2}h\nu/kT}}{1 - e^{-h\nu/kT}} \quad (91)$$

For any real vibration $h\nu/kT$ is never small. However, for the "vibration" of the activated complex which leads to reaction, ν may be taken to be very low and $h\nu/kT$ taken to be $\ll 1$. Then (91) becomes

$$f^{\text{vib}} = \frac{kT}{h\nu} \quad (92)$$

and, using equation (90), the concentration of activated complexes is given by

$$[AB^\ddagger] = [A][B]K^\ddagger = \frac{kT}{h\nu} \frac{Q_{AB^\ddagger}^0}{Q_A^0 Q_B^0} e^{-E_0/RT} [A][B] \quad (93)$$

where $Q_{AB^\ddagger}^0$ is the partition function for $AB^\ddagger$ from which the factor corresponding to the "reaction vibration" has been removed. The rate coefficient for reaction can now be written

$$k = \frac{\nu[AB^\ddagger]}{[A][B]} = \frac{kT}{h} \cdot \frac{Q_{AB^\ddagger}^0}{Q_A^0 Q_B^0} \cdot e^{-E_0/RT} \quad (94)$$

The treatment as a vibration of the degree of freedom along the reaction co-ordinate is somewhat unsatisfactory, especially in view of the special assumptions

that need to be made about the frequency. It is possible, however, to treat the degree of freedom as a translation, and no special conditions need be imposed. Let the distance along the reaction co-ordinate which corresponds to the activated complex be δ , as in Fig. 3. In this region the potential energy curve is flat, and motion corresponds to a translation, so we may extract from $Q_{AB^\ddagger}^0$ the partition function for translation in one dimension for a particle mass $m^\ddagger$ of $(2\pi m^\ddagger kT)^\frac{1}{2} \delta/h$, viz.

$$Q_{AB^\ddagger}^0 = \frac{(2\pi m^\ddagger kT)^\frac{1}{2} \delta}{h} \cdot Q_{AB^\ddagger}^{\prime\prime 0} \quad (95)$$

Hence

$$[AB^\ddagger] = [A][B] \cdot \frac{(2\pi m^\ddagger kT)^\frac{1}{2} \delta}{h} \cdot \frac{Q_{AB^\ddagger}^{\prime\prime 0}}{Q_A^0 Q_B^0} \cdot e^{-E_c/RT} \quad (96)$$

Now the kinetic theory of gases gives the average velocity at which particles move in *one direction* as

$$\bar{c} = \left(\frac{kT}{2\pi m^\ddagger} \right)^\frac{1}{2} \quad (97)$$

The frequency at which activated complexes pass over to products is the average velocity of particles moving from left to right divided by the distance δ (a frequency is the number of times a second that an oscillating system passes any given point in *one direction*). Thus

$$v = \left(\frac{kT}{2\pi m^\ddagger} \right)^\frac{1}{2} \frac{1}{\delta} \quad (98)$$

and the rate coefficient for reaction, k , is given by

$$\begin{aligned} k &= \frac{v[AB^\ddagger]}{[A][B]} = \left(\frac{kT}{2\pi m^\ddagger} \right)^\frac{1}{2} \cdot \frac{1}{\delta} \cdot \frac{(2\pi m^\ddagger kT)^\frac{1}{2} \delta}{h} \cdot \frac{Q_{AB^\ddagger}^{\prime\prime 0}}{Q_A^0 Q_B^0} e^{-E_c/RT} \\ &= \frac{kT}{h} \cdot \frac{Q_{AB^\ddagger}^{\prime\prime 0}}{Q_A^0 Q_B^0} \cdot e^{-E_c/RT} \end{aligned} \quad (99)$$

This expression is identical to (94), since $Q_{AB^\ddagger}^{\prime\prime 0}$ and $Q_{AB^\ddagger}^{\prime 0}$ are in fact the same.

The above derivation, given by Laidler (Bibliography), is equivalent to that given originally by Eyring⁵⁶, although Eyring worked in terms of the momentum and reduced mass of the activated complex. Evans and Polanyi⁵⁷, in their derivation, considered, in effect, the average velocity in both directions along the reaction co-ordinate, but also allowed for the fact that only one half of the complexes

move towards products. Their final expression for the rate coefficient was therefore the same as Eyring's.

Laidler and Polanyi⁵⁸ have shown that the equilibrium treatment yields a valid result by the artificial device of considering a hypothetical system in which the potential energy curve rises again after the flat region (δ of Fig. 3). In this case the flat region corresponds to an ordinary state AB, and an equilibrium concentration of AB is established. By extracting the appropriate factor from the partition function for AB as before, the concentration of AB is given by

$$[AB] = [A][B] \cdot \frac{kT}{hv} \cdot \frac{Q'_{AB}}{Q_A^0 Q_B^0} \cdot e^{-E_c/RT} \quad (100)$$

Since the state AB is in equilibrium with A+B, the rate at which species pass through the state from left to right is equal to the rate at which they pass through from right to left. The latter rate is equal to $v[AB]$ so that the rate at which species enter the region δ from the left is equal to

$$[A][B] \cdot \frac{kT}{h} \cdot \frac{Q'_{AB}}{Q_A^0 Q_B^0} \cdot e^{-E_c/RT} \quad (101)$$

Since v is independent of the subsequent form of the potential energy surface, (101) would also give the rate at which species entered the region δ even if the potential energy curve decreased again on the right of the flat region (as in Fig. 3). If every species that reaches the activated complex configuration moves on to products, then (101) gives the rate of reaction. This result is identical with the result normally obtained by applying the equilibrium assumption directly to the activated complex. What is important is that the *algebraic expression* is identical: a classical equilibrium between reactants and activated complexes does not exist (and, as Laidler and Polanyi⁵⁸ point out, if activated complexes were added to the system moving from left to right the "equilibrium" would not be disturbed).

4.3.3 Partition functions

The problem of evaluating rate coefficients from the statistical-thermodynamic form of the Transition State Theory requires calculation of partition functions for the reactants and activated complex. This section gives derivations of the partition functions for various degrees of freedom.

A total partition function per particle, Q , is defined by

$$Q = \sum_i g_i e^{-\epsilon_i/kT} \quad (102)$$

where ϵ_i is the energy of the i th level of a quantised energy level system and g_i

its degeneracy. According to the usual approximation for separating energies, the total partition function can be factorised into separate components for the several degrees of freedom. The symbol f with appropriate superscripts is used here to represent the factorised components of the partition functions.

$$Q = f^{\text{trans}} f^{\text{vib}} f^{\text{rot}} f^{\text{elec}} \quad (103)$$

(Note that Q is volume dependent or not according to whether f^{trans} is divided by V or not: all Q in the rate equations are volume independent). A term f_0 is sometimes included to correspond to the zero point energy of the system, ε_0 , and to give all the energy modes a common reference point. The nature of the summation differs in the various cases, and they are dealt with separately.

(a) *Translational partition function*

The quantised energy levels for a particle in a box of dimensions $a \times b \times c$ is given by

$$\varepsilon_i = \frac{h^2}{8m} \left(\frac{l_i^2}{a^2} + \frac{m_i^2}{b^2} + \frac{n_i^2}{c^2} \right) \quad (104)$$

where l, m, n are quantum numbers describing the motion in the x, y and z directions. Thus

$$f^{\text{trans}} = \sum_i g_i \exp \left\{ - \frac{h^2}{8mkT} \left(\frac{l_i^2}{a^2} + \frac{m_i^2}{b^2} + \frac{n_i^2}{c^2} \right) \right\} \quad (105)$$

The separation between the energy levels is very small for translational energy, and the summation of (105) over g_i -fold degenerate levels can be replaced by an integration over all values of l, m and n from 0 to ∞ . The translational partition function is then given by

$$f^{\text{trans}} = \iiint_0^\infty \exp \left\{ - \frac{h^2}{8mkT} \left(\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2} \right) \right\} dl dm dn \quad (106)$$

The triple integral can be resolved into the product of three integrals of the form

$$\int_0^\infty e^{-a^2 x^2} dx = \sqrt{\frac{\pi}{2a}} \quad (107)$$

The partition function in three dimensions is, therefore, given by

$$f^{\text{trans}} = \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} \cdot abc = \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} V \quad (108)$$

where V is the volume. [The corresponding expression for translation in one dimension is clearly the result of a single integration

$$f^{\text{trans}} = \frac{(2\pi mkT)^{\frac{1}{2}}}{h} a, \quad \text{etc.} \quad (109)]$$

The partition function derived is for a *single* particle. For a system of N non-interacting indistinguishable particles the total translational partition function is:

$$f_N^{\text{trans}} = \frac{(f^{\text{trans}})^N}{N!} \quad (110)$$

All further derivations are for single particles, and overall partition functions for assemblies of particles must be calculated according to (110).

(b) Vibrational partition function

The vibrational energy levels for a harmonic oscillator of frequency ν are given by

$$\varepsilon_i = (v + \frac{1}{2})h\nu/kT \quad (111)$$

where v is the vibrational quantum number. Thus

$$f^{\text{vib}} = \sum_{v=0}^{\infty} e^{-(v+\frac{1}{2})h\nu/kT} = \frac{e^{-\frac{1}{2}h\nu/kT}}{1 - e^{-h\nu/kT}} \quad (112)$$

The total vibrational partition function for a molecule will be the product of the partition functions for the $3N-6$ normal vibrations ($3N-5$ for a linear molecule). More complex summations are possible which take account of the anharmonicity of vibration.

If the partition function f_0 is included in the product of (103) then f'^{vib} is defined such that

$$f'^{\text{vib}} = (1 - e^{-h\nu/kT})^{-1} \quad (113)$$

(c) Rotational partition function and statistical factors

The form of the rotational partition function depends on the rotational sym-

metry of the molecule. For a rigid linear rotator

$$\varepsilon_J = B_0 J(J+1) \quad (114)$$

and the degeneracy is given by

$$g_J = 2J + 1 \quad (115)$$

It is customary to divide g_J by a "symmetry number" which is the number of equivalent rotational orientations that can be obtained for the molecules. However, several groups of workers have shown that the use of symmetry numbers in Transition State Theory formulations can give rise to erroneous rate coefficients. For this reason the symmetry number is omitted from the rotational partition functions derived in this section; an alternative method for obtaining the correct statistical factors to apply, due to Bishop and Laidler⁵⁹, is given at the end of the section.

The rotational constant, B_0 , is equal to $h^2/8\pi^2 I$ where I is the moment of inertia of the molecule. Thus

$$f^{\text{rot}} = \sum_{J=0}^{\infty} (2J+1) e^{-J(J+1)h^2/8\pi^2 I k T} \quad (116)$$

For systems where only a few rotational levels are excited, it is probably best to carry out the summation. If, however, $h^2 \ll 8\pi^2 I k T$, then we can write f^{rot} as an integral

$$f^{\text{rot}} = \int_0^{\infty} e^{-J(J+1)h^2/8\pi^2 I k T} \cdot (2J+1) dJ \quad (117)$$

$$= \frac{8\pi^2 I k T}{h^2} \quad (118)$$

The problem of treating symmetric and asymmetric tops quantum mechanically is of some difficulty, and it is far easier to use a classical approximation equivalent to (117). For a polyatomic molecule with moments of inertia I_A , I_B and I_C about the principal axes

$$f^{\text{rot}} = \frac{(8\pi^2 k T)^{\frac{3}{2}} (\pi I_A I_B I_C)^{\frac{1}{2}}}{h^3} \quad (119)$$

In general, for substances in the gas phase at normal temperatures, the differences between quantum mechanical and classical rotational partition functions is small.

The problem of the correct statistical factors to be used in transition state formulations must now be considered. Schlag^{60,61} has recently proposed treatments which use a group-theoretical approach, and has shown clearly that the use of symmetry numbers can yield erroneous results. Bishop and Laidler⁵⁹ have also presented a rather simpler treatment of the statistical factors, and, since the results seem to be in general identical with Schlag's, their method is described here.

Statistical factors s_f and s_r are defined for forward and reverse reactions in a process such as



These factors are the numbers of different sets of products which could be formed if all identical atoms in the reactants were labelled. (It is noted that if both reactants are the same chemical species then the statistical factor is half the number of different sets.) An example may make the method clear. Consider the reaction



The hydrogen atoms on the methane may be labelled H^1 , H^2 , H^3 and H^4 so that the following four sets of products may be formed: $\text{H}^1\text{Cl} + \text{CH}^2\text{H}^3\text{H}^4$, $\text{H}^2\text{Cl} + \text{CH}^1\text{H}^3\text{H}^4$, $\text{H}^3\text{Cl} + \text{CH}^1\text{H}^2\text{H}^4$ and $\text{H}^4\text{Cl} + \text{CH}^1\text{H}^2\text{H}^3$. Thus the statistical factor for the forward reaction, s_f , is four. The reverse process can be treated in a similar way, taking, say, $\text{H}^1\text{Cl} + \text{CH}^2\text{H}^3\text{H}^4$. The products are the two mirror image forms of $\text{CH}^1\text{H}^2\text{H}^3\text{H}^4 + \text{Cl}$, so that s_r is two. Bishop and Laidler show that for *any* reaction such as (120) the *ratio* s_f/s_r is the ratio of rotational symmetry factors for reactants and products.

The application of the method to the formulation of the rate equation is as follows. If the forward and reverse statistical factors for the process



are $s_f^\ddagger$ and $s_r^\ddagger$ respectively, then as a result of the property of quotients of symmetry numbers the concentration of $\text{AB}^\ddagger$ should be written

$$[\text{AB}^\ddagger] = [\text{A}][\text{B}] \cdot \frac{kT}{h\nu} \cdot \frac{s_f^\ddagger}{s_r^\ddagger} \cdot \frac{Q_{\text{AB}^\ddagger}''^0}{Q_{\text{A}}''^0 Q_{\text{B}}''^0} \cdot e^{-E_a/RT} \quad (122)$$

where Q''^0 are total partition functions from which the symmetry numbers have been omitted. This omission will already be present in the rotational partition functions if the expressions (118) or (119) are used. It will be recalled that Laidler

and Polanyi⁵⁸ obtained the rate at which reacting species entered the region δ in our Fig. 3 by consideration of the hypothetical equilibrium case where a stable species AB is formed rather than the unstable activated complex $AB^\ddagger$. It was shown that the rate is $v[AB]$ in the simple case, so it is $s_r v[AB]$ in the reaction system with statistical factor greater than unity. Thus

$$s_r v[AB] = [A][B] \frac{kT}{h} \cdot s_f \cdot \frac{Q_{AB}''^0}{Q_A''^0 Q_B''^0} \cdot e^{-E_c/RT} \quad (123)$$

in the present case, and this would also be the rate at which reactants entered the transition state when the unstable $AB^\ddagger$ is formed. Hence the rate coefficient for reaction is

$$k = \frac{kT}{h} s_f^\ddagger \frac{Q_{AB^\ddagger}''^0}{Q_A''^0 Q_B''^0} \cdot e^{-E_c/RT} \quad (124)$$

It is therefore apparent that the ordinary formulation using symmetry numbers in the rotational partition functions gives a rate coefficient that is too low by the factor $s_f^\ddagger$, and is in error for all reactions in which $s_f^\ddagger \neq 1$.

(d) Electronic partition function

The energy difference between the ground electronic state and the first excited state is frequently very large compared with kT , so that only the ground state makes an appreciable contribution to the electronic partition function. If the zero of energy is chosen to be the ground electronic state, then the electronic partition function is unity. However, in the general case we must write

$$f^{\text{elec}} = \sum_j g_j e^{-\epsilon_j/kT} \quad (125)$$

where ϵ_j and g_j are the electronic energy and degeneracy of the j th electronic state. A number of species have relatively low lying excited electronic energy levels, while other are degenerate in their ground states [*e.g.* $O_2(^3\Sigma_g^-)$, $NO(^2\Pi)$, etc.], and for these species electronic partition functions must be calculated.

(e) Nuclear partition function

The energy of the ground nuclear state is taken to be unity, so that for species in their ground nuclear states the important contribution to the partition functions comes from the nuclear degeneracy $(2I+1)$ where I is the total nuclear spin of the species.

TABLE 4

PARTITION FUNCTIONS FOR THE SEVERAL DEGREES OF FREEDOM TOGETHER WITH NUMERICAL ORDER OF MAGNITUDE

<i>Energy form</i>	<i>Expression for partition function</i>	<i>Order of magnitude</i>
Translation in 3-dimensions	$f^{\text{trans}} = (2\pi mkT/h^2)^{3/2}$ (volume independent)	10^{24} – 10^{27}
Vibration	$f^{\text{vib}} = e^{-\frac{1}{2}h\nu/kT} (1 - e^{-h\nu/kT})^{-1}$ for each vibration. This includes zero-point energy.	1 – 10^2
Rotation (Classical forms)	$f^{\text{rot}} = (8\pi^2 kT)^{3/2} (\pi I_A I_B I_C)^{1/2} / h^3$ for three unequal moments of inertia. $f^{\text{rot}} = 8\pi^2 I kT / h^2$ for linear molecule	10 – 10^3
Electronic	$f^{\text{elec}} = \sum_{j=0}^{j^*} g_j e^{-\epsilon_j/kT}$ (j^* is highest populated state)	1
Nuclear	$f^{\text{nuc}} = (2I+1)$	1

To conclude this section a table (Table 4) is presented of the analytical expressions for partition functions, together with approximate numerical values obtained by substitution of reasonable magnitudes for the various molecular parameters needed.

4.3.4 Predicted pre-exponential factors

The pre-exponential factor for any reaction can now, in principle, be calculated from equation (94) or (99) and the appropriate values of the partition functions. Unfortunately, however, it is not possible to calculate accurate values for the partition functions of the activated complex unless the exact configuration of the activated complex and the form of the potential energy surface at the transition state are known. On the other hand, good estimates may be made; and rough guides may be obtained for pre-exponential factors just by using the orders of magnitude quoted in Table 4. Before carrying out the exercise of predicting numerical pre-exponential factors for a number of different types of reaction, it is of interest to see to what form the analytical expression for the rate coefficient reduces if the reactants are two atoms A and B (*i.e.* hard spheres). A and B will both have only translational partition functions, while the activated complex, $AB^\ddagger$, will behave as a diatomic molecule with a moment of inertia $I = \sigma_{AB}^2 \mu$ (σ , μ are respectively, the radius and reduced mass of the complex). The reaction co-ordinate is clearly the vibration along the line of centres, so no partition func-

tion need be written for this vibration. Substitution of the expressions for the partition functions into (94) then gives

$$k = \frac{kT}{h} \frac{\left(\frac{2\pi m_A + m_B}{h^2} kT\right)^{\frac{1}{2}} \frac{8\pi^2 \sigma_{AB}^2 \mu kT}{h^2}}{\left(\frac{2\pi m_A}{h^2} kT\right)^{\frac{1}{2}} \left(\frac{2\pi m_B}{h^2} kT\right)^{\frac{1}{2}}} \cdot e^{-E_c/RT}$$

$$= \sigma_{AB}^2 \left(\frac{8\pi kT}{\mu}\right)^{\frac{1}{2}} e^{-E_c/RT} \quad (126)$$

which is identical to the value given by the collision theory (section 3.2). Thus the Transition State Theory and the Collision Theory are consistent if the hard sphere condition is imposed on the reactants.

The order of magnitude of the rate coefficient for the two atom reaction could also be obtained from the values quoted in Table 4. If the partition functions for the same degree of freedom are regarded as identical for the different species, then the rate coefficient could be written:

$$k = \frac{kT}{h} \frac{f'^{\text{trans}} (f^{\text{rot}})^2}{(f^{\text{trans}})^2} \cdot e^{-E_c/RT} = \frac{kT}{h} \frac{(f^{\text{rot}})^2}{f'^{\text{trans}}} \cdot e^{-E_c/RT} \quad (127)$$

Taking $f'^{\text{trans}} = 10^{24}$ and $f^{\text{rot}} = 10$, and with $kT/h \sim 10^{13} \text{ sec}^{-1}$, we obtain a value for the frequency factor of about $10^{-9} \text{ cm}^3 \cdot \text{molec}^{-1} \cdot \text{sec}^{-1}$, or about $10^{12} \text{ l. mole}^{-1} \cdot \text{sec}^{-1}$. This is the same value as was calculated for the gas-kinetic collision rate coefficient, $Z_{AB}/n_A n_B$, in section 3.4. Comparison of the Transition State Theory and Collision Theory results may be extended by calculation of expected P factors for various types of reaction. For example, if the reaction between two non-linear molecules, containing n_1 and n_2 atoms respectively, proceeds *via* a non-linear complex, then the complex will possess translation in three dimensions, three rotations and $3(n_1 + n_2) - 6$ vibrations, of which one is along the reaction coordinate. The powers of the partition functions to be found in the pre-exponential part of the rate coefficient expression are, therefore,

$$\frac{f'^{\text{trans}} (f^{\text{vib}})^{3(n_1 + n_2) - 7} (f^{\text{rot}})^3}{f'^{\text{trans}} (f^{\text{vib}})^{3n_1 - 6} (f^{\text{rot}})^3 \cdot f'^{\text{trans}} (f^{\text{vib}})^{3n_2 - 6} (f^{\text{rot}})^3} = \frac{(f^{\text{vib}})^5}{f'^{\text{trans}} (f^{\text{rot}})^3} \quad (128)$$

The partition functions for the two atom reaction are in the ratio $(f^{\text{rot}})^2/(f'^{\text{trans}})$ so that the pre-exponential factor for the non-linear reaction will be in the ratio $(f^{\text{vib}}/f^{\text{rot}})^5$ to that for the hard sphere model, and this ratio is the probability factor, P . Using the orders of magnitude in Table 4, $P \sim 10^{-5}$, a result in keeping with the experimental data for the reaction rates of complex molecules. Probability

factors for other reactions involving species of intermediate complexity may be calculated in a similar way, and, in general, they are similar to the experimentally determined values. The expression for P usually involves the ratio $(f^{\text{vib}}/f^{\text{rot}})$ raised to some power, and the form suggests an interpretation of low P factors in terms of the loss of rotational degrees of freedom of the reactants and their replacement by vibrations.

The temperature dependence of the pre-exponential term is clearly determined by the final ratio of partition functions, and will therefore be related both to the complexity of reaction and to the exact form adopted for the partition functions.

4.4 MORE GENERAL TREATMENTS OF THE TRANSITION STATE THEORY

During the last decade or so, a number of attempts have been made to formulate the Transition State Theory without recourse to all of the assumptions made by Eyring⁵⁶ or Evans and Polanyi⁵⁷. A number of these have merely used the activated complex as a basis for statistical theory. For example, Light and his collaborators have produced statistical theories of reaction kinetics⁶² which have now been extended to include reactions with activation energy⁶³. The breakup of a strong-coupling complex is computed statistically, and the theory applied to a number of reactions (such as $\text{K} + \text{HBr}$, $\text{H} + \text{Cl}_2$, $\text{H} + \text{HBr}$, *etc.*) to determine rate coefficients, reaction cross sections and information about rotational, vibrational and electronic excitation of the products. In general the results are consistent with experimental data.

Of more direct concern to a discussion of the Transition State Theory are the attempts to investigate the effect of non-equilibrium distribution on calculated rates. Prigogine *et al.*^{64,65} made some of the earliest investigations of this kind by a derivation of the perturbation of the Boltzmann distribution as a result of reaction. Present⁵⁵ subsequently made a more reliable calculation which suggested that if the ratio E_c/RT is sufficiently large (so that the more energetic reactants are not too rapidly removed) the simple collision theory yields results which are not greatly in error. (For $E_c/RT = 5$, the collision theory is in error by 8 %.) Mahan⁶⁶ has performed similar calculations for radical combination processes (small activation energy). A collision cross section was calculated which indicated that translationally hot radicals combine more rapidly than average "thermal" radicals. The cross section was then used to calculate the disturbance of the velocity equilibrium in reactions of this kind, and it was concluded that the equilibrium assumption was justified in photochemical systems, where the radical concentration is low, although if the radical concentration approaches 0.1 mole fraction, the assumption may no longer be valid. Ross *et al.*^{67,68} have also attempted perturbation solutions of the Boltzmann equilibrium to obtain an equation for the rate of bimolecular reaction. Ross and Eu^{69,70} have, in addition,

developed a theory of collisions to derive an activated complex theory of reaction rates.

Marcus⁷¹ has given generalised forms of the Transition State Theory using both quantum mechanical and classical treatments. A rate expression is derived without introduction of the assumption that there is a Cartesian reaction co-ordinate, and it is shown that this expression reduces to the ordinary form if the Cartesian assumptions are subsequently made. The new equation for the transmission coefficient contains internal centrifugal terms, and a rotational interaction can be introduced into the formalism. The absence of the Cartesian assumption allows calculations to be made with a potential surface having separated variables, since the theory can deal with a separated wave equation for the reaction co-ordinate which is curvilinear (*cf.* ref. 72). A simpler and more general derivation of the final equations was subsequently given⁷³ which permitted the introduction of analytical mechanics. Marcus has also given⁷⁴ a quasiequilibrium expression which relates sums over reaction cross sections to properties of activated complexes. One of the most interesting features of the latter work is the comparison of the results with the kinematic calculations of Karplus *et al.*⁴³ (*cf.* section 3.7) on the $\text{H} + \text{H}_2$ reaction. The equations developed were used⁷⁵ to formulate a statistical dynamical model for the total chemical reaction cross section expressed as a function of the relative velocity and the vibrational and rotational states of the reactants. Encouraging agreement with the $\text{H} + \text{H}_2$ results of Karplus *et al.* was obtained⁷⁶ in the low to moderate velocity range, although at very high velocities the comparison between the predictions of Marcus' theory and the kinematic calculations indicated the occurrence of some vibrational non-adiabaticity.

4.5 POTENTIAL ENERGY SURFACES

4.5.1 General considerations

Some knowledge of the potential energy surface is needed in either collision or transition state theories of chemical kinetics. The simple collision theory requires only a value for the height of the energy barrier to be surmounted, while the transition state theory needs information also about the shape of the potential energy surface in the neighbourhood of the saddle point in order to calculate the partition functions for the activated complex. In principle, calculation of tunnelling corrections also presupposes a knowledge of the form of the surface near the saddle point, while the more sophisticated calculations of reaction rate, such as the kinematic treatments, need detailed information about the whole surface joining reactants and products. Direct calculation of a potential energy surface from experimental data is not, at present, possible, although experimental information of one kind or another can be used to test the validity of a hypothetical

surface: this matter is discussed in section 4.5.5. The problem is therefore one of constructing the surface from quantum mechanical principles of systems involving several particles. Unfortunately, such approaches usually require drastic approximations which at once makes their validity questionable and reduces their general applicability. Empirical methods and semi-empirical methods are less satisfying intellectually, although they frequently have the advantage of greater tractability. For these reasons it seems useful to divide the treatment of potential energy surface calculations into three sections, 4.5.2–4, according to whether the approach is non-empirical, semi-empirical or empirical.

The potential energy of a many-particle system can only be represented as a function of co-ordinates of the particles on a many-dimensional hypersurface. However, many of the theoretical calculations to be discussed concern the three body problem connected with exchange reactions of the type $A + BC \rightarrow AB + C$ so that the atoms are confined to a single plane and their positions relative to each other can be expressed in terms of the distances between each of the atoms. If the angle of approach of A to BC is defined, it is now possible to define the relative positions by two variables, so that the potential energy surface may be represented in two dimensions by a contour map. For more complex systems contour maps may be drawn of sections through the potential energy hypersurface. Fig. 4 is a contour map for the reaction system formally represented by the formation of a linear complex in the reaction system $A + BC \rightarrow AB + C$ and the interatomic distances r_{AB} , r_{BC} define the relative positions of A, B and C. A section taken through the potential energy surface parallel to the r_{BC} axis and taken at large separation of A and BC (*i.e.* large r_{AB}) would be the ordinary potential energy

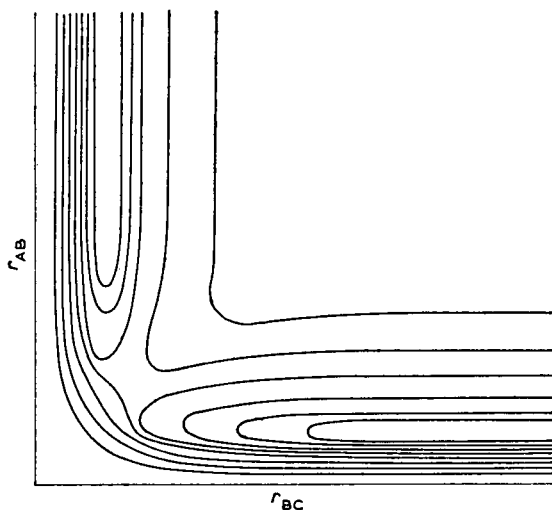


Fig. 4. Contour diagram for the potential energy of a linear system of three atoms A, B and C as a function of their relative positions defined by r_{AB} and r_{BC} .

curve for BC (and the similar section taken parallel to the r_{AB} axis would give the potential energy curve for AB).

Some comments by Marcus about the validity of expressing potential energies in the way described are put forward at the end of section 4.5.8.

The first kinematic calculations using potential energy surfaces (*cf.* section 3.7) were made by point by point calculation of the successive co-ordinates of a sliding mass point. In order that the relative kinetic energy of the three atoms should be represented by the kinetic energy of a single particle moving on the potential surface, it is necessary that the axes of the contour map be skewed. Although the concept of the sliding point mass is no longer especially valuable now that the calculations are performed by high speed computers, many contour maps have been presented in the past with skewed axes, and it is therefore proposed that the angle of skew required to satisfy the kinetic energy condition be derived. The calculations refer to a colinear system. Suppose three atoms A, B and C of mass m_A , m_B and m_C lie along a line so that the centres of A and C are separated by the distances r_{AB} , r_{BC} respectively from B. The total internal kinetic energy (T) with respect to the centre of mass of the system is given by:

$$T = \frac{1}{(m_A + m_B + m_C)} \{m_A(m_B + m_C)\dot{r}_{AB}^2 + 2m_A m_C \dot{r}_{AB} \dot{r}_{BC} + m_C(m_A + m_B)\dot{r}_{BC}^2\} \quad (129)$$

Now we require to choose the angle at which r_{AB} and r_{BC} must be plotted with respect to each other so that two conditions are filled:

(i) there must be no cross-terms if the kinetic energy is expressed in terms of the *perpendicular* axes of the system, x and y , and

(ii) the coefficients of x^2 and y^2 in such a system must be equal. That is, we wish to express T as $\frac{1}{2}m(x^2 + y^2)$ where m is the mass of the sliding point. If θ be the angle between r_{AB} and the $r_{BC}(=x)$ axes, then $r_{AB} = x - y \tan \theta$ and $r_{BC} = cy \sec \theta$ (c is a factor introduced to modify the scale of the sloping co-ordinate). Substitution of the expressions for r_{AB} and r_{BC} in terms of x , y , c and θ into (129), and application of the restrictions (i) and (ii) leads to the conclusions

$$\sin \theta = \left\{ \frac{m_A m_C}{(m_A + m_B)(m_B + m_C)} \right\}^{\frac{1}{2}} \quad (130)$$

$$c = \frac{m_C}{m_A + m_B} \quad (131)$$

and

$$m = \frac{m_A(m_B + m_C)}{(m_A + m_B + m_C)} \quad (132)$$

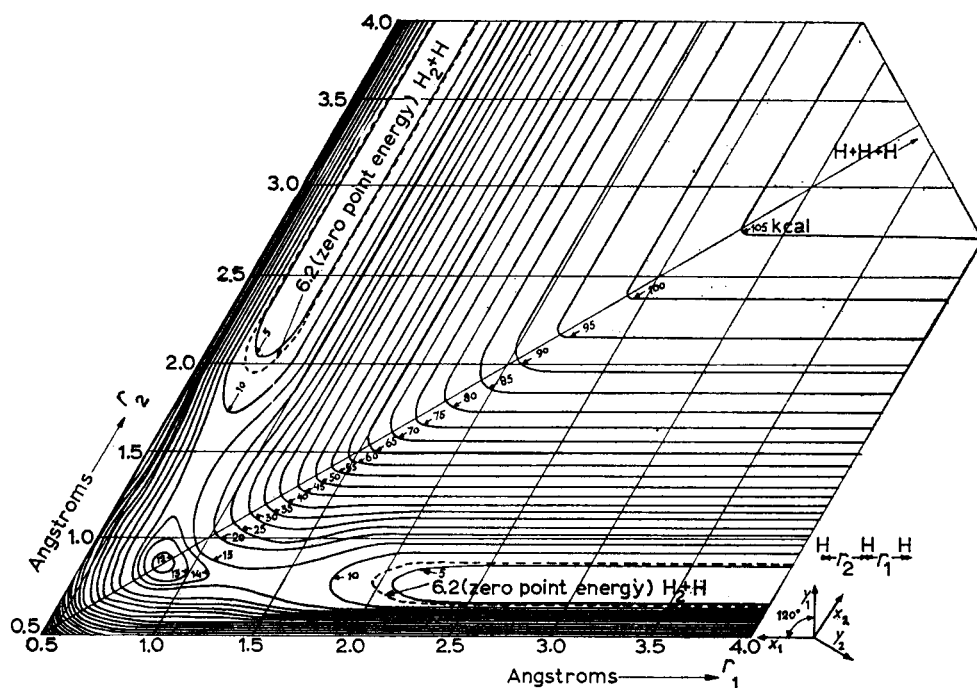


Fig. 5. Skewed axis contour map for the co-linear H-H-H system (after Eyring *et al.*⁷⁷).

Fig. 5 shows a skewed potential energy contour map for the surface for co-linear H-H-H, taken from the paper by Eyring *et al.*⁷⁷.

4.5.2 Non-empirical treatments

London⁷⁸ made one of the earliest attempts to apply a simple valence-bond treatment to the H_3 system, and suggested that the energy of the system could be expressed in terms of the three Coulombic interaction integrals Q_A , Q_B and Q_C and the three exchange integrals α , β and γ , for the three pairs of atoms

$$E = Q_A + Q_B + Q_C \pm [\frac{1}{2}(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\alpha - \gamma)^2]^{\frac{1}{2}} \quad (134)$$

(The contribution of overlap interaction was neglected, and the equation given without proof. Coolidge and James⁷⁹ showed that the derivation of (134) required that a number of approximations be made.)

Sugiura⁸⁰ had given the Coulombic and exchange integrals for the system, and Eyring and Polanyi⁸¹ used these values to compute a potential energy surface for linear configurations of H_3 on the basis of (134). It should now be possible to compare the difference in the binding energy between the separated reactants

and H_3 at the saddle point (E'_c) with experimentally determined activation energies for the $H + H_2$ reaction. A recent recalculation (see ref. 58, p. 7) gives this energy difference as about $8.8 \text{ kcal. mole}^{-1}$. (There is a "basin" in the potential energy surface where the exact saddle point should be: this is almost certainly a result of the approximations employed, as discussed in section 4.3.1.) The experimental activation energy (say $7-8 \text{ kcal. mole}^{-1}$, from reference 9 or 33) must be corrected for comparison with E'_c from the theoretical surface, since the latter does not allow for the differences in zero-point energy between $H + H_2$ and H_3 . When this correction is made, a reasonable value for the experimental value of the barrier height would be around 8 or $9 \text{ kcal. mole}^{-1}$. This apparent agreement must not, however, be regarded as a very good confirmation of the approximations adopted, since the approximations themselves are larger than the energies calculated. The fortuitous nature of the result was demonstrated by Coolidge and James⁷⁹ who carried out the calculations of the energy according to the London method at the configuration corresponding approximately to the basin in the potential surface ($r_{AB} = r_{BC} = 0.9 \text{ \AA}$) and obtained a result for the potential energy of $4.0 \text{ kcal. mole}^{-1}$ above the $H + H_2$ level, in good agreement with Eyring and Polanyi. Coolidge and James then recalculated the potential energy after making allowance for terms resulting from overlap, from the permutation of more than two electrons, and the effect on the various two-atom exchange terms of having a third atom present. The potential energy at $r_{AB} = r_{BC} = 0.9 \text{ \AA}$ was now about $33 \text{ kcal. mole}^{-1}$ above $H + H_2$, leaving no doubt that the more exact calculation gave a far worse estimate of the height of the energy barrier.

Quantum mechanical calculations, of the kind described above, for many bodies, are probably best solved by making use of the Variation Theorem. Hirschfelder *et al.*^{82, 83} made the first attempts at a variational treatment of the linear $H-H-H$ system. The "best" linear symmetrical configurations were based on homopolar, and then on homopolar and polar, combinations of three $1s$ atomic orbitals. Hirschfelder⁸⁴ later extended the treatment to include non-linear, but still symmetrical, configurations. The results of these variational calculations were rather unsatisfactory: the investigation of non-linear systems had suggested that the linear configuration would, in fact, give the lowest energy barrier. On the other hand, the lowest binding energy calculated by Hirschfelder *et al.*⁸² was about $25 \text{ kcal. mole}^{-1}$ calculated relative to $H + H_2$ (the $H + H_2$ level was calculated according to the same approximations).

The advent of high speed computers has allowed the use of more involved variational treatments, but the results have proved disappointing. Kimball and Trulio⁸⁵ set up an LCAO-MO wave function from five $1s$ orbitals for linear symmetric H_3 ; their rather complete variational treatment gave a value of $19.9 \text{ kcal. mole}^{-1}$ binding energy relative to $H + H_2$ calculated by their treatment. It might be thought that this sophisticated treatment had brought the calculated binding energy considerably closer to the experimental value of $8-9 \text{ kcal. mole}^{-1}$. How-

ever, the $\text{H} + \text{H}_2$ energy calculated is itself in considerable error, and referred to the *experimental* $\text{H} + \text{H}_2$ energy, the binding energy is about 37 kcal. mole⁻¹. Boys and Shavitt^{85a} have performed an even more involved variational calculation, and considered asymmetric and bent configurations. The linear configuration was found to be most stable, and the binding energy in the symmetric H_3 was 29.2 kcal. mole⁻¹ referred to experimental values of $\text{H} + \text{H}_2$. Thus even the best calculated values of the barrier height are a factor of three higher than the values determined experimentally from reaction rates.

4.5.3 Semi-empirical surfaces

The rather unsatisfactory results obtained from non-empirical determinations of potential energy surfaces for even the simplest reaction systems such as $\text{H} + \text{H}_2$ have led to the parallel development of "semi-empirical" treatments in which the form of the surface is based on the quantum mechanical methods, but in which various parameters can be adjusted to fit experimental observations.

The most famous of the semi-empirical surfaces is the London-Eyring-Polanyi (LEP) surface. Eyring and Polanyi⁸¹ first took the form of the potential energy curve for any H-H pair (obtainable from spectroscopic data) and equated the energy with the total binding energy for the pair (*i.e.* $Q_A + \alpha$, $Q_B + \beta$, $Q_C + \gamma$). Now Sugiura's values⁸⁰ for the integrals suggested that at internuclear separations greater than 0.8 Å, the fraction of Coulombic to total energy

$$\rho_A = \frac{Q_A}{Q_A + \alpha} \quad (135)$$

was constant (and about 0.10 to 0.15). By taking a constant value of ρ in conjunction with the Morse form of the potential energy curve for H-H pairs, it is possible to calculate the Coulomb and exchange energies at any internuclear separation, and a potential energy surface may be constructed. This semi-empirical LEP surface predicts a basin at $r_{AB} = r_{BC} \sim 0.8$ Å, although, as we have seen in section 4.5.1, experimental observations suggest that H_3 is not metastable. The depth of the basin increases with an increase in ρ . Table 5 shows the height of the energy barrier, E_c (referred to the LEP calculated $\text{H} + \text{H}_2$ energy), and the depth, E_d , of the basin at three values of ρ (*cf.* ref. 58).

Sato⁸⁶ has pointed out that not only does the LEP surface possess the basin at $r_{AB} = r_{BC} = 0.8$ Å, but that it possesses a further basin at $r_{AB} = r_{BC} = 0.3$ Å (which at $\rho = 0.12$, for example, is 75 kcal. mole⁻¹ deep!), and he has proposed modifications to the LEP method to remove these features. Sato's technique was to obtain separated values for Coulombic and exchange components such as Q_A and α by taking the energy of the attractive H_2 state as $Q_A + \alpha$ and that of the

TABLE 5

HEIGHT OF ENERGY BARRIER (E_c) AND DEPTH OF BASIN (E_d) ON THE SEMI-EMPIRICAL LEP SURFACE AT THREE RATIOS OF COULOMBIC TO TOTAL ENERGY

ρ	E_c (kcal. mole ⁻¹)	E_d
0.07	20	0
0.14	14	2-3
0.20	7	5-6

repulsive state as $Q_A - \alpha$. Since the forms of the attractive and repulsive energy curves are well known for H_2 it is possible to write out the London equation (134) without reference to the arbitrary parameter ρ . The valence-bond energies for the hydrogen molecule are

$$\frac{Q_A \pm \alpha}{1 + S_A^2} \quad (136)$$

where S_A is the overlap integral for the pair of hydrogen atoms, and Sato suggested that some allowance for the overlap integrals could be incorporated into the London equation by writing

$$E = \frac{1}{1 + S^2} \{Q_A + Q_B + Q_C \pm [\frac{1}{2}\{(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\alpha - \gamma)^2\}^{\frac{1}{2}}]\} \quad (137)$$

for the energy of the H_3 species. The parameter S is now to be adjusted to give the best fit with experimental data. Table 6 shows the barrier heights calculated by both Eyring and Sato procedures for four hydrogen atom abstraction reactions. The figures are those given in Sato's paper⁸⁶, and the Sato calculations are based on $S^2 = 0.18$. The experimental activation energies quoted should presumably

TABLE 6

EXPERIMENTAL AND CALCULATED ACTIVATION ENERGIES FOR SOME HYDROGEN ABSTRACTION REACTIONS⁸⁶

Reaction	Activation energy (kcal. mole ⁻¹)		
	Experimental	Eyring	Sato
$H + H_2 \rightarrow H_2 + H$	5-7	7.9	5.4
$H + HCl \rightarrow H_2 + Cl$	4.5	11.4	4.7
$H + HBr \rightarrow H_2 + Br$	1.2	10.4	0.7
$H + HI \rightarrow H_2 + I$	1.5	7.7	0.3

be corrected for the difference in zero-point energy between reactants and activated complex. For $S^2 = 0.18$ the potential surface has no basin at the configuration corresponding to the transition state, and in this respect is more in keeping with present experimental data than the LEP surface. However, Weston⁸⁷ has criticised the Sato surface on the grounds that there is no justification for using a constant value of S^2 , especially when the value chosen is about one-third of the true value at the transition state configuration. Weston's further criticism that the Sato barrier is too thin and therefore predicts a high degree of quantum-mechanical tunnelling may be less cogent now that tunnelling effects have been recognised in the $\text{H} + \text{H}_2$ reaction³³.

Polanyi *et al.*⁸⁸ have introduced even more empiricism into the LEP calculations by allowing the three overlap integrals to be variable. They write their expression

$$E = \frac{Q_A}{x} + \frac{Q_B}{y} + \frac{Q_C}{z} + \left\{ \frac{\alpha^2}{x^2} + \frac{\beta^2}{y^2} + \frac{\gamma^2}{z^2} - \frac{\alpha\beta}{xy} - \frac{\beta\gamma}{yz} - \frac{\alpha\gamma}{xz} \right\}^{\frac{1}{2}} \quad (138)$$

where x, y, z are $1 + S_A^2, 1 + S_B^2$ and $1 + S_C^2$. The overlap parameters have been adjusted to give surfaces for kinematic calculation of energy distributions of products in exothermic reactions of the $\text{A} + \text{BC} \rightarrow \text{AB} + \text{C}$ type, and the predictions correlated with experimental results of the product energy distribution (see section 4.5.5).

The kinematic calculations of Karplus *et al.*⁴³ have already been described (section 3.7). The potential energy surface for H_3 used by these workers was based

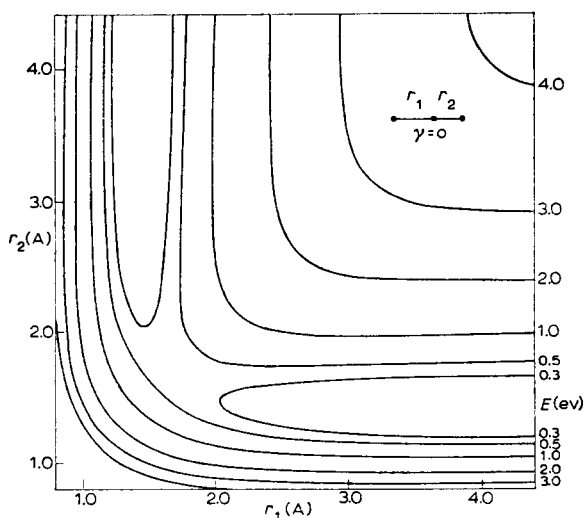


Fig. 6. Potential energy surface for linear symmetrical H_3 calculated by Porter and Karplus⁸⁹.

on a modified LEP surface devised by Porter and Karplus⁸⁹. A combination of theoretical and semi-empirical treatments was used. The Coulombic term was taken to be $Q_A + Q_B + Q_C$ as before, but the exchange integrals were assumed to have the form $\alpha + \Delta$, where Δ is a residual term. A refined calculation from attractive and repulsive H_2 energy curves gave values for Q_A , Q_B , Q_C and α , β , γ . Overlap integrals and Δ were calculated from $1s$ atomic orbitals, as was a term for the exchange over the three atoms. The most favourable configuration for the activated complex was found to be linear and symmetrical with $r_{AB} = r_{BC} = 0.90$ Å. The barrier height under these conditions is $8.58 \text{ kcal. mole}^{-1}$, in good agreement with experimental data. Fig. 6 shows the surface for linear, symmetric H_3 as calculated by Porter and Karplus⁸⁹.

4.5.4 Empirical potential energy surfaces

Johnston and Parr⁹⁰ have developed Johnston's Bond Energy Bond-Order (BEBO) method (see Johnston, Bibliography), which relies on two empirical rules relating bond length, r , and bond energy, D , for a bond of order n to the length r_s and energy D_s for the single bond. The relations are

$$r = r_s - 0.26 \ln n \quad (139)$$

and

$$D = D_s n^p \quad (140)$$

where p is a constant. For the $H + H_2$ reaction it is assumed that the *total* bond order is unity, so that the orders of the two bonds in H_3 , n_{AB} and n_{BC} , are such that $n_{BC} = 1 - n_{AB}$. Application of the relations (139) and (140) now allows evaluation of the energy of the H_3 system at any set of separations (a term for the repulsion of the outer pair of hydrogen atoms is estimated from the repulsive energy curve for H_2). The result of this simple calculation is relatively good: for the H_3 system, the barrier height is found to be about $10 \text{ kcal. mole}^{-1}$.

Empirical potential energy surfaces have also been devised from manipulations of the Morse function for the potential energy of the diatomic H_2 system. Surfaces of this kind are described by Wall and Porter⁹¹, by Blais and Bunker⁹² and by Karplus and Raff⁹³.

4.5.5 Tests of potential energy surfaces: energy distributions in products

There are a number of ways in which the validity of a hypothetical potential energy surface may be tested. In its simplest form, a test may consist merely of a comparison between experimental and calculated activation energies and, as we have seen in section 4.5.2, the agreement even at this level between theory and

experiment is not good. An alternative approach is to adjust the parameters of a semi-empirical surface to give a good agreement for barrier energy, and to test some other property. The barrier shape and thickness will affect the predicted rate of quantum mechanical tunnelling, while the form of the whole surface will affect the overall rate of reaction predicted by kinematic or generalised activated complex procedures. Unfortunately, however, the experimental results themselves are not sufficiently unequivocal to allow definite testing of the potential energy surface. On the other hand, investigations into the energy distribution in *product* species have recently proved most fruitful in attempts to test potential surfaces. Spectroscopic techniques used under proper conditions are able to determine accurately the energies of product molecules as they are formed, and velocity analysis techniques in molecular beam experiments can also yield precise information about the energy distribution of products. This section deals, therefore, with tests of potential energy surfaces based on determinations of excitation in products. The other methods whereby a potential energy surface may be tested are discussed implicitly in sections 3.7, 4.4 and 4.5.6.

The "atomic flame" reactions of sodium with the halogens were the earliest experiments in which the excitation of products was used to infer the nature of the potential energy surface for reaction. Evans and Polanyi⁹⁴ studied the chemiluminescent emission of sodium resonance radiation in the reaction of sodium vapour with iodine. The processes leading to excitation were suggested to be



($\text{NaI}^\dagger$ represents a molecule of *vibrationally* excited sodium iodide, while Na^* is an *electronically* excited, 2P sodium atom). Although the nature of the energy transfer process (143) was not clear at the time (and has only recently been elucidated⁹⁵) it was apparent that both (142) and (143) were highly efficient. Reaction (142) is only 1 kcal. mole⁻¹ more exothermic than the energy needed to excite sodium resonance radiation. It follows, therefore, that not only must (143) occur without loss of energy, but also that all the energy liberated in (142) must go into vibration of the newly formed Na-I bond. In order that there shall be no loss of energy in rotational excitation, the activated complex must be a co-linear system; and, further, there must be virtually no repulsion between the newly formed particles in the final state of the activated complex as otherwise translational excitation will result. Evans and Polanyi⁹⁴ showed what kinds of potential energy surface met these conditions, and a calculation of the surface for the reaction with atomic *chlorine* demonstrated the absence of repulsion as the activated complex moved over to products. More recently, crossed molecular beam experiments have confirmed this kind of behaviour in a more direct manner for

the reaction of alkali metals with halogens. Datz and Minturn⁹⁶, for example, conclude from the scattering pattern from crossed beams of $\text{Cs} + \text{Br}_2$ that about 99 % of the energy released in the reaction goes into internal excitation of CsBr . The reaction is said to proceed by a "stripping" mechanism in which the Br atoms separate immediately after the Cs atom comes within its range of interaction; the momentum of the CsBr product is therefore just the sum of the initial momenta of the Cs and one of the Br atoms. It is not known how the internal energy is apportioned between vibration and rotation.

Polanyi *et al.*^{88,97} have categorised potential energy surfaces for exothermic exchange reactions of the type $\text{A} + \text{BC} \rightarrow \text{AB} + \text{C}$ into those which result in (i) *attractive* energy release (in which the atom A approaches BC at the ordinary BC separation), (ii) *repulsive* energy release (in which AB at its normal separation retreats from C) and (iii) *mixed* energy release (an intermediate case in which A approaches BC while BC extends). The reaction of alkali metals with the halogens clearly belong to the class of processes for which there is an attractive energy release. The degree of excitation has been put on a much more quantitative footing as a result of the kinematic computer calculations⁸⁸. Eight potential energy hypersurfaces were considered covering the range of possibilities between purely repulsive and largely attractive behaviour. The surfaces themselves were derived from the modified LEPS surface expressed by equation (138). Product energy distributions (with respect to vibration, rotation and translation) were obtained as a function of initial position, impact parameter and kinetic energy for all eight combinations of light and heavy masses for A, B and C. Polanyi *et al.*⁹⁸⁻¹⁰⁰ obtained experimental data on the vibrational excitation of product hydrogen chloride in the reaction



from observation of the infrared chemiluminescence from the excited $\text{HCl}^\dagger$. The reaction is about 45 kcal. mole⁻¹ exothermic and the activation energy is about 2.5 kcal. mole⁻¹, so that there is 47.5 kcal. mole⁻¹ liberated after the barrier. This is very approximately the energy difference between the zero point and $v = 6$ for HCl. Excitation of levels up to the sixth does occur, and the probability of formation of a product molecule in any given level relative to $v = 3$ was measured¹⁰⁰. The results are given in Table 7. Absolute intensity measurements suggest that about 7 % of the total energy available is converted into vibration. A high degree of rotational excitation was observed⁹⁹ and it would appear that the balance of the total energy is converted into rotation and translation. The type of behaviour observed for the $\text{H} + \text{Cl}_2$ reaction therefore corresponds to largely repulsive behaviour. Kuntz *et al.*⁸⁸ quote unpublished experimental results of Airey, Pacey and Polanyi for the $\text{H} + \text{Br}_2$ reaction which indicate a similar inefficiency for vibrational excitation in product HBr .

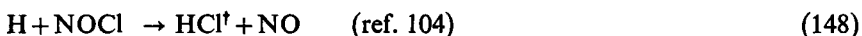
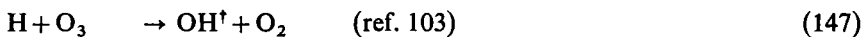
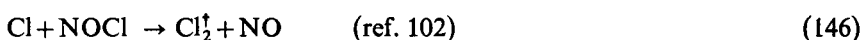
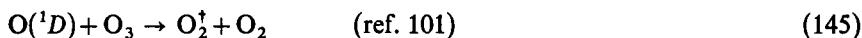
TABLE 7

RELATIVE PROBABILITY OF PRODUCT FORMATION IN DIFFERENT VIBRATIONAL LEVELS
FOR THE REACTION $\text{H} + \text{Cl}_2 \rightarrow \text{HCl}^\dagger + \text{Cl}$ (from reference 100)

<i>Vibrational level</i>	<i>Probability</i>
0	20
1	3
2	2
3	1
4	0.3
5	0.04
6	0.004

The experimental results quoted above indicate that a large fraction of the exothermicity of reaction is converted into internal excitation for the reactions of metals with halogens (*e.g.* $\text{Na} + \text{I}_2$, $\text{Cs} + \text{Br}_2$), and it would appear from the results of Evans and Polanyi⁹⁴ that this internal energy is largely vibrational (the molecular beam experiments⁹⁶ do not clearly show how much of the internal excitation is vibrational and how much rotational). On the other hand, only a small proportion of the available energy appears as vibration in the reaction of atomic hydrogen with halogens (*e.g.* $\text{H} + \text{Cl}_2$, $\text{H} + \text{Br}_2$). Kuntz *et al.*⁸⁸ showed that there was a tendency for the exchange reactions to give less vibrational excitation of products on surfaces with appreciable repulsive character if the attacking atom, A, were light. The greatest variation in the vibrational excitation with the nature of the potential energy surface was found for the attack of a light atom on a molecule BC in which both B and C were heavy. The final conclusion was that a surface with a large proportion of repulsive character provided the most likely qualitative explanation for the relatively low efficiency of vibrational excitation in the hydrogen atom-halogen systems.

A number of other reactions are known to give vibrationally excited products: for example, the processes



all give rise to vibrationally excited products, and in the latter two cases infrared chemiluminescence may be detected. The total yields of vibrationally excited products is in doubt, especially in the first two reactions, and detailed kinematic calculations for systems of the type $\text{A} + \text{BCD} \rightarrow \text{AB} + \text{CD}$ appear not to have been made so far. One general conclusion which seems to be valid for this kind of

reaction is that such vibrational excitation as there is appears in the newly formed A-B bond, and not in the fragment CD.

The remaining tests of potential energy surfaces to be described here use experimental data from molecular beam studies. Reference has been made already (section 4.3.1) to the scattering observed in the $D + H_2$ ⁵⁴ and $M + RI$ ^{52, 53} reactions, and to the inference drawn that the activated complex is short lived. The reaction of potassium with methyl iodide has also been the subject of theoretical calculation by both Blais and Bunker⁹² and by Karplus and Raff^{93, 105}. Although Blais and Bunker restricted their calculations to two dimensions, they were able to show that distributions of product scattering angle were peaked such that the product KI was emitted predominantly backward and the CH_3 forward with respect to the initial K direction. Furthermore, for an assumed potential energy surface of reasonable shape, most of the energy of reaction should go into internal vibrational and rotational excitation of KI. These results were found to be relatively insensitive to the presence or absence of rotation or vibration of CH_3I before collision and to minor changes in the assumed potential of interaction. The rotational energy distribution in KI corresponded closely to complete conversion of the initial angular momentum to product rotation. Karplus and Raff⁹³ subsequently extended the calculations to three dimensions and found that the potential assumed by Blais and Bunker, although giving results apparently in concordance with the observed energy distribution in the products, nevertheless led to significant disagreement with the experimental results for the total and differential reaction cross-section. It was found that the over-estimate of the long-range interaction between the reactants was the probable source of the error, and that introduction of an appropriate three-body attenuation term into the expression for the potential led to reasonable results. The three dimensional calculations have been followed up by Karplus and Raff¹⁰⁵ with the primary object of obtaining an understanding of the limitations on the form of the potential energy surface provided by the available experimental information. Four potential energy surfaces were examined: (i) a modified "Blais-Bunker" surface, (ii) a surface similar to (i) in which the three-body attenuation term was added, (iii) a surface similar to (i) in which the K- CH_3 repulsive term was replaced by an "exponential-six" interaction with the centre of mass of CH_3I as origin, and (iv) a special surface which was an artificial representation of the three-body potential for the (K, I, CH_3) system as a superposition of covalent and ionic structures. Comparisons were made of the total reaction cross-section, the rotational and vibrational state distribution of the product molecule, the reaction energy distribution, the differential cross-section and several further parameters. Surfaces (i) to (iii) all predicted that a large proportion of the energy of reaction would appear as internal excitation of the newly formed KI molecule, and that the centre-of-mass angular distribution for KI would be strongly peaked in the backward direction, although there were some quantitative differences between the results using the various surfaces. The

artificial surface, (*iv*), disagreed with the other three in that it predicted that half of the energy of reaction should appear as kinetic energy. The first three surfaces gave results which suggested that the replacement of a potassium atom by an atom of a heavier alkali metal should result in the appearance of a larger fraction of the reaction energy in internal degrees of freedom, while a corresponding increase in the mass of the alkyl group would produce little or no kinematic effect.

The experimental results with which the predictions of Blais and Bunker, or Karplus and Raff, are to be compared are those of Herschbach and his collaborators. The results are summarised in reference 52 for a series of reactions $M + RI$ where $M = K, Rb$ and Cs and $R = CH_3, C_2H_5, etc.$ The general conclusions are very similar to the theoretical predictions outlined in the preceding paragraph, the main difference being that the experimental results suggest that the translational energy decreases with an increase in mass of the alkyl group (an effect not predicted by the kinematic considerations).

Two further exothermic reactions of atomic potassium have been studied by molecular beam techniques with the object of testing the results against hypothetical potential energy surfaces. These reactions are



and



The internal energy of the products of the reaction with HBr was recently determined from a product recoil velocity analysis by Grosser *et al.*¹⁰⁶. At two initial relative kinetic energies, it was found that the difference between initial and final relative kinetic energies was small, so that most of the heat of reaction must go into internal excitation of the newly formed KBr. The observed angular and velocity distribution of KBr implied that the reactive scattering in the centre-of-mass system was confined predominantly to low angles: that is, the product KBr "preferred" the direction of the incident K atom. However, further interpretation of the results was limited as a result of unfavourable kinematic factors. Experimental information for the $K + Br_2$ reaction is relatively extensive^{107, 108}. Warnock *et al.*¹⁰⁹ have recently reported a computer analysis of the product velocity distribution experiments, and give references to previous work on the reaction. Studies have been carried out with both energy-selected¹⁰⁷ and thermal¹⁰⁸ beams, and suggest that most of the energy of reaction appears as internal excitation of the product. The technique used by Warnock *et al.* to compare the experimental velocity analyses with predictions based on calculations from hypothetical potential energy surfaces was to use a process of "inversion" of the data. Assumed functional forms for the differential reaction cross-section and product internal excitation function were used to compute contour maps of relative

number density or flux; the experimental velocity analyses were then compared with cuts taken through the contour map at various laboratory scattering angles. The "best" trial distribution functions were obtained by iterative computation, and the final forms compared with theoretical distribution functions derived from trajectory analyses using semi-empirical potential energy surfaces. The results obtained were in fair accord with distribution functions obtained by Karplus and by Polanyi (unpublished) for certain hypothetical surfaces.

It would seem apparent that considerable advances will be made in the near future by studies of the kind outlined above. With sufficient refinement, studies of the product energy distributions may eventually allow rather complete description of the potential energy surface for an elementary chemical reaction.

4.5.6 *Quantum mechanical tunnelling*

In classical mechanics, the probability that a particle can overcome an energy barrier is zero if the particle has less energy than the barrier height, and unity if it has more. However, in quantum mechanics there is a finite probability that particles with energy less than the barrier height will appear on the other side of it. A wave reaching a barrier will be attenuated by the barrier; the wave will emerge on the far side of the barrier with its original period but with reduced amplitude. The largest deviations from classical behaviour are to be expected for particles of low mass, and it is well known that the motions of electrons cannot be treated classically. Bell suggested¹¹⁰ that as a result of their relatively small mass, protons might also be subject to non-classical behaviour. At room temperature, protons moving with thermal velocities have De Broglie wavelengths of 0.1 to 1.0 Å associated with them: barriers to chemical reaction probably have widths of the order of a few Angstroms so that proton tunnelling might be of importance. To a lesser extent, heavier particles may also tunnel through energy barriers although the expected effect becomes increasingly small as the particle mass increases.

The quantum mechanical versions of the trajectory studies (insofar as "trajectory" has a quantum mechanical meaning) presented in sections 3.7 and 4.4 make allowances for quantum mechanical tunnelling, and the rates of reaction predicted by such calculations need no further correction for non-classical behaviour. However, the calculations have been made only for a few, rather specific, reactions, and their validity depends on the form adopted for the potential energy surface on which the trajectory analysis is made. It has been more usual to apply a correction to the "classical" reaction rate which allows for tunnelling, and the form of this correction is what concerns us in this section. Most of the treatments which have been adopted involve a calculation of the tunnelling correction for a one-dimensional barrier of more or less artificial shape. A problem arises in this case, since there are a number of close-lying alternative paths on the reaction surface,

each of which is probably curved, and tunnelling may occur from any of them. Both Johnston and Rapp³¹ and Mortensen and Pitzer⁴⁸ have carried out multi-dimensional tunnelling studies which suggest that the one-dimensional approach is inadequate; unfortunately Mortensen and Pitzer give evidence that the one-dimensional treatment *underestimates* the extent of tunnelling, while Johnston and Rapp suggest that it *overestimates* the importance of tunnelling. (For some comments about multidimensional reaction paths by Marcus⁷² see the final paragraph of section 4.5.8.) A further criticism may be levelled at the usual method by which correction factors are computed, and that is the assumption of a thermal equilibrium distribution of energies in the particles reaching the barrier. For those systems in which equilibrium cannot be assumed, the special statistical or kinematic treatments must be used.

If the "permeability" of the barrier (the probability that a particle will cross the barrier) is given the symbol $G(W)$ where the variable W is the energy of the particle, then for a system of particles possessing a Boltzmann energy distribution, the tunnel correction Q_t is given by¹¹¹

$$Q_t = e^{-\epsilon_c/kT} \int_0^\infty \frac{1}{kT} e^{-W/kT} \cdot G(W) dW \quad (151)$$

Q_t is to be multiplied by the expression for the reaction rate to give the true rate. Bell¹¹² has given a set of expressions for a wide range of parabolic barriers. An analytical expression is derived for a simple parabolic barrier of half-width a and height E_c : for a particle of mass m the tunnelling correction is given by

$$Q_t = \frac{1}{2} u_t / \sin \frac{1}{2} u_t, \quad \text{where} \quad u_t = \frac{h}{a\pi kT} \left(\frac{\epsilon_c}{2m} \right)^{\frac{1}{2}} \quad (152)$$

If tunnelling were the *only* cause of temperature dependence of the Arrhenius parameters, A and E_a , then from (152) the relations

$$E_a = E_c - RT \{1 - \frac{1}{2} u_t \cot \frac{1}{2} u_t\} \quad (153)$$

and

$$A = A_c \cdot \frac{\frac{1}{2} u_t}{\sin \frac{1}{2} u_t} \exp \{ \frac{1}{2} u_t \cot \frac{1}{2} u_t - 1 \} \quad (154)$$

may be derived, where E_c is the (molar) classical barrier height and A_c the classical pre-exponential factor. Wigner¹¹³ has also given an approximate expression for the tunnelling correction for a parabolic barrier in terms of the imaginary frequency, ν_i , of vibration at the top of the barrier

$$Q_t = 1 + \frac{1}{24} \left(\frac{h\nu_i}{kT} \right)^2 \quad (155)$$

This is the first term of the more exact expression derived by Bell. Eckart¹¹⁴, as long ago as 1930, devised an artificial barrier which may correspond more closely than the parabolic barrier to the actual shape of the potential barrier in a classical reaction. Fig. 7 shows the form of the Eckart barrier, and the potential energy, V , may be expressed by

$$V = \frac{Ce^u}{1+e^u} + \frac{De^u}{(1+e^u)^2}; \quad u = 2\pi x/L \quad (156)$$

where

$$C = E_c - E_1; \quad D = (E_c^{\frac{1}{2}} + E_1^{\frac{1}{2}})^2; \quad L = \frac{2\sqrt{2}\pi}{E_c^{-\frac{1}{2}} + E_1^{-\frac{1}{2}}} \left(\frac{\partial^2 V}{\partial x^2} \right)^{-\frac{1}{2}}.$$

E_c and E_1 are identified in Fig. 7, and x is the reaction co-ordinate. $(\partial^2 V / \partial x^2)$ is calculable from the form of the potential surface adopted, and substitution of the Eckart expression for the potential energy into the Schrödinger equation allows values to be obtained for Q_i . Johnston *et al.*^{31, 115}, Shavitt¹¹⁶ and Shin¹¹⁷ have given values of Q_i for a number of different parabolic and Eckart barriers.

The best-known example of a tunnelling process in the gas-phase is the proton tunnelling in ammonia¹¹⁸, which gives rise to the inversion doubling observed in some of the lines of the ammonia vibrational spectrum. A double-minimum potential field exists, with a barrier through which tunnelling can occur to cause splitting of the vibrational levels. The accuracy with which energies can be measured by spectroscopic means is, of course, far greater than the secondhand evidence that can be obtained from rates of chemical reactions. In any case, the complete lack

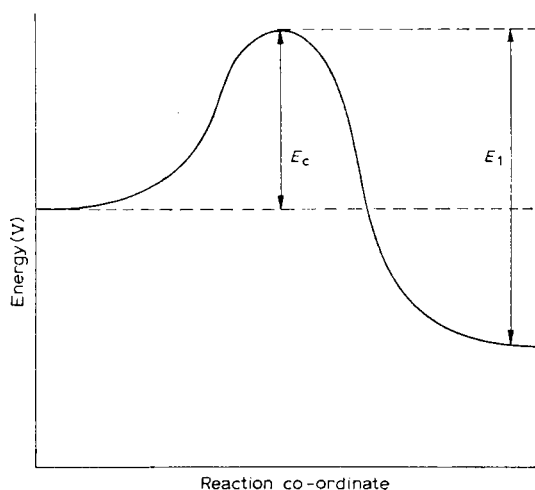


Fig. 7. The Eckart barrier.

of reliance that can be placed on theoretically predicted barrier heights shows that it is not possible to test a tunnelling correction by comparison of a measured rate and the "classical" predication. Experimental tests of tunnelling theories therefore tend to fall into two classes: experiments in which a search is made for curvature in an Arrhenius plot, and experiments in which the rates of processes involving similar reactants are compared.

The general behaviour of an Arrhenius plot for a reaction in which there is quantum mechanical tunnelling was discussed in section 3.6.4 in terms of Fig. 2. All treatments of tunnelling give a correction factor whose importance increases at low temperatures [*e.g.* equations (152) and (155)], and it is at low temperatures, therefore, that curvature of the Arrhenius plot should be most apparent. Curvature of Arrhenius plots which must be ascribed to tunnelling is fairly well established in solution processes; the greater accuracy of many condensed phase observations and the lower temperatures at which reaction may be carried out both help to make the identification of tunnelling effects relatively unambiguous. On the other hand, curvature of Arrhenius plots for gas-phase experiments is much less easy to demonstrate. The *ortho-para* hydrogen conversion, $\text{H} + \text{H}_2 \rightarrow \text{H}_2 + \text{H}$, is a reaction in which tunnelling would be expected to play a significant part since it involves the transfer of hydrogen atoms. However, until quite recently, the experimental data available showed no evidence of deviation from linear Arrhenius behaviour. Both Weston⁸⁷ and Shavitt¹¹⁶ reviewed the evidence available in 1959. For example, in Weston's work, a parabola was fitted to the profile of a Sato-type potential energy surface, and the tunnelling correction calculated by Bell's method. The calculated correction term ranged from 52.6 at 295 °K to 1.23 at 1000 °K. Tunnelling of this magnitude would certainly give rise to curvature of the Arrhenius plot, and since the experimental results showed little signs of curvature it was concluded that either the parabolic correction terms gave an overestimate of the tunnelling, or that the barrier profile given by the Sato surface was too thin. In view of the claim by Mortensen and Pitzer⁴⁸ that the one-dimensional treatment underestimates the amount of tunnelling, the second explanation seemed more reasonable. The situation has altered to some extent, however, as a result of the recent re-examination of the $\text{H} + \text{H}_2$ kinetics by Schulz and LeRoy³³. In the temperature range 300 °K to 444 °K, these workers find a marked curvature of the Arrhenius plot, and they were able to fit the temperature dependence of the rate constant by assuming quantum mechanical tunnelling through a parabolic barrier of imaginary frequency, ν_i , equal to 1213 cm^{-1} (more properly $1213 i \text{ cm}^{-1}$). The height of the barrier was $9.7 \text{ kcal. mole}^{-1}$ measured between zero-point energy levels of reactants and activated complex, and if the tunnelling treatment is valid, it would seem that this value, corrected for the zero point energies, is the one to compare with predicted values of the barrier height E_c . One interesting point that Schulz and LeRoy make is that assumption of Eckart-type barriers could lead to linear Arrhenius behaviour even in the presence of signi-

ficant tunnelling, so that linearity of an Arrhenius plot does not necessarily indicate that tunnelling is unimportant. (Conversely, of course, deviation from Arrhenius behaviour does not unequivocally show the presence of tunnelling, since a number of other causes—for example, anharmonicity or vibration-rotation interaction—could be responsible for the deviations.)

The second procedure that may be used to demonstrate the presence of tunnel effects is to compare the rates of similar reactions. In particular, the use of isotopically substituted reactants can lead to changes in reaction rate which can be associated in part with tunnelling. *Kinetic isotope effects* arise from a number of causes¹¹⁹. First, the zero-point energies of both reactants and activated complex will be changed by the substitution: except in rare cases these changes will not cancel out, so the barrier height will be altered by the substitution. Secondly, the component terms of the partition functions in the pre-exponential function will be changed. To some extent, since the changes in translational and rotational partition functions are in the same direction for both reactants and activated complex, the main effect on the pre-exponential term results from changes in the vibrational partition function. Finally, the extent to which tunnelling occurs will depend on the masses of the particles reacting. In all respects, hydrogen-deuterium substitution tends to have the largest effect, especially since tunnelling is likely to be most important in reactions involving light particles. The diagnostic tests for tunnelling are, qualitatively: (i) that the ratio of rate coefficients, say k_H/k_D , for the two processes is greater than the value calculated from classical theory, and it should be temperature dependent; (ii) the difference in experimental activation energies, $E_a^D - E_a^H$, is greater than the difference predicted on zero-point energy considerations alone; (iii) the ratio of experimental pre-exponential factors, A_D/A_H , should be much greater than unity—this is, perhaps, the best test for tunnelling since classical theory would usually give a value close to unity for the ratio.

Johnston *et al.*^{32, 120} have studied the reactivity of CF_3 with a number of hydrocarbons in an attempt to establish the validity of both potential energy surfaces and tunnelling corrections. For example, Sharp and Johnston³² studied the pair of reactions



and



A Sato surface was constructed, and the value of S^2 [equation (137)] was estimated by fitting it to give the correct rate for the exchange reaction



The variation of the ratio of rate coefficients for (157) and (158) was found to be quite different from that expected in the absence of tunnelling. By using the Johnston and Rapp method³¹ for calculation of the tunnelling correction, good agreement was obtained with the experimental ratio of rate coefficients, on condition that a complete vibrational analysis of the activated complex was employed which involved the variation of potential energy with more co-ordinates than just the reaction co-ordinate. One-dimensional calculations did not give such good agreement with experiment as the multidimensional ones.

The reactions of halogen atoms with hydrogen or deuterium have also been used in studies of tunnelling in gas-phase reactions¹²¹⁻¹²⁵. In the latest series of experiments¹²⁴ tritium has also been used, and the rates of reaction of Cl atoms with H₂, HD, D₂, HT, DT and T₂ were measured over the temperature range -30 °C to +70 °C. The isotope effects observed could only be explained if tunnelling were present, and it was found that the best agreement (within 15 %) of the calculated values with experiment was obtained by the use of a generalised Sato model for the potential together with the Johnston-Rapp treatment³¹ of tunnelling through an asymmetrical barrier. Empirical sets of four force constants describing the transition state H-H-Cl were also given, and their use in the calculation of isotope effects gave excellent agreement with experimental values. In most cases a multidimensional treatment of tunnelling gives better correlation with experimental results than does the one-dimensional calculation. However, in experiments on the pair of reactions¹²⁵



(*Cl is a radioactive chlorine atom), although tunnelling was important at the low temperatures used, the Johnston-Rapp treatment was unable to give corrections which accounted for the experimental results. On the other hand, the Bell correction gave satisfactory correction terms for a number of possible barrier profiles.

The discussion of this section will have made it apparent that what are sometimes known as "Tests of the Transition State Theory" may in fact be testing the potential energy surface used, or the form of the tunnelling corrections applied. Orders of magnitude of frequency factors have been calculated a number of times, and, if the vibration frequencies and geometries of activated complexes are sufficiently carefully chosen, good agreement is obtained with experiment (see, for example, the calculations of Herschbach *et al.*¹²⁶). On the other hand, attempts to calculate activation energies and the exact form of the variation of rate coefficient with temperature have been rather less successful.

4.5.7 The transmission coefficient, κ

The transmission coefficient, κ , is the probability that a system which has reached the transition state will continue to the final state. It is unity if systems cross a classical barrier only once in their passage from initial to final states; in the general case, some of the systems which cross the transition state will cross it again in the opposite direction. In classical theory, κ is near unity at low temperatures as it is very improbable that a system with barely enough energy to cross the barrier will find its way back¹²⁷. At higher temperatures, however, there may be a decrease in the magnitude of κ . Departures of κ from unity are to be expected as a result of quantum mechanical effects. Quantum mechanical tunnelling can give rise to a situation in which the rate of reaction is *apparently* greater than the rate at which activated complexes are formed at the top of the classical energy barrier. For systems in which tunnelling is important, therefore, a transmission coefficient greater than unity can be obtained. The correction Q_i introduced in section 4.5.6 allows for tunnelling effects, so for the purposes of the present section, κ refers to a transmission coefficient which excludes tunnelling corrections. A particle which has wave properties associated with it may be reflected both on its way up and on its way down a potential barrier separating final and initial states, and κ can be less than unity from this cause. The generalised treatments of reaction rates (sections 3.7 and 4.4) should allow for these reflections, and the rate coefficients predicted from the most complete of these treatments need no further adjustment. For the ordinary statistical-thermodynamic formulation of the equilibrium transition state theory the magnitude of κ is still required. Hirschfelder and Wigner¹²⁸ first showed that the transmission coefficient is a rapidly fluctuating function of the energy of the system, so that if the thermal energy spread is sufficiently great to cover several periods of the fluctuation, an average transmission coefficient may be defined which agrees closely with the classical value. For a one-dimensional barrier the corrections are small, although in a multidimensional system the transmission coefficient is affected by the interchange of vibrational and translational energy. It was concluded that for a large number of reactions involving species in their ground electronic states, κ is close to unity.

In section 3.4, two special causes of small P factors were discussed. The reactions concerned were (i) the bimolecular recombination of atoms and (ii) non-adiabatic reactions, and the small P factor can be identified in each case with a low probability of the activated complex reaching the final state, rather than a low probability that the activated complex is formed. Examples of reactions were given which were electronically or vibrationally non-adiabatic. It is proposed to discuss further some considerations respecting electronically non-adiabatic processes. The confusion about terminology pointed out in section 3.4 will be avoided here by not using the term "non-adiabatic", and instead referring to the actual conditions which give rise to high or low values of κ .

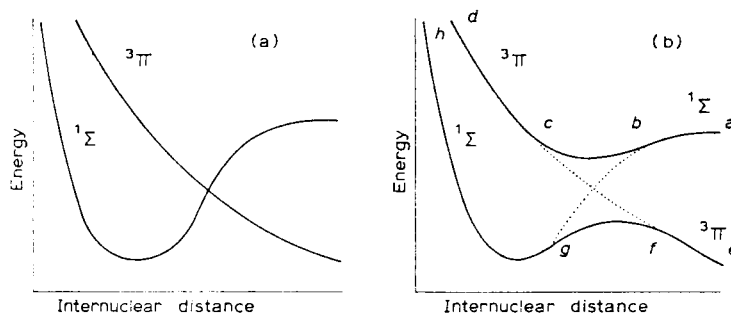


Fig. 8. Crossing potential energy curves.

The representation of the “crossing” of surfaces is difficult in two dimensions, and an example is given here of the “crossing” of two potential energy curves (which may be a section through a surface). Fig. 8 shows a pair of curves: they are designated $^3\Pi$ and $^1\Sigma$, and in Fig. 8(a) they are represented as zero-order crossing curves, while in Fig. 8(b) they are represented as a pair of higher order non-crossing curves. The important point is that if the curves are regarded as non-crossing, then the spectroscopic designation of the curves changes at the point which corresponds with crossing in Fig. 8(a).

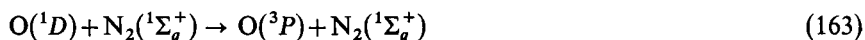
An actual example of a system with curves similar to those in Fig. 8 is nitrous oxide. Ground state $\text{N}_2\text{O}(^1\Sigma^+)$ correlates with ground state $\text{N}_2(^1\Sigma_g^+)$ and excited atomic oxygen, $\text{O}(^1D)$, while the $^3\Pi$ repulsive state of N_2O correlates with $\text{O}(^3P)$ and $\text{N}_2(^1\Sigma_g^+)$.

The extent of the separation of the two curves at the “crossing” point in Fig. 8(b) depends on the symmetry properties of the two states. If the states are of the same “species” (*e.g.* they have the same values of Λ and S), then the resonance separation, equal to the matrix element $\int \psi_1 H \psi_2 d\tau$ where ψ_1, ψ_2 are the wavefunctions for the two states, is large (several kcal. mole⁻¹). On the other hand, if the states are of different species, as they are in Figure 8, then the resonance splitting is small. For a small resonance splitting, the probability of transition in either direction between upper and lower surfaces is near unity, and the curves behave as though they really crossed. Conversely, the probability of transition between curves separated by a large resonance splitting is small. It is found that for states of different symmetry species, the resonance splitting depends on spin-spin and spin-orbital interactions, and it is possible to derive selection rules for transitions between two crossing curves or surfaces¹²⁹. For example, where the quantum numbers Λ and S have a meaning, $\Delta\Lambda = 0, \pm 1$ and $\Delta S = 0$ are selection rules for radiationless transitions. One aspect of these considerations is that reaction systems for which there are “crossing” potential energy surfaces tend to resist any change in S : this is the Wigner spin-conservation rule¹⁹. If the two potential surfaces are of the same species the resonance splitting is large and no

transition between upper and lower surfaces is expected; however, for the two surfaces to be of the same species, they must have the same value of S . Surfaces with *different* values of S are of different species, and transitions occur at the (small) splitting point with a conservation of S . In our example of nitrous oxide, the probability of the process



is therefore small¹³⁰. The relationship between the probability of transition from one surface to another and the transmission coefficient may be illustrated for the quenching process



If we take Fig. 8 to refer to nitrous oxide, then there are two ways in which (163) can occur *via* N_2O . The point a in Fig. 8(b) corresponds to the reactants, and the point e to the products. Point e can be reached (i) by the system remaining on $abcd$ during a vibration and crossing from c to f in a subsequent vibration, or (ii) by direct crossing from b to g and subsequent passing to e . If the probability of crossing is P , then the overall probability of (i) is the probability of first remaining on $abcd$, $(1-P)$, multiplied by the probability of crossing, P . Similarly, the probability of (ii) is also $P(1-P)$, so the overall probability of the quenching process (163) is $2P(1-P)$, and

$$\kappa = 2P(1-P) \quad (164)$$

An interesting inference to be drawn from the form of (164) is that quenching is unlikely either for very high or for very low values of P , and κ has a maximum of $\frac{1}{2}$ when $P = \frac{1}{2}$. The more complex relationships between κ and transition probabilities in systems where there are several potential energy surfaces have been discussed by Laidler¹³¹ and by Magee¹³².

The magnitude of P has been given by Landau¹³³ and Zener¹³⁴ for crossing between states of different species as

$$P = \exp[-4\pi^2\varepsilon_{12}/hV|s_1-s_2|] \quad (165)$$

ε_{12} is the resonance splitting, V the velocity with which the system passes the point of closest approach of the surface, and $|s_1-s_2|$ the magnitude of the difference of slopes with which the curves would cross in the zero-order approximation (*i.e.* the difference of slopes of cf and hg). No treatment seems to have been given for transitions between curves of the same species. The Landau-Zener equation, (165), indicates that $P = 1$ when $\varepsilon_{12} = 0$, and for systems in which ε_{12} is small,

the probability that the system will *remain* on a given curve, P' ($= 1 - P$), is given by

$$P' = \frac{4\pi^2 \varepsilon_{12}^2}{hV(s_1 - s_2)} \quad (166)$$

The value of κ calculated¹³⁰ for reaction (162) is 10^{-4} .

4.5.8 The reaction co-ordinate

The use of a one-dimensional representation for the path of the reacting species over an energy barrier, such as the representation in Fig. 3, suggests that the reaction system may be treated as confined to motion along just the one path which has been called the reaction co-ordinate. Some of the discussion of earlier sections has indicated that the one-dimensional treatment is inadequate, particularly when the one-dimensional energy barriers are used to calculate tunnelling corrections. The nature of the reaction co-ordinate is required in the ordinary formulation of the transition state theory, since it determines which vibrational term is to be omitted from the vibrational partition function for the activated complex. For the purpose of obtaining a rough estimate of reaction rates the exact choice of reaction co-ordinate may not be of great importance, since vibrational stretching frequencies are mostly of the same order of magnitude. (Laidler and Polanyi⁵⁸ note that in *cis-trans* isomerizations, where the angle of twist is the reaction co-ordinate, the extraction of a normal vibrational frequency from the partition function calculation will give an erroneous rate coefficient.) The correct procedure for obtaining accurate values for rate coefficients is to attempt to identify the reaction co-ordinate with a normal mode of vibration: normal-mode analysis of the vibrations will then indicate the frequency of the vibration to be omitted. This situation exists, for example, in the $\text{H} + \text{H}_2$ reaction where linear $\text{H}-\text{H}-\text{H}$ is thought to be the activated complex configuration. The H_3 complex has four normal modes of vibration: symmetric and asymmetric stretching and the doubly degenerate bending modes. The symmetrical nature of any reasonable potential energy surface for H_3 suggests at once that the reaction co-ordinate may be identified with the normal-mode asymmetric stretching vibration, ν_2 . As the reactant H atom approaches colinearly with H_2 , the most distant H atom of the molecule recedes. The actual frequencies of ν_3 and ν_1 can then be calculated by using force constants estimated from the shape of the potential energy surface around the top of the barrier along the chosen reaction co-ordinate. In the case of a polyatomic activated complex, if the reaction co-ordinate corresponds to a linear combination of several normal modes, then the frequencies of the vibrations to remain in the partition function are obtained from a normal-mode analysis with the combination held fixed. A rather different ap-

proach may be needed in cases where the reaction co-ordinate cannot be identified with a normal mode. For example, the “repulsive” energy surfaces, believed to be of importance in the reactions of atomic hydrogen with halogens (see section 4.5.6), are characterised by behaviour in which for a reaction $A + BC \rightarrow AB + C$, AB at its normal separation retreats from C. The reaction co-ordinate is therefore the B–C distance, and the analysis of frequencies must be carried out with the B–C distance held fixed. The remaining vibrations are the two bending modes and the motion of A along the line of centres in B–C.

The assumption that the potential energy is such as to permit separation of the variables r_{AB} and r_{BC} from all the others is an approximation. Marcus⁷² has shown, however, that *in the immediate vicinity* of a potential energy maximum or of a saddle point, the major topographical features of a “non-separable” potential energy surface can be imitated by those of a surface permitting separation of variables. According to this local approximation, an orthogonal curvilinear co-ordinate system is introduced which permits separation of the variables, the origin of the co-ordinate system being selected so that there exists a pair of curvilinear co-ordinate curves intersecting, and cotangential with the paths of steepest ascent and descent, at the saddle point. The theory of reaction rates developed by Marcus allows the use of curvilinear reaction paths. The curvature of the separable curvilinear reaction co-ordinate is found to be less than that of the path of steepest ascent. In classical terms, this means that a trajectory passing through the saddle point proceeds along the sides of the valley, rather than following the path of steepest ascent, as a result of the centrifugal force experienced by the particle as it follows the curved path. (The centrifugal force is least at the saddle point where the velocity is smallest.) The effect occurs also in quantum mechanical formulations, and one of the consequences of the curvature of the reaction path is that the potential barrier is “thicker” than that predicted from the paths of steepest ascent, so some correction may be needed to tunnelling terms.

5. Unimolecular reactions

5.1 INTRODUCTION

A unimolecular reaction may be represented by the elementary process



and the rate of reaction in (167) will obey a first-order kinetic law

$$\frac{-d[A]}{dt} = k[A] \quad (168)$$

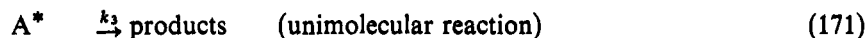
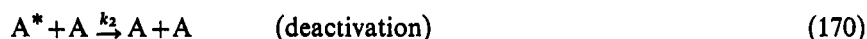
There are, however, a large number of reactions which are kinetically first-order under suitable experimental conditions but which are not elementary processes in the sense that only a single step is involved in the reaction. These reactions involve a unimolecular transformation in their final stage, and are always treated together with any discussion of unimolecular reactions.

For a unimolecular reaction to take place, the molecule which undergoes reaction must have sufficient energy to allow the necessary rearrangement of the chemical bonds. The required energy can derive from several sources: it may be energy received by the absorption of light or ionizing radiation (as in photochemical or radiochemical processes), or it may be the excess energy with which the molecule has been formed in an exothermic chemical reaction ("chemical activation"). The source of the energy in thermal reactions must be molecular collision. Lindemann¹³⁵ made the original suggestion that unimolecular reaction could take place under the influence of collisions, and Hinshelwood¹³⁶ showed that if molecular collisions could both activate and deactivate a reactant molecule, then collisional activation could be reconciled with first-order kinetic behaviour. This first theory of unimolecular reactions explained the more general kinetic features of experimental observations, although it did not allow for the varying rate of the unimolecular step with differing degrees of excitation of the reactant. Subsequent developments of the theory have been made in order to account for the more detailed behaviour of unimolecular reactions.

Kassel¹³⁷ and Rice and Ramsperger¹³⁸ extended the Lindemann-Hinshelwood theory for activated molecules of differing levels of excitation on the assumption that energy in the energised molecule could flow freely between some or all of the normal modes of excitation in the molecule. The initials of the proponents are often used to designate this theory—RRK (sometimes RRKH to include the parentage of the theory from Hinshelwood's earlier ideas). Slater¹³⁹ proposed an alternative theory of unimolecular reactions in which energy flow in the activated molecule was forbidden. Instead, the intrinsic rate of decomposition of activated molecules was predicted by superposition of the oscillator amplitudes on the assumption that the oscillators were strictly harmonic. That is to say, the activated molecule decomposes only when the phase relationships lead to sufficient distortion of the bond to allow the chemical transformation. The application of the transition state theory to unimolecular processes is subject to two major considerations. First, the equilibrium assumption may be quite invalid under low pressure conditions where the rate of reaction of activated molecules may be greater than the rate at which they can be activated by the collision process. Secondly, it is difficult to see exactly what is the transition state of the system. The potential energy of interaction of the two fragments of a decomposing molecule will not have a well-defined maximum (see, however, the discussion of *free-energy* maxima in section 4.3.1). Marcus^{140, 141} has developed a quantum theory of unimolecular reactions which uses the methods of the transition state theory to extend the RRK approach.

5.2 THE LINDEMANN-HINSHELWOOD THEORY

The principle steps of activation, deactivation and unimolecular transformation of the Lindemann-Hinshelwood theory^{135, 136} are



Solution of the steady-state equations for $[A^*]$ gives

$$[A^*] = \frac{k_1[A]^2}{k_2[A] + k_3} \quad (172)$$

so that the rate of reaction is given by

$$-\frac{d[A]}{dt} = k_3[A^*] = \frac{k_1 k_3 [A]^2}{k_2[A] + k_3} \quad (173)$$

At sufficiently high pressures, $k_2[A] \gg k_3$, and equation (173) becomes, approximately

$$-\frac{d[A]}{dt} = \frac{k_1 k_3}{k_2} [A] \quad (174)$$

The Lindemann-Hinshelwood mechanism therefore predicts first-order kinetics for thermal unimolecular reactions under conditions where the concentration of activated molecules is controlled by the gas-phase bimolecular activation-deactivation processes (169) and (170), rather than by the rate at which A^* is removed in (171).

At low pressures, the concentration of A^* will be determined by the rate of activation and the rate of unimolecular removal of A^* . That is, $k_2[A] \ll k_3$, and

$$-\frac{d[A]}{dt} = k_1[A]^2 \quad (175)$$

Thus, under low pressure conditions, the rate of reaction is kinetically second-order in $[A]$, even though the step leading to loss of A is unimolecular. The explanation of first-order behaviour at high pressures, and the turning over to second-order kinetics at low pressures was one of the main achievements of the Lindemann-Hinshelwood theory. Experimental results were found to behave

approximately in the manner predicted, and by observing the pressure at which first-order gave way to second-order behaviour, some estimate could be made of the relative magnitudes of k_2 and k_3 . A "first-order" rate coefficient, k' , may be written for the processes, such that

$$k' = - \frac{1}{[A]} \frac{d[A]}{dt} = \frac{k_1 k_3 [A]}{k_2 [A] + k_3} \quad (176)$$

although k' will be a function of $[A]$. At low pressures, indeed, k' will show an approximately linear dependence on $[A]$, while at high pressures it will be nearly independent of $[A]$. As an example of the way in which experimental observations qualitatively support the predictions of the Lindemann-Hinshelwood scheme, Fig. 9 shows the variation of k' with pressure for some results obtained¹⁴² for the reaction



It is seen that the general features of the pressure dependence of k' are as anticipated, and that k' tends towards a limiting value at high pressures (the actual experimental results extend to pressures of several atmospheres, and k' remains virtually constant). At high pressures, therefore, the reaction is very nearly first-order.

The pre-exponential factors of k_1 and k_2 might be expected to be approximately identical, so that the high pressure value of k' (which we will refer to as k_∞)

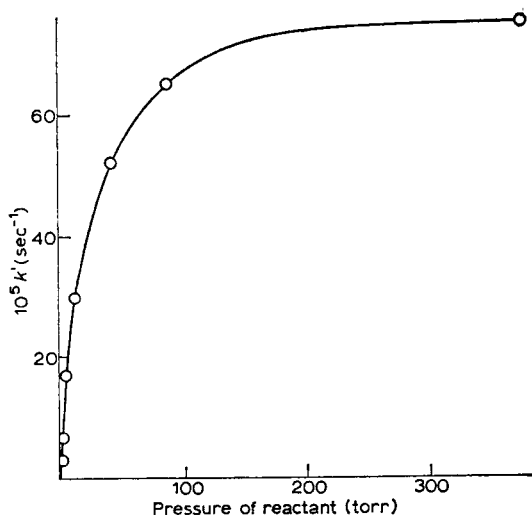


Fig. 9. The variation of k' with pressure in the isomerization of CD_3NC to CD_3CN ¹⁴².

should be consistent with an Arrhenius-type expression

$$k' = Ae^{-E_a/RT} \quad (178)$$

where A is the pre-exponential factor of k_3 , and E_a a composite activation energy for k_1 , k_2 and k_3 . Although, in general, experimental rate determinations show, to a greater or lesser extent, deviations from the simple Arrhenius law, they can be fitted sufficiently closely to a linear plot to yield values of A . It has frequently been pointed out that the pre-exponential factor for unimolecular reaction has a tendency to lie near $10^{13.5} \text{ sec}^{-1}$, and frequency distribution curves or histograms have often been presented to bear out this contention. Fig. 10 is a histogram of numbers of reactions known to have A factors lying within various ranges of $\log A = 0.5$, taken from recent tables of experimental A factors¹⁴³ for isomerization and fission reactions.

The pre-exponential factor for unimolecular reaction has a high probability of having a value close to the vibration frequency of the bond involved in the reaction (*i.e.* 10^{13} – 10^{14} sec^{-1}), and this result is the one expected on the basis of either RRK or Slater theories, as will appear in due course. (The vibration frequency does not arise as such in the Lindemann–Hinshelwood formulation, although it was, of course, understood that k_3 had the dimensions of frequency.) The existence of reactions possessing A factors considerably in excess of vibration frequencies—there are, for example, 6 reactions with $A > 10^{17} \text{ sec}^{-1}$ given by Benson and DeMore¹⁴³—requires consideration. The explanation frequently suggested is that the ratio of A factors for k_1 and k_3 is *not* unity. The treatment

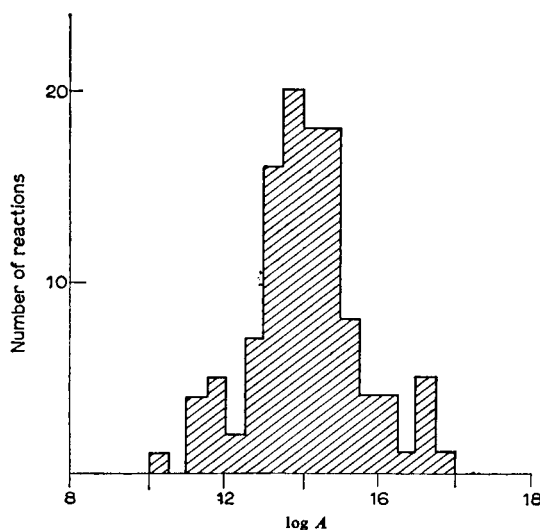


Fig. 10. Histogram of numbers of unimolecular reactions having given pre-exponential factors.

given in section 3.6.3 for activation in many degrees of freedom indicates that if a total energy greater than E_c can lead to excitation of A^* , no matter where that energy is stored in A , then the rate of excitation will be

$$\frac{e^{-E_c/RT} \left(\frac{E_c}{RT} \right)^{\frac{1}{2}S-1}}{(\frac{1}{2}S-1)!} \cdot Z_{AA} \quad (59)$$

Since the collision rate coefficients for (169) and (170) will be identical, the ratio of pre-exponential factors, A_1/A_2 will be given by

$$\frac{A_1}{A_2} = \frac{\left(\frac{E_c}{RT} \right)^{\frac{1}{2}S-1}}{(\frac{1}{2}S-1)!} \quad (179)$$

As was pointed out in section 3.6.3 for a reaction with a reasonable activation energy (related to E_c), this ratio could reach 10^5 or 10^6 for 10 "square terms" (S). The absolute value of k_1 may be calculated from experimental results by inversion of (176), *viz.*

$$\frac{1}{k'} = \frac{k_2}{k_1 k_3} + \frac{1}{k_1 [A]} \quad (180)$$

A plot of $1/k'$ against $1/[A]$ should be linear and have a slope of k_1 . In an early experiment on the isomerization of cyclopropane, Chambers and Kistiakowsky¹⁴⁴ calculated that to fit their values of k_1 , S must be about 26 (*i.e.* 13 vibrations must be involved) on the basis of a collision rate estimated from a molecular diameter of 3.9 Å for cyclopropane. Cyclopropane has 9 atoms in the molecule, which means that there are 21 vibrational modes. Between one-half and two-thirds of the total available number of vibrations must be excited, therefore, in order to account for the value of the observed rate coefficient, k_1 . Although Chambers and Kistiakowsky fitted their results at different temperatures to a linear Arrhenius plot, later experiments on various unimolecular processes have been fitted to the temperature dependence of E_a suggested by equation (60) in section 3.6.3, using S as a variable parameter. In most cases the best agreement with experiment was obtained with $S/2$ set equal to between one-half and two-thirds of the number of vibrational degrees of freedom, thus bearing out in a general way the conclusions of Chambers and Kistiakowsky.

If, in fact, equation (180) represented adequately the variation of k' with $[A]$, the value of $1/k'$ at $1/[A] = 0$ would give the ratio $k_2/k_3 k_1$, and since the value of k_1 is also known, a value for the relative efficiencies of quenching or reaction, k_2/k_3 , could be obtained. Estimates of the quenching rate might be made from the

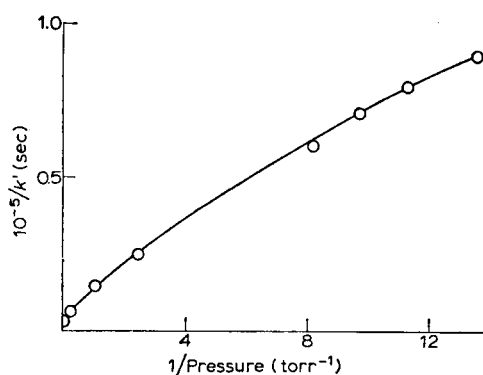


Fig. 11. Plot of $1/k'$ against inverse pressure for the isomerization of CD_3NC ¹⁴².

results of energy transfer experiments, and each rate coefficient would be known absolutely. How far equation (180) is an adequate representation is shown in Fig. 11.

Fig. 11 is a plot of $1/k'$ against $1/\text{pressure}$ for some of the experimental results¹⁴² on the isomerization of CD_3NC to CD_3CN . Although over any small region, especially at high values of $1/p$, a straight line might be drawn through the points, it is seen that the plot is really curved. The curvature can be taken to mean that the value of the intercept for $1/p = 0$ varies with pressure, and is greatest at low pressures. This means that if k_2/k_1 is independent of pressure, k_3 is a function of pressure. The pressure dependence of k_3 reveals the weakness of the Lindemann-Hinshelwood theory. According to the reaction scheme (169) to (171), there are only two kinds of A molecule: those with and those without enough energy to undergo unimolecular reaction (171). No provision is made for a molecule with a considerable amount of energy beyond the critical amount to react more rapidly than a molecule with just barely the necessary energy. The inference to be drawn from Fig. 11, that k_3 is lower at lower pressures, is in accordance with an energy dependence of k_3 . A more complete theory of unimolecular reactions will therefore differentiate between all levels of excitation of the reactant, and ascribe a separate value of k_3 to the reactivity of molecules in each level. In addition, the values of k_1 and k_2 should now be separated out for the production and deactivation of each level of excitation of A.

It is possible to derive an expression for the rate of reaction, using the idea that k_3 is energy dependent, which shows the same pressure dependence of kinetic order as that predicted by the simple Hinshelwood scheme. Let us suppose that there exists an active state of A, A_a^i , such that A_a^i has a non-zero chance of being converted into a "reactive" species A_i^* on collision. Then the rate of formation of A_i^* is given by

$$+ \frac{d[\text{A}_i^*]}{dt} = k_c[\text{A}][\text{A}_a^i] \quad (181a)$$

$[k_c]$ is here the "collision rate coefficient" $= Z_{AA_a}/[A][A_a] = 4\sigma^2(RT/\pi M)^{1/2}$. If the cross-section, σ^2 , for activation and deactivation is identical, then the rate at which A_i^* is deactivated in reactions of type (170), $(-d[A_i^*]/dt)_d$, is given by

$$\left(-\frac{d[A_i^*]}{dt}\right)_d = k_c[A_i^*][A] \quad (181b)$$

while the rate at which A_i^* is removed by unimolecular reaction, $(-d[A_i^*]/dt)_r$, is equal to

$$\left(-\frac{d[A_i^*]}{dt}\right)_r = k_3^i[A_i^*] \quad (182)$$

k_3 now being split into k_3^i for each energy level, E_i , of A_i^* . Solution of the steady state equations for $[A_i^*]$ gives

$$[A_i^*] = \frac{k_c[A][A_c^i]}{k_c[A] + k_3^i} \quad (183)$$

so that the overall rate of reaction can be expressed by

$$-\frac{d[A]}{dt} = \sum_i k_3^i[A_i^*] = \sum_i \frac{k_c k_3^i[A][A_c^i]}{k_c[A] + k_3^i} \quad (184)$$

The value of $[A_c^i]$ may be calculated from the Maxwell-Boltzmann law if thermal equilibrium may be assumed, *viz.*

$$[A_c^i] = [A] \frac{g_i e^{-E_i/RT}}{\sum_j g_j e^{-E_j/RT}} \quad (185)$$

For the high pressure condition, $k_3^i \ll k_c[A]$, equation (184) may therefore be rewritten

$$-\frac{d[A]}{dt} = [A] \frac{\sum_i k_3^i g_i e^{-E_i/RT}}{\sum_j g_j e^{-E_j/RT}} \quad (186)$$

Similarly, at low pressures, where $k_3^i \gg k_c[A]$

$$-\frac{d[A]}{dt} = k_c[A]^2 \cdot \frac{\sum_i g_i e^{-E_i/RT}}{\sum_j g_j e^{-E_j/RT}} \quad (187)$$

The kinetic orders of (186) and (187) are thus the same as those of equations (174) and (175).

It is of interest to calculate the "experimental" activation energy for the high pressure case. We may write

$$E_a = RT^2 \frac{d \ln k_\infty}{dT} = \frac{\sum_i E_i k_3^i g_i e^{-E_i/RT}}{\sum_i k_3^i g_i e^{-E_i/RT}} - \frac{\sum_j E_j g_j e^{-E_j/RT}}{\sum_j g_j e^{-E_j/RT}} \\ = \left(\text{average energy of molecules with } E > E_i \right) - \left(\text{average energy of all molecules} \right) \quad (188)$$

The equation should be compared with equation (63) of section 3.6.3.

One very important observation must be made about the general derivation of the kinetic orders. Although the assumption of thermal equilibrium may be justified in the high pressure case, it may very well *not* be valid at low pressures. Indeed, the derivation depends on the fact that the concentration of A_i^* is determined by removal processes not involving collisional deactivation, and if k_3^i is greater for more energetic molecules, the rate of depletion of energetic molecules may be greater than the rate of production needed to maintain the equilibrium concentration.

Before moving on to discuss the application of the transition state theory to unimolecular reactions, and to more general treatments, it seems appropriate to point out the dangers of assuming that all first-order processes involve either a true unimolecular step or an activation-deactivation mechanism. A chain mechanism, not involving activation, can give rise to first-order kinetics. The first-order decomposition of dinitrogen pentoxide was, for a long time, regarded as a "unimolecular" process, although it became apparent that the kinetics were of the first order down to the lowest pressures at which the experiments were conducted. Ogg¹⁴⁵ and Mills and Johnston¹⁴⁶ have shown that, in fact, the reaction proceeds *via* a chain mechanism involving the radical NO_3



Solution of the steady state equation for $[\text{NO}_3]$ leads to an expression

$$-\frac{d[\text{N}_2\text{O}_5]}{dt} = \frac{2k_6 k_4 [\text{N}_2\text{O}_5]}{k_5 + 2k_6} \quad (192)$$

which is first-order at all pressures. A considerable number of thermal reactions, formerly thought to be unimolecular, has been shown to proceed *via* a chain mechanism. The nitric oxide inhibition experiments devised by Hinshelwood¹⁴⁷ appear to show that a number of pyrolyses proceed *via* both a chain and a unimolecular mechanism. Even the thermal decomposition of azomethane, which was for many years regarded as a good example of a unimolecular reaction, has been shown¹⁴⁸ to involve short chains initiated and terminated at the surface of the reaction vessel.

5.3 APPLICATION OF THE TRANSITION STATE THEORY TO UNIMOLECULAR REACTIONS

There are two main obstacles to the use of the transition state model for unimolecular reactions. First, there is the difficulty in defining the transition state for a unimolecular process, and secondly there is the question of the validity of the equilibrium assumption (although for the high pressure condition such an assumption may be applicable). In the introduction to this discussion of unimolecular reactions, section 5.1, and in section 4.3.1, it was pointed out that the transition state for dissociative processes may be defined by a maximum in a free-energy surface. An alternative suggestion is that the effective energy of the reaction system should be taken to be the sum of the potential energy and the rotational energy, *i.e.*

$$\varepsilon_{\text{eff}} = \varepsilon + \varepsilon_J \quad (193)$$

The rotational energy for any value of J depends on the moments of inertia, and thus ultimately on the internuclear distance between a pair of atoms which separate during a unimolecular dissociation. The rotational contribution to the total energy therefore rises to a maximum and then finally disappears as dissociation occurs. If, in fact, it is the rotational energy maximum that defines the transition state, then it is clearly of importance to know the value of ε_J under these conditions.

Substitution of the approximate classical rotational partition function (119) for three unequal moments of inertia into the usual expression for rate coefficient (99) yields

$$k = \kappa \frac{kT(I_A^\ddagger I_B^\ddagger I_C^\ddagger)^\frac{1}{2} f'_{\text{vib}}}{h(I_A I_B I_C)^\frac{1}{2} f'_{\text{vib}}} e^{-E_c/RT} \quad (194)$$

The relative simplicity of this equation is a direct result of the nature of a unimolecular process. The masses of the initial molecule and the activated state are identical, and since the translational partition functions are raised to the first

power in both numerator and denominator of the expression they cancel, as do the constants in the rotational partition functions.

A further simplification may be introduced if only one of the moments of inertia, I_A say, is increased significantly on stretching of the bond to be broken. If the expression for the vibrational partition functions (113) is now inserted into (194), the equation becomes

$$k = \kappa \frac{kT}{h} \left(\frac{I_A^\ddagger}{I_A} \right)^{\frac{1}{2}} \frac{\prod_{3N-7} [1 - e^{-h\nu^\ddagger/kT}]}{\prod_{3N-6} [1 - e^{-h\nu/kT}]} e^{-E_c/RT} \quad (195)$$

For normal bonds at room temperatures, $h\nu \gg kT$ for both initial and activated states, and the vibrational partition functions are unity. Thus

$$k = \kappa \cdot \frac{kT}{h} \left(\frac{I_A^\ddagger}{I_A} \right)^{\frac{1}{2}} e^{-E_c/RT} \quad (196)$$

Although the calculation of the exact value of $I_A^\ddagger$ may be difficult, it is probable that $I_A^\ddagger$ is of the same order of magnitude as I_A , and the pre-exponential factor becomes $\kappa(kT/h)$. At room temperature kT/h is about 10^{13} sec^{-1} , so that for $\kappa = 1$, the "normal" frequency factor is obtained (see section 5.2).

If the vibration frequency of the bond to be broken is abnormally low, then just the first term in the expansion for $e^{-h\nu/kT}$ need be taken in evaluating the partition function. Under these circumstances, (195) with $I_A^\ddagger \sim I_A$ simplifies to

$$k = \kappa \nu_b e^{-E_c/RT} \quad (197)$$

where ν_b is the frequency of the breaking bond. For a temperature around 300°K , the form of the expression given in (197) may be valid for $\nu_b < 10^{12} \text{ sec}^{-1}$.

The transition state theory does not give any explanation of abnormally large A factors in unimolecular reactions. Some increase in the pre-exponential factor above 10^{13} sec^{-1} can be achieved if $I_A^\ddagger > I_A$, although it is hardly possible to find a *reasonable* value for $I_A^\ddagger$ which will give a factor of 10^{16} or 10^{17} .

One of the most valuable applications of the transition state theory to unimolecular reactions has been made by Marcus^{140, 141}. The theory is in principle a quantum mechanical transition state reformulation of the RRK free-flow theory, and includes both quantum and anharmonic effects. In view of the importance of Marcus' theory, and its assumption of a free flow of energy, it seems that the theory merits a short section of its own which should follow the discussion of free-flow theories. The Marcus approach is therefore developed in section 5.5.

Another important treatment of an equilibrium theory of unimolecular reactions has been given by Giddings and Eyring¹⁴⁹ for both dissociation and isomeriza-

tion processes; the values to be expected for κ in the unimolecular step are discussed. For example, they point out that since κ is small in a bimolecular association process (see section 3.4), the transmission coefficient in the reverse, unimolecular dissociation, process will also be small. Following the presentation of Giddings and Eyring, we may write the general reaction scheme as



A^* is a molecule of energy ε about to undergo unimolecular reaction *via* a transition state $A^\ddagger$. If $K_e = k_i/k_{-i}$, then the fraction of molecules, F_e , with energy ε per unit energy range is given by

$$F_e = a_e K_e \quad (200)$$

where a_e is a non-equilibrium constant, *viz.*

$$a_e = \frac{k_{-i}[A]}{k_{-i}[A] + k_e} \quad (201)$$

The overall rate coefficient for reaction is then given by

$$k' = \int_{\varepsilon=0}^{\infty} F_e k_e d\varepsilon = \int_0^{\infty} \frac{k_e k_i [A]}{k_e + k_{-i} [A]} d\varepsilon \quad (202)$$

This expression possesses the correct dependence of k' on $[A]$. It is now possible to write a transition state formulation of k' in which the non-equilibrium effects can be included in the value of the transmission coefficient. If κ is the ordinary transmission coefficient, then the pre-exponential part of the expression must be multiplied by a factor γ given by

$$\gamma = \kappa \int_{\varepsilon=0}^{\infty} k_e a_e K_e d\varepsilon \bigg/ \int_{\varepsilon=0}^{\infty} k_e K_e d\varepsilon \quad (203)$$

The evaluation of k_e by the transition state theory has been considered by Rosenstock *et al.*¹⁵⁰. They have shown that for any system in which $\kappa = 1$

$$k_e = \frac{1}{h} \int_{\varepsilon=0}^{\varepsilon-\varepsilon_c} \frac{\rho^\ddagger(\varepsilon, \varepsilon_c, \varepsilon_t)}{\rho(\varepsilon)} d\varepsilon_t \quad (204)$$

where $\rho, \rho^\ddagger$ are energy densities, and the additional energy variable, ε_t , is the

translational energy of the system passing across the saddle point of the transition state. If every vibration in A_e^* and $A^\ddagger$ is a classical harmonic oscillator, then (204) may be written in the form

$$k_e = \frac{\prod_{i=1}^s v_i}{\prod_{i=1}^s v_i^\ddagger} \left(\frac{\epsilon - \epsilon_c}{\epsilon} \right)^{s-1} \quad (205)$$

where s is here the *number of oscillators*, and v_i are the vibrational frequencies.

The final result for the case at low pressure, where thermal equilibrium is not maintained, is obtained by formulating the rate coefficients k_i and k_{-i} in terms of the transition state theory and summing over all states. Giddings and Eyring¹⁴⁹ give the expression

$$k' = \frac{1}{hQ_A} \int_0^\infty a_e \rho(\epsilon) e^{-\epsilon/kT} d\epsilon \int_0^{\epsilon - \epsilon_c} \frac{\rho^\ddagger(\epsilon, \epsilon_c, \epsilon_t)}{\rho(\epsilon)} d\epsilon_t \quad (206)$$

(where Q_A is the partition function for A) for the rate coefficient, k' , at any pressure.

The relation of the expressions obtained by the transition state theory to the results of the RRK free-flow theory will be shown in section 5.4.

5.4 FREE FLOW THEORIES

The interpretation of the kinetics of unimolecular processes in terms of the free flow of energy between the various degrees of freedom in the molecule was considered first by Kassel^{137, 151} and Rice and Ramsperger^{138, 152}. Kassel has summarised the work in his book (see Bibliography). For the present purposes, the form of the theory used by Kassel is adopted. Let there be s oscillators, each with the same frequency ν ; the energy of a state defined by the quantum number i is then taken to be $\epsilon = i h \nu$.

It is now necessary to calculate the probability, p_i , that of this energy an amount $\epsilon_c (= m h \nu)$ shall be concentrated in any one oscillator. The coupling between the oscillators is regarded as sufficiently weak to allow each oscillator to be independent, but at the same time sufficiently strong that the total energy may be freely distributed about the various degrees of freedom. It may be shown that

$$p_i = \frac{i!(i - m + s - 1)!}{(i - m)!(i + s - 1)!} \quad (207)$$

Kassel then assumed that the rate coefficient for the unimolecular decomposition

process (171), k_3^i , is proportional to p_i . It may be shown that

$$p_i \rightarrow (1 - m/i)^{s-1} \quad \text{as } i \rightarrow \infty \quad (208)$$

and since the ratio m/i may be expressed in terms of the molar energies E and E_c ($\epsilon_c/\epsilon = E_c/E = ih\nu/mh\nu$), then

$$k_3^i = A(1 - E_c/E)^{s-1} \quad (209)$$

(where A is the constant of proportionality) if the system behaves as a system of non quantised oscillators (*i.e.* if i is very large).

An expression has already been obtained for the rate of a thermal first order reaction in terms of k_3^i (section 5.2, equations 184 to 187). In the high pressure case

$$k' = k_\infty = \frac{\sum_i k_3^i g_i e^{-E_i/RT}}{\sum_j g_j e^{-E_j/RT}} \quad (210)$$

The statistical weight g_i is given by

$$g_i = \frac{(i+s-1)!}{i!(s-1)!} \quad (211)$$

(g_j is given by a similar expression with i replaced by j). Substitution of g_i , g_j and k_3^i into (210) gives

$$k_\infty = A e^{-E_c/RT} \quad (212)$$

without use of the approximation (208).

To obtain the rate coefficient, k' , at a general pressure, the expression derived from (184) must be used, *viz.*

$$k' = \sum \left(\frac{k_c k_3^i [A_c^i]}{k_c [A] + k_3^i} \right) \quad (213)$$

$[A_c^i]$ may be obtained by a classical statistical procedure for excitation in several degrees of freedom, and the function of $[A_c^i]$ is represented by equation (57) of section 3.6.3 (with $S/2$ replaced by s because of the new definition of s). If the differential form of (61) with $\Gamma(s) = (s-1)!$ is used together with the classical value of k_3^i given in equation (209), and the summation of (213) replaced by integration,

we obtain

$$k' = \frac{A}{(s-1)!(RT)^s} \int_{E_c}^{\infty} \frac{(E-E_c)^{s-1} e^{-E/RT} dE}{1 + (A/k_c[A]) \cdot (1-E_c/E)^{s-1}} \quad (214)$$

It may be shown that at high pressures, the value obtained (214) is identical with the result of the non-approximate calculation (212). At low pressures, (214) becomes, on performing the integration

$$k' = \frac{(E_c/RT)^{s-1}}{(s-1)!} \cdot k_c[A] \cdot \left[1 + \frac{(s-1)}{E_c/RT} + \frac{(s-1)(s-2)}{(E_c/RT)^2} + \cdots + \frac{(s-1)!}{(E_c/RT)^{s-1}} \right] e^{-E_c/RT} \quad (215)$$

Thus it can be seen that if E_c/RT is large, the experimental activation energy is given by

$$E_a = RT^2 \frac{d \ln k'}{dT} = E_c - (s-1)RT \quad (216)$$

This expression is the same as the activation energy obtained by Hinshelwood for activation in many degrees of freedom (see equation (60), bearing in mind that $S/2$ is called s in the Kassel notation). However, the Hinshelwood theory predicts the same activation energy at all pressures, while the Kassel theory suggests that the measured activation energy will vary from E_c at high pressures (equation (212)) to the lower value predicted by (216) at low pressures. The lowering of the effective activation energy at low pressures indicates that the predicted *pressure* dependence of k' will be different for the two theories, a sharper change of k' with pressure being expected on the basis of the Kassel theory.

Kassel's classical theory¹³⁷ was followed by two quantum-mechanical versions¹⁵¹, which give substantially the same results as the classical theory. For example, in one of the theories the s classical oscillators are replaced by quantal oscillators of frequency ν . The form of k_3^i is retained as

$$k_3^i = A p_i \quad (217)$$

and the high pressure rate coefficient $k' = k_{\infty}$ is obtained in the quantum mechanical case by writing

$$[A_c^i] = [A](1-\alpha)^s \alpha^i p_i \quad (218)$$

where $\alpha = e^{-h\nu/kT}$. Thus, using the high pressure form of (184) together with (218) and summing over i from m to infinity, we obtain

$$k_{\infty} = A\alpha^m = Ae^{-E_c/RT} \quad (219)$$

The general rate coefficient is obviously given by the general form of (184) after substitution of $[A_c^i]$ from (218), and k_3^i , p_i from (217) and (207).

The discussion of this section has been concerned with a model allowing a free-flow of energy, and Kassel's treatment has been used as an illustration. Rice and Ramsperger^{138, 152} developed their theory at about the same time as Kassel produced his formulation, and their theory differs from Kassel's only in that it requires the energy ϵ_c to accumulate in one degree of freedom rather than in one oscillator (= two degrees of freedom), for dissociation to occur. The outcome of the difference is that k_{∞} is not quite of the form (212), although the variation of k' with pressure expressed as k'/k_{∞} is very similar to the result obtained from the Kassel theory. The actual comparison of experimental results with the various theoretical predictions will be presented, after the Slater and Marcus theories have been discussed, in section 5.7.

It may finally be of interest to compare the RRK result for the unimolecular rate coefficient, k_3^i , with the result given by Giddings and Eyring¹⁴⁹ in their application of the transition state theory to unimolecular reactions. The transition state formula for k_e (205) and the RRK formula for k_3^i (209) may be identified, and since $(\epsilon - \epsilon_c)/\epsilon = (E - E_c)/E$, it may be shown that

$$A = \frac{\prod_{i=1}^s v_i}{\prod_{i=1}^s v_i^{\dagger}} \quad (220)$$

5.5 MARCUS' QUANTUM MECHANICAL TRANSITION STATE FORMULATION

The errors inherent in any theory of unimolecular reactions which treats the vibrations as classical and which neglects the importance of anharmonicity in the coupling of vibrations led Marcus to develop^{140, 141} a formulation of the RRK theory using the transition state method. Two specific advantages of the Marcus formulation are (i) that the curve-fitting parameter often used in the RRK formulation is absent, and (ii) that the theory is applicable under conditions where anharmonicity or quantum effects are important. This latter point is of particular significance, since it covers just those situations where the Slater theory to be developed in section 5.6 is not applicable.

A recent extension¹⁵³ of Marcus' theory has allowed both for centrifugal effects and for reaction path degeneracy. For the present purposes, however, it is

intended to set out the results of only the earlier work. The major changes effected by the modifications are first the inclusion of detailed rotational energies in the energy-balance equation, and secondly the introduction of a term for reaction-path degeneracy which in some places requires the use of symmetry numbers rather than rotational partition functions.

The original derivation given by Marcus¹⁴¹ uses a reaction scheme



The notation is that used by Marcus, and the subscripted rate coefficients should not be confused with those used previously, although they correspond roughly with those used for equations (169) and (170). A^* is an "active" molecule of A , and A^+ is the transition state; activation and deactivation is by means of a third body, M , which may or may not be A itself. The rate coefficients are all functions of the energy of the initial A^* molecule. It is assumed that each active molecule is randomly distributed about all configurations subject to energy and angular momentum conservation, and the degrees of freedom of the active molecule are classified as "active", "adiabatic", or "inactive", according to their role in intermolecular energy transfer. The "adiabatic" degrees of freedom are taken to remain in a given quantum state during the decomposition of the molecule, and do not, therefore, contribute much energy to the breaking bond; it was shown from an argument involving angular momentum conservation that the two rotational degrees of freedom involving the larger moments of inertia were approximately adiabatic. Energy in the "inactive" degrees of freedom is assumed to be transferred to the breaking bond only when the molecule is essentially in the transition state, and A^* must possess a "non-fixed" energy E^* such that E^* is greater than the sum of the activation energy and the "non-fixed" energy of the "inactive" modes (a "non-fixed" energy is energy which does not contain the energy "fixed" as zero-point energy).

The probability that A^* is in some critical configuration A^+ is now calculated by the methods of statistical mechanics. The vibrational energy levels of A^+ are treated quantum mechanically, as are the levels of A which are *not* levels of A^* : for simplicity of calculation, it was assumed that the number of energy levels for a given energy range in A^* could be computed classically (since the energies involved are relatively large). A weighted frequency factor, describing passage over the potential energy surface characterising the activated complex, is now multiplied by the probability that A^* was A^+ to give the rate coefficient k_a as a function of

the energy of the active molecule. The result is

$$k_a = \frac{P_1^+}{P_1} \frac{P_R^+}{h} \frac{1}{(r/2)!} \sum_{E_v^+ \leq E^+} \left\{ \frac{[(E^+ - E_v^+)/RT]^{r/2} P(E_v^+)}{N^*(E_a + E^+ + E_0)} \right\} \quad (224)$$

This is the original result¹⁴¹, but expressed using the more recent nomenclature, *viz.*

P_1^+ , P_1 are the partition functions of the *adiabatic* rotations of A^+ and A .
 P_R^+ is the partition function of the r *active* rotations of A^+ .

E^+ is the "non-fixed" energy of A^+ (see above).

E_v^+ is the difference between the zero-point energy of A^+ and the energy of the v th vibrational level in A^+ ; it is $P(E_v^+)$ fold degenerate.

E_0 is the zero-point energy of A .

E_a is the difference between the energy of A in its lowest state (all modes) and A^+ in its lowest state.

N^* (energy) is the number of energy states per unit energy of the active modes of A^* at the stated energy.

Treatment of $[A^*]$ by steady-state methods gives a value for the "unimolecular" rate coefficient, k' , if it is assumed that the fraction of energetic molecules having *insufficient* energy to react is the equilibrium value at all pressures (an assumption which is not always justifiable). The equation derived for k' is

$$k' = \frac{kT}{h} \frac{P_1^+ P_R^+}{P_1 P_2} \frac{e^{-E_a/RT}}{(r/2)!} \int_0^\infty \sum_{E_v^+ \leq E^+} \frac{[(E^+ - E_v^+)/RT]^{r/2} P(E_v^+) e^{-E^+/RT}}{1 + k_a/k_2 P} d(E^+/RT) \quad (225)$$

and it is shown that the high pressure value of k' , k_∞ , is given by

$$k_\infty = \frac{kT}{h} \cdot \frac{P_1^+ P_2^+}{P_1 P_2} \cdot e^{-E_a/RT} \quad (226)$$

where P_2^+ and P_2 are the partition functions for *all* active modes of A^+ and A , respectively. This equation is equivalent to (194) given in section 5.3. The quantum mechanical treatment of energy levels manifests itself in equation (225) and (226) by the presence of the quantum mechanical partition functions for A and A^+ , P_2 and P_2^+ , and the degeneracy $P(E^+)$.

Calculation of a unimolecular rate coefficient at any pressure P now requires that a value for $N^*(E_a + E^+ + E_0)$ should be obtained, and that a systematic method for enumerating the number of energy levels be evolved to evaluate $\Sigma[(E^+ - E_v^+)/RT]^{r/2} P(E_v^+)$. If t rotations of A^* are active as well as the vibrations, then any non-fixed energy u may have a non-fixed rotational energy x separated

from it, and $N^*(u + E_0)$ may be expressed as

$$N^*(u + E_0) = \int_0^{u + E_0} N_v^*(u - x + E_0) N_r^*(x) dx \quad (227)$$

where

$$N_r^*(x) = \frac{(x/RT)^{\frac{1}{2}t-1} P_R}{\Gamma(\frac{1}{2}t) RT} \quad (228)$$

(P_R is the partition function for the t active rotations of A^*). A classical treatment of the number of energy states per unit energy is much simpler than the exceedingly laborious quantum treatments, but generally gives a gross overestimate of $N_v^*(u - x + E_0)$. Marcus and Rice¹⁴⁰ have shown that the semi-classical expression

$$N_v^*(u - x + E_0) = (u - x + E_0)^{s-1} / \Gamma(s) \prod_{i=1}^s h\nu_i^* \quad (229)$$

for a system of s oscillators of frequencies ν_i^* , is a reasonable approximation. Substitution of (229) and (228) into (227) gives

$$N^*(u + E_0) = \frac{P_R(u + E_0)^{s + \frac{1}{2}t - 1}}{(RT)^{\frac{1}{2}t} \Gamma(s + \frac{1}{2}t) \prod_{i=1}^s h\nu_i^*} \quad (230)$$

It is suggested¹⁵⁴ that when some vibration frequencies of A^* are very high in relation to the others, a better approximation than (229) is obtained by separating the frequencies into two classes, and computing the degeneracies of the high frequency group by a quantum mechanical treatment.

A number of modifications have been made to these expressions for N^* . For example Rabinovitch and Diesen¹⁵⁵ propose a corrected form

$$N_{\text{corrected}}^*(u - x + E_0) = (u - x + aE_0)^{s-1} / \Gamma(s) \prod_{i=1}^s h\nu_i^* \quad (231)$$

where a is an empirical correction factor. Whitten and Rabinovitch¹⁵⁶ have obtained exact sums of harmonic vibrational energy level densities for 45 molecules at various energies by the use of a digital computer. The results have been used to characterise the correction factor a with regard to its magnitude, its energy dependence and its dependence on the nature of the molecular frequency pattern concerned. The use of the approximation has the great advantage of being very rapid and of not requiring any calculational detail; the actual molecular fre-

quencies are employed. An extension of the method has been given to obtain an approximation for rotation-vibration energy level sums¹⁵⁷. It is shown that the approximation used by Marcus¹⁴¹, equivalent to equation (227), for the separation of vibrational and rotational sums leads to an error of less than 5 % for molecules having $s < 40$ and that the error decreases with increasing energy.

Wieder and Marcus¹⁵⁴ have reduced the quantum form of the theory to its classical limit, and show that the equations (224) and (225) reduce to the classical Kassel form (see section 5.4). Two important particular cases may be quoted: the ratio of rate coefficients on classical and quantum theories at the high pressure limit and as $p \rightarrow 0$. At the high pressure limit

$$\frac{k_{\infty}^c}{k_{\infty}^q} = \left(\frac{Q_2^+}{Q_2} \right) \bigg/ \left(\frac{Q_2^+}{Q_2} \right)^q \quad (232)$$

(The superscripts c and q refer to classical and quantum quantities, and Q_2^+ , Q_2 correspond to P_2^+ , P_2 , but are measured from the potential energy minimum rather than ground vibrational state). Using an approximation for the low pressure rate coefficients, it is shown that the ratio of the leading terms in expansions of k_0^c and k_0^q is given by

$$\frac{k_0^c}{k_0^q} = \frac{Q_2^q}{Q_2^c} \cdot \frac{e^{-E_0^+/RT}}{(1 + E_0^+/U)^{s+\frac{1}{2}-1}} \quad (233)$$

where E_0^+ is the zero point energy of A^+ , and U is the difference in potential energy of the *least unstable* configuration of A^+ and the *most stable* configuration of A .

Although the value of k_0^q may be calculated without a knowledge of the quantitative properties of A^+ , the more general test of the RRKM theory—a plot of k'/k_{∞} against pressure—does require this knowledge. The properties of A^+ are in general deduced from a knowledge of the high-pressure frequency factor, guided by certain considerations about the type of activated complex. Each reaction should be treated individually, and the reader is referred to the several papers which present RRKM treatments for specific chemical processes. The predicted RRKM rate coefficients for some reactions are quoted (but not derived) in the comparison of experimental results with the various theories presented in section 5.7.

5.6 SLATER'S THEORY

Slater^{139, 158, 159} has suggested a theory of unimolecular reactions whose basic assumptions are entirely different from those of the RRK theory. The postulates of the theory and some of the main conclusions are given in this section. (The "Slater Theory" discussed here should not be confused with "Slater's New

Approach" which will be touched upon briefly in section 5.8).

Collisional energisation of a molecule is assumed to result in a statistical distribution of the energy between the degrees of freedom: *energy does not flow between the degrees of freedom during subsequent vibrations of the molecule*. A molecule undergoes reaction when a certain critical co-ordinate undergoes a sufficient extension: that is to say, reaction occurs if and when the phase relationships of the various vibrations allow the critical bond to be broken. Let the critical co-ordinate be given the symbol q_1 , and the normal co-ordinates be Q_i , then if α_{1i} is the transformation matrix of q_1 to Q_i

$$q_1 = \sum_i \alpha_{1i} Q_i \quad (234)$$

We may write for a system in which n' normal vibrations take part in activation

$$q_1 = \sum_{i=1}^{n'} \alpha_{1i} \varepsilon_i^{\frac{1}{2}} \cos 2\pi(\nu_i t + \psi_i) \quad (235)$$

where ε_i and ν_i are the energy and frequency of the i th vibration, and ψ_i is a phase angle. The energies and phases can change only on collision, and the time t is calculated from the moment of the last collision. It follows that, on Slater's theory, not only must the total energy in the molecule be greater than a critical amount E_c , but the critical co-ordinate must reach a value q_c corresponding to E_c , so that

$$|\alpha_{1i}| \varepsilon_i^{\frac{1}{2}} \geq q_c \quad (236)$$

The probability of unimolecular conversion therefore depends on the frequency of molecular collisions and the properties of the function (235). Slater's theory shows¹³⁹ that at the high pressure limit

$$k_\infty = \bar{\nu} e^{-E_c/RT} \quad (237)$$

where $\bar{\nu}$ is a weighted r.m.s. frequency of normal vibrations of the molecule given by

$$\bar{\nu} = \left[\frac{\sum \alpha_{1i}^2 \nu_i^2}{\sum \alpha_{1i}^2} \right]^{\frac{1}{2}} \quad (238)$$

Equation (237) is very similar to equation (212) of the Kassel theory with the frequency factor A replaced by the weighted mean frequency $\bar{\nu}$. Indeed, Slater points out that his theory gives a value for $k_3(E)$, the rate coefficient of uni-

molecular conversion of an energised species, energy E , such that

$$k_3(E) = \bar{\nu} \left(\frac{E - E_c}{E} \right)^{n'-1} \quad (239)$$

The equation is formally identical to Kassel's [equation (209)] with A set equal to $\bar{\nu}$. However, A is an arbitrary constant which represents the frequency of interchange of energy between oscillators, a process not allowed in the Slater theory. Thus the two relations are different in concept, although the mathematical similarity ensures the same forms (237) and (212) for the limiting rate at high pressures. Slater derives (239) from an equilibrium distribution of energies, and the value of $k_3(E)$ may not be applicable at low pressures. Slater has¹⁶⁰, however, used the theory to obtain a value for the general rate. The approximate formula derived is

$$k' = \frac{\bar{\nu} e^{-b}}{m!} \int_0^\infty \frac{x^m e^{-x} dx}{1 + x^m \theta^{-1}} \quad (240)$$

where

$$b = \varepsilon_c/kT$$

$$x = \varepsilon_i/kT$$

$$m = \frac{1}{2}(n' - 1)$$

$$\theta = m! \omega / \bar{\nu} (4\pi b)^m \prod_{i=1}^{n'} \mu_i$$

$$\mu_i = \alpha_i / (\sum_i \alpha_i^2)^{\frac{1}{2}}$$

$$\omega = k_c p$$

At high pressures, $x^m \theta^{-1} \ll 1$ and $k_\infty \rightarrow \bar{\nu} e^{-b}$ which is identical with the exact result (237). At low pressures

$$k' = k_c p (4\pi b)^m \prod_{i=1}^{n'} \mu_i \quad (241)$$

approximately, which indicates that the overall reaction is kinetically of second order. One of the consequences of the form of (241) is that the activation energy at low pressures is given by

$$E_a = RT^2 \frac{d \ln k'}{dT} = E_c - \frac{1}{2}(n' - 1)RT \quad (242)$$

Comparison of (242) with the expression (216) for E_a on the Kassel theory shows that the two activation energies are identical only if

$$n' = 2s - 1 \quad (243)$$

Slater has suggested that the non-correspondence of n' and s results from the different criteria of a critical energy and critical vibration amplitude in the Kassel and Slater theories.

The variation of the ratio of general to limiting high pressure rate coefficients with pressure is frequently used as a test of theories of unimolecular reaction. From (237) and (240) it is seen that

$$I_m(\theta) = (k'/k_\infty)_n = \frac{1}{m!} \int_0^\infty \frac{x^m e^{-x} dx}{1 + x^m \theta^{-1}} \quad (244)$$

A curious result of the Slater theory is obtained for a diatomic species which has $n = 1$ and thus $m = 0$. The *exact* form of k'/k_∞ from (244) is given by

$$\frac{k_\infty}{k'} = 1 + \frac{\bar{\nu}}{\omega} \quad (245)$$

Since, in gases, $\bar{\nu} \gg \omega$, k' cannot become a first order rate coefficient, and a diatomic molecule (represented by simple oscillators) will not undergo unimolecular decomposition.

Slater¹⁶⁰ has considered the possibility that although the assumption of no energy flow might be valid at moderately low pressures, at *very* low pressures the RRR predictions might nevertheless be valid. This situation would arise if there were a *slow* flow of energy between normal modes in the reactant, which could only make a significant contribution to energization of the critical bond if the time between collisions were sufficiently long.

5.7 EXPERIMENTAL TESTS OF THEORIES OF UNIMOLECULAR REACTION

In principle, the validity of the main theories of unimolecular reaction can be tested by means of "experimental" (*i.e.* computer) studies of trajectories for reaction. Such studies have been made, and will be mentioned very briefly in section 5.8. The tests of the theories with which we are concerned here involve laboratory chemical experiments.

There are, of course, several levels at which the tests might be made. Predicted and experimental high pressure first-order rate coefficients may be compared in the first instance. These comparisons may be extended to a low pressure case, and

finally the entire shape of the "fall-off" curve (usually plotted as $\log(k'/k_\infty)$ versus pressure) may be compared with experiment. Fig. 11 is, in effect, a fall-off test of the Lindemann-Hinshelwood theory, and the shortcomings of the theory are well revealed. In point of fact, the experimental data is taken from work of Schneider and Rabinovitch^{142, 161, 162} on the isomerization of isocyanides carried out to test the RRKM theory. The reactions which have been studied are



Two tests are implicit in this work, and they will be examined in reverse order. Curve fitting of the fall-off gave the best fit for the Slater theory with $n = 5.5$, 6.1 and 5.6 for reactions (246), (247) and (248) respectively. The Slater theory predicts that there should be no change of n on perdeuteration of CH_3NC since the original symmetry is retained. On the other hand, monodeuteration removes the degeneracy present in some vibrational modes, and it is shown¹⁶² that n should increase by *at least* one for CH_2DNC . The experimental results suggest that n is virtually constant, and argue against the validity of the Slater theory. Tests of the RRKM theory may now be made. First, the variation of k'/k_∞ with pressure may be examined for each isocyanide. Fig. 12 shows the results obtained by

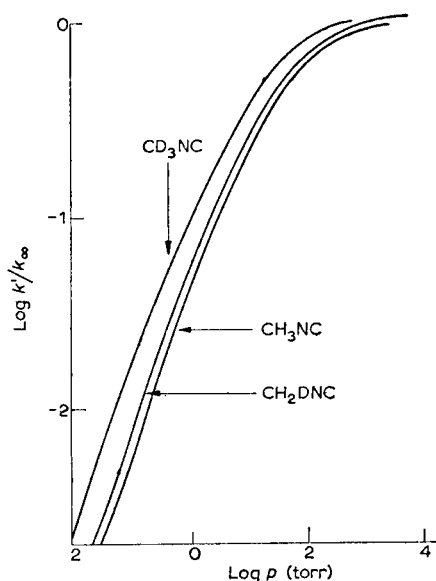


Fig. 12. Logarithmic "fall-off" plots for the isomerization of methyl isocyanides (from Schneider and Rabinovitch¹⁶²).

Schneider and Rabinovitch¹⁶² plotted on a logarithmic scale against pressure. The fall-off curves shift to progressively lower pressures with increasing deuteration, as expected on the basis of the RRKM theory. A more exact test could be made of the rates of isomerization of either the mono- or the perdeuterated compound *relative* to the rate of isomerization of CH_3NC .

At high pressures, the isotopic rate ratio, called $k_{\text{H}}/k_{\text{D}}$, was found to be slightly greater than unity, in agreement with the predicted rates. A non-equilibrium inverse isotope effect is predicted by the RRKM theory for low pressures, and the experimental value of $k_{\text{H}}/k_{\text{D}}$ decreases at low pressures in the manner expected. The inversion of the isotope effect is a result of opposing statistical isotope effects which become more important than the primary isotope effect at low pressures¹⁶³.

Another reaction system in which the RRK and Slater theories have been differentiated by means of behaviour on isotopic substitution is the thermal isomerization of cyclopropanes¹⁶⁴. The value of n determined from fall-off experiments in cyclopropane and cyclopropane- d_2 is 13–14 in both cases. In Slater's theory, the highest value n can take for light cyclopropane (which has seven doubly degenerate vibrational modes) is 14; for cyclopropane- d_2 there are no degenerate vibrations, and n could reach 20 or 21. The actual invariance of n as determined from the fall-off is evidence against the postulate of Slater's theory.

An important series of fall-off tests of the RRKM theory has been made by Wieder and Marcus¹⁵⁴ for unimolecular reactions of cyclopropane, cyclobutane, methyl cyclopropane, but-2-ene, ethyl chloride, nitrous oxide and nitrogen pentoxide. The tests were made with energy densities calculated by the approximate formula (229), and by the grouping procedure involving quantum mechanical treatment of the high frequency groups. Fig. 13 shows the results obtained for the isomerization of methylcyclopropane to butenes (several butenes are produced, but the frequency factors and activation energies are similar in the various cases). The

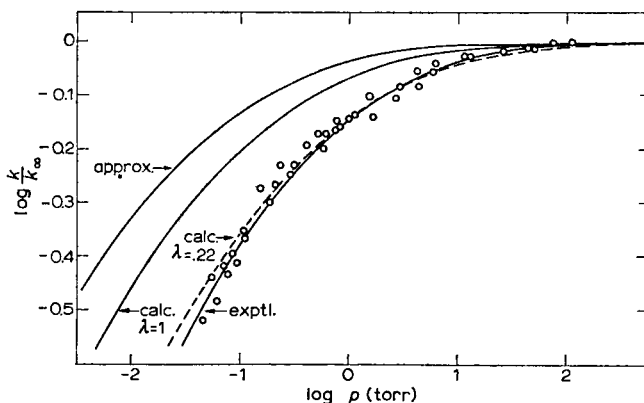


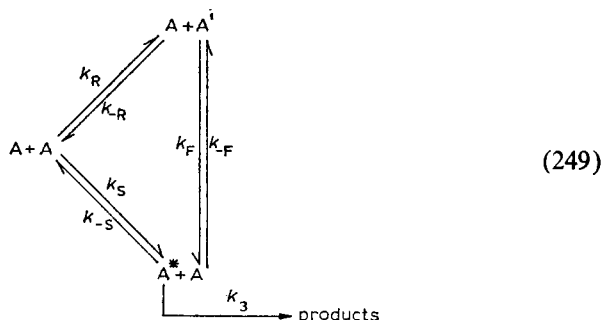
Fig. 13. Fall-off test of the thermal isomerization of methylcyclopropane (Wieder and Marcus¹⁵⁴).

figure shows curves calculated for the two counting procedures as "approx." and "calc.". A better fit to the experimental results is obtained if the collisional deactivation efficiency, λ , is set equal to 0.22 rather than unity.

Application of the Slater theory to the prediction of unimolecular rates necessitates a complete vibrational analysis of the reacting molecule. Slater himself undertook such an analysis for cyclopropane¹⁶⁵, in the first complete test of his theory. It is, of course, important to decide which co-ordinate is the critical one, and the decision rests on choosing a correct model for the mechanism of reaction. In the main part of Slater's test, it was assumed that isomerization occurred when the vibrations carried any hydrogen atom too near to a carbon of another methylene group. On this model, the frequency factor found for the high pressure rate coefficient was $4 \times 10^{14} \text{ sec}^{-1}$ compared with an early experimental value of Chambers and Kistiakowsky¹⁶⁶ of about $15 \times 10^{14} \text{ sec}^{-1}$. Adoption of a second model in which isomerization followed the over-stretching of a carbon-carbon bond yielded a high pressure frequency factor of only $6 \times 10^{13} \text{ sec}^{-1}$. The shape of the predicted $\log(k'/k_\infty)$ versus $\log p$ curve for the first reaction model was compared with the experimental curve obtained by Pritchard *et al.*¹⁶⁷. Although the theoretical curve was of the correct shape, it was displaced by a factor of $3\frac{1}{2}$ times on the pressure scale from the experimental curve. However, the absolute relation between θ and pressure was calculated from the activation energy of Chambers and Kistiakowsky, and assuming unity collisional deactivation efficiency. Error in either of these factors could lead to appreciable error in the absolute position of the theoretical curve.

Gill and Laidler¹⁶⁸ have examined the experimental data for the low-pressure second-order rate coefficients of a number of unimolecular processes in the light of the RRK and Slater theories. This work was complementary to a complete vibrational analysis of the hydrogen peroxide¹⁶⁹ and ozone¹⁷⁰ molecules. For hydrogen peroxide, the choice of the O-O distance as the critical co-ordinate leads to a rate of energization on the Slater theory which is about 10^{-5} of the experimental value, and a maximum possible rate of reaction which is 10^{-2} of the experimental rate. On the other hand, for ozone the Slater rates of energization appear to be of the correct order of magnitude, while the RRK rates are too high unless fewer degrees of freedom are employed than are actually in the molecule. The Slater theory was found to be consistent in a similar way with the experimental results for the reactions of nitrogen pentoxide, cyclopropane and ethyl chloride, while the RRK theory predicted the rates more correctly for ethane and nitrous oxide decomposition as well as for the hydrogen peroxide reaction. The results were explained on the hypothesis that the Slater theory is correct as far as the breakdown of A^* is concerned, but not always correct with regard to the rate of formation of A^* . If flow of energy between normal modes can take place, a molecule A' energised in the RRK sense, but not having the distribution of energy needed to be energised in the Slater sense, may become an A^* . Gill and Laidler¹⁶⁸

propose an extension of the Lindemann-Hinshelwood scheme which allows of this process



k_R, k_{-R} are RRK activation, deactivation rate coefficients, k_S, k_{-S} the Slater rate coefficients, and k_F, k_{-F} the rate coefficients for flow of energy between A' and A^* . A steady state treatment shows that if k_F, k_{-F} are small, the ordinary Slater situation obtains. Further, if the pressure is high

$$-\frac{d[A]}{dt} = \frac{k_3 k_S [A]^2}{k_3 + k_{-S} [A]} \quad (250)$$

under all conditions, even if k_F is large. However, at low pressures

$$-\frac{d[A]}{dt} = (k_S + k_R) [A]^2 \quad (251)$$

and since, probably, $k_R \gg k_S$, this expression reduces to

$$-\frac{d[A]}{dt} = k_R [A]^2 \quad (252)$$

Thus at low pressures the rate should be given by the RRK formula for the rate of energization. The experimental results suggest, on this modified theory, that k_F, k_{-F} are more rapid for N_2O , H_2O_2 and C_2H_6 than for O_3 , N_2O_5 , C_3H_6 and C_2H_5Cl . Gill and Laidler advance reasons for believing that in small molecules the flow of energy will tend to be more rapid than in large ones.

Contribution to the activation and deactivation processes can be made by an added gas which is otherwise inert chemically with respect to the reaction system. Experimental results show qualitatively that addition of an unreactive gas can restore first order kinetic behaviour to a system which was previously at a low enough pressure to display second order behaviour. Quantitative data obtained from experiments of this kind tends, however, to give information about the

relative energy transfer efficiencies for the added gas and reactant rather than provide tests of anything more than the basic Lindemann activation-deactivation hypothesis. There do exist, of course, a number of techniques whereby the absolute efficiency for vibrational energy transfer of a gas may be determined, and a discussion of this topic appears elsewhere in *Comprehensive Chemical Kinetics* (see last Chapter, Vol. 3). In instances where these efficiencies are known unequivocally, a more direct test of unimolecular reaction theory is possible. The main "use" of added gases appears, however, to be in allowing rate determinations of unimolecular processes to be made at high pressures. For example, Schneider and Rabinovitch¹⁶² conserved CH_2DNC by using a high pressure of chemically inert gas to simulate the high pressure conditions.

Lyon¹⁷¹ has recently suggested a simple test of unimolecular reaction theory. If a molecule undergoes sequential unimolecular isomerization at high pressures, then the isomerization will become direct at sufficiently low pressures. A typical reaction scheme might be



Since the lifetime of B^* is inversely proportional to pressure in (255) and pressure independent in (256), the overall reaction selectivity will shift with decreasing pressure from $\text{A} \rightarrow \text{B}$ to $\text{A} \rightarrow \text{C}$. This shift may be calculated from RRK theory without any empirical information or assumptions concerning the molecules or transition states if A, B and C are virtually identical molecules. As an example of such a system, Lyon has studied the isomerization of 1-proto-perdeuterocyclopentadiene as a function of pressure. The observed low-pressure selectivity shift is in good agreement with the prediction.

Tests of unimolecular reaction theory based on fall-off curves suffer from the disadvantage that they test simultaneously a whole series of assumptions. For example, the curves of Fig. 13 are a test of the RRKM theory, but at the same time they test the state counting procedures, and, since λ is introduced as a variable, the efficiency of collisional processes. One solution to the problem might be to start with a molecule in an initially well-defined activated state. Ideally the species would be monoenergetic and the energy would be distributed in a known fashion about the various degrees of freedom, so that the rate coefficient k_3^I could be predicted on any theory of unimolecular reaction and compared with experiment. A technique which produces activated molecules which come some of the way to meeting these requirements is that of "chemical activation".

Chemical activation studies have yielded some valuable information about

unimolecular reaction theory, and a few examples of the many applications of the technique are described here. A review of chemical activation processes has been given by Rabinovitch and Flowers¹⁷² and part of the discussion is devoted to the relation of chemical activation studies to reaction theory.

Rabinovitch *et al.*¹⁷³ reported one of the first chemical activation studies of a unimolecular reaction. Methylene (CH_2) was produced by the photolysis of ketene at two wavelengths, and was added to *trans*-ethylene- d_2 . The addition was stereospecifically *cis*-, and the "hot" cyclopropane molecules suffered geometric and structural isomerizations at rates depending on the wavelength of photolysis of ketene. According to RRK or Slater theories

$$k_3^i = A(1 - E_c/E)^{r-1} \quad (257)$$

where r is s or n in the Kassel or Slater models [*cf.* equations (209) and (239)]. Agreement between (257) and the observed experimental rate coefficient could be obtained only if quite unrealistic values were adopted for the energy, E , of the hot cyclopropane. Rabinovitch *et al.* write a form of the Marcus expression, using the "semi-classical" counting procedure, which allows direct comparison with (257)

$$k_3^i = \text{constant} \left(\frac{E - E_c + E_0^+}{E + E_0^*} \right)^{r-1} \quad (258)$$

where E_0^+ , E_0^* are vibrational zero point energies of A^+ , A^* as before. The exponent r is now the total number of vibrational modes which interact with the bond breaking co-ordinate, degenerate modes being counted fully. The values of k_3^i predicted from (258) were found to be satisfactory with the constant set equal to the observed high pressure frequency factor, and with reasonable values for E , E_c and the zero-point energies. Chemical activation systems such as this one may be the most suitable cases in which to apply the Marcus semi-classical approximation, since the requirement that $E \gg E_c$ may be met in these instances.

An experimental distinction between free-flow and Slater mechanisms was made by Butler and Kistiakowsky¹⁷⁴. Methylene was produced by the photolysis of either ketene or diazomethane at a number of wavelengths, and allowed to react with either cyclopropane or propylene



The hot methylcyclopropane produced in (259) or (260) rearranges by a unimolecular process into a mixture of butenes unless it is stabilised by collision. The

lifetime of the methylcyclopropane depends greatly on whether it is produced from cyclopropane or propylene and on the source of methylene, while the *composition* of the butenes does not depend on any of these factors. This latter result is taken to mean that the energy of the "hot" molecules migrates freely among the normal modes of vibration of methylcyclopropane molecules, and is therefore evidence strongly in favour of a free-flow rather than Slater theory. A similar result was obtained by Frey¹⁷⁵ and by Setser and Rabinovitch¹⁷⁶ for the lifetimes of "hot" dimethylcyclopropanes formed by the addition of methylene from different sources to *cis*- and *trans*-but-2-ene. It was shown¹⁷⁶ that the activated molecules were formed in a distribution which was nearly monoenergetic.

Rabinovitch and his collaborators have subsequently studied the unimolecular conversions of a large number of chemically activated species, and, in general, the results give good agreement with RRKM theory for a suitable choice of critical configuration, and with all degrees of freedom active. The technique typically involves analysis of decomposition (D) and stabilisation (S) products of the activated species. An average rate coefficient for unimolecular reaction, k_a , is then defined by

$$k_a = \omega \frac{[D]}{[S]} \quad (261)$$

where ω is the collision frequency for deactivation ($k_c p$) with an assumed efficiency, λ , for deactivation of unity. Since the chemically activated species still possesses some energy spread, represented by a distribution function $f(E)$, (261) may be re-written

$$\frac{k_a}{\omega} = \frac{[D]}{[S]} = \frac{\int_{E_c}^{\infty} \frac{k_3(E)}{k_3(E) + \omega} f(E) dE}{\int_{E_c}^{\infty} \frac{\omega}{k_3(E) + \omega} f(E) dE} \quad (262)$$

The unimolecular decomposition of "hot" secondary butyl radicals provides a good example of the type of study described. References to the original papers are given in the review by Rabinovitch and Flowers¹⁷². Activated secondary butyl radicals may be produced with varying degrees of excitation by the addition of atomic hydrogen to (i) but-1-ene, (ii) *cis*-but-2-ene and (iii) *trans*-but-2-ene, *viz.*



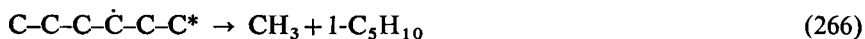
Furthermore, since the activation energy for (263) is low, (about 2 kcal. mole⁻¹), the reaction can be carried out at reduced temperatures (−78 °C in the studies reported) in order to reduce the average energy of the *sec*-butyl radicals as well as

their energy spread. The energy possessed by the radicals produced in (263) is only just that required for the reverse dissociation to butene and atomic hydrogen, and the reverse of (263) is thought to be unimportant. Thus the important reactions open to hot *sec*-butyl radicals are



and measurement of suitable reaction products gives the ratio [D]/[S]. A similar series of hot deuterio-*sec*-butyl radicals may also be formed by the addition of atomic deuterium to the three butenes. The known thermochemistry of the systems can now be used to calculate the average energy of the radicals produced at reduced or room temperatures. The ratio of the limiting high and low pressure forms of k_a , viz. $k_{a\infty}/k_{a0}$, provides a measure of the energy spread described by the distribution $f(E)$. For monoenergetic systems, the ratio is, of course, unity, and for the butyl radical system under consideration, the ratio was found to be less than 2 at room temperature and less than 1.3 at -78°C . The ratio is orders of magnitude smaller than the equivalent ratio for thermal activation. A comparison of the experimental values of k_a with RRKM theory for the twelve different average energies is given by Rabinovitch *et al.*¹⁷⁷ and the agreement is shown to be good. Another noteworthy feature of the unimolecular reaction of butyl radicals is that the hot radicals formed from but-1-ene and but-2-ene decompose at comparable rates. In the first case the butyl radical formed decomposes by rupture of the C-C bond furthest from the point at which H added, and in the second case the rupture is of a bond attached to the carbon to which H added. The apparent lack of importance of the initial location of excess energy suggests strongly that there is a free flow of energy.

Competitive unimolecular decomposition processes have also been studied with chemically activated systems¹⁷⁸. A general formulation of the rates was derived, together with various relationships between the rates and between the energetics of the competing processes. The experimental system employed was activated 3-hexyl radicals from the addition of atomic hydrogen to *trans*-hex-3-ene at 300°C . Two unimolecular decomposition reactions occur



and the relative and absolute rates of the two processes were determined. A quantum statistical model was used to predict the rate coefficients of (266) and (267) as a function of energy, for use in the general relations. Good agreement between the theoretical and experimental data was obtained over the pressure range 0.004 to 90 torr.

Hassler and Setser¹⁷⁹ have carried out a very extensive test of the RRKM theory for the HCl elimination reactions of hot C_2H_5Cl , $1,1-C_2H_4Cl_2$ and $1,2-C_2H_4Cl_2$. The experimental rates for both thermal and chemically activated systems were compared with rates predicted from RRKM calculations employing several models for the transition state. Chemically activated mono- and di-chloroethanes were formed by the various recombination reactions of methyl and chloromethyl radicals with about 90 kcal. mole⁻¹ of excess energy. The distribution function is so narrow that less than 10 % spread in k_a exists between high and low pressure values at 300 °K. Table 8 shows the experimental and calculated values of the Arrhenius A factors for thermal activation at 800 °K, and the k_a values for 300 °K in the chemically activated system. The model of the transition state adopted to give the calculated results is an asymmetric four centre complex with torsional motion in the molecule treated as a vibration.

TABLE 8

EXPERIMENTAL AND CALCULATED VALUES OF ARRHENIUS A FACTORS (THERMAL, 800°K) AND k_a (CHEMICAL ACTIVATION, 300 °K) IN HCl ELIMINATION REACTIONS¹⁷⁹

Halide	Average excess energy in chemically activated system (kcal. mole ⁻¹)	10 ⁻¹³ A		10 ⁻⁹ k_a	
		exptl. (sec ⁻¹)	calc.	exptl. (sec ⁻¹)	calc.
C_2H_5Cl	91.5	0.97	1.14	2.6	1.7
$1,1-C_2H_4Cl_2$	92.3	1.24	2.75	11	4.9
$1,2-C_2H_4Cl_2$	88.7	1.5	2.69	0.18	0.82

Sufficient examples of the use of chemically activated systems in tests of reaction theory have been given to show the power of the technique. It is apparent that chemical activation studies can also give valuable evidence about collisional deactivation. This topic lies just outside the scope of the present section, and the reader is therefore referred to the review by Rabinovitch and Flowers¹⁷² for a brief summary of this application of activation experiments.

5.8 CONCLUSION

The summary of unimolecular reaction theory which has been given has intentionally concentrated on those theories which have found favour in comparisons with experimental work. The discussion of section 5.7 suggests that, so far as the theories considered are concerned, free-flow theories are more consistent with the experimental results than are purely harmonic theories, and of these theories, the quantum-statistical model of Marcus appears to give the most reliable calculated

rates. However, a number of more or less formal unimolecular reaction theories have appeared over the last decade or so, and it seems appropriate to mention some of them.

Slater¹⁸⁰ first questioned the hypothesis of "random incidence of dissociation" [see Chapter 9, of Slater's book (Bibliography)]. The theories described hitherto have assumed that the dissociation of individual "interesting" molecules is random (although the *average* lifetime before dissociation is precise). In the "new approach to rate theory", Slater points out that if the precise classical state of an interesting molecule were known, then it would be possible to calculate the precise time, s say, it required to reach the dissociation configuration. At a concentration where the collision frequency is ω , the absolute chance of dissociation (*i.e.* of *no* collision) would be $e^{-\omega s}$, and it should be possible to calculate a reaction rate constant as the integral of this probability over the collisional rate of production of interesting molecules in the various relevant states. The mathematical equipment for handling the $3N$ co-ordinates and $3N$ momenta of a system of N particles requires the use of the *phase-space* concept, and much of the more recent work is expressed in terms of phase-space boundaries and gaps. Bunker (Bibliography, p. 51 *et seq.* of his book) gives a useful summary of the methods used. Slater uses the technique to re-derive the rate equations for the harmonic oscillator model, and finds that at high and low concentrations the results are identical with those derived previously assuming random behaviour. Thiele¹⁸¹ has also re-examined the Slater, Kassel and Giddings-Eyring theories in the light of Slater's "new approach" and made some suggestions about further development of a non-linear theory. A quantum model adopted by Wilson and Thiele¹⁸² has been elaborated by Thiele¹⁸³ to provide a formal quantum theory for unimolecular reactions.

A second assumption made in all the theories of activation and deactivation is the "strong-collision" assumption. There is some reason to suppose¹⁸⁴ that collisions may be inefficient in certain cases, so that the rate of activation controls the overall rate of the unimolecular process. Discussions of weak collisions have been given by Buff and Wilson¹⁸⁵ and Hoare¹⁸⁶. Using Landau-Teller probabilities to obtain rates of collisional energy transfer Buff and Wilson investigated the "weak collision" unimolecular rates and found that the fall-off region was more extended than for strong collisions.

Two further theories of unimolecular reaction have appeared recently. One is to be found in a series of papers by Levine¹⁸⁷, who discusses quantum mechanical dissociation rates in terms of Green's operator formalism. Valence and Schlag¹⁸⁸ have obtained theoretical unimolecular rate coefficients for many-level systems, and examined the case where any or all of the reactant levels have non-equilibrium effects.

Finally, brief mention must be made of computer kinematic or trajectory studies for unimolecular reactions. Several such studies have been made¹⁸⁹⁻¹⁹³:

all show that the normal-mode approximation used is a bad one, and that anharmonicity has a very important influence on the vibrational motion and on the coupling of rotation to it. There seems, therefore, to be no real reason for retaining Slater's strictly harmonic model. In general, the computer values for $k_3(E)$ are smaller than the RRKM ones; part of the difference can be ascribed to a larger value of the energy density of states, N^* , obtained in the trajectory studies. Although there is some evidence that non-random lifetimes may arise under some circumstances, the trajectory tests suggest that non-randomness will be unimportant for large molecules.

6. Termolecular and third-order processes

The last class of chemical process with which we shall deal is that of third-order reactions. Termolecular reactions are relatively rare, and the best authenticated examples are the reactions of nitric oxide with hydrogen, deuterium, oxygen, bromine and chlorine, and the recombination of atoms or small radicals. Examples of the two groups of reactions are



In view of the great importance of the recombination reactions in a variety of gas phase processes, it is necessary to examine briefly elementary reactions which are kinetically third order.

Bimolecular recombination of two atoms or radicals naturally produces a species which initially possesses the entire energy of the newly formed bond. In the sense described in section 5.7, this species is therefore "chemically activated", and can undergo spontaneous unimolecular dissociation unless it is stabilised. So far as the formation of diatomic molecules in processes such as (269) is concerned, this energy must remain in the only bond present, and dissociation is likely to occur in about one vibration, unless a collision with a "third-body" takes place during this vibrational lifetime. Stabilization of a vibrationally excited molecule by a vibrational radiative process is extremely unlikely (and impossible if the diatomic molecule is homonuclear). Even if the molecule first formed is *electronically* as well as vibrationally excited, electronic emission processes will not usually help to stabilise the molecule, since at the very best the electronic radiative lifetime will not be much shorter than 10^{-8} sec; as against a vibration period of about 10^{-13} sec. Collisional stabilization is therefore the most probable way in which the newly formed molecule can be stabilized. If, however, the species formed in a recombination reaction possesses a large number of bonds, then a different situation arises. The discussion of section 5 suggests that vibrational energy can flow freely

between the various vibrational degrees of freedom in a polyatomic molecule, so that the energy concentrated initially in the newly formed bond can begin to distribute itself about several bonds during the course of a vibrational period. It is true that the molecule is still able to undergo unimolecular dissociation, but since the energy available is only just equal to the critical energy for dissociation and is distributed about many degrees of freedom, the lifetime of the molecule will be considerable. In this case, the molecule will suffer many deactivating collisions, and the rate controlling process will be the rate of recombination rather than recombination plus deactivation, and the kinetics of recombination will be second order. At sufficiently low pressures, of course, such a process will ultimately become third order if *wall* deactivation steps are not admitted.

For recombination reactions of species larger than, say, ethyl radicals, the process seems to be adequately described for all experimental conditions as bimolecular, while the recombination of atoms, as in (269) is certainly termolecular. The recombination of methyl radicals affords an example of intermediate behaviour, and the molecularity of the process has often been discussed. The number of bonds available is not sufficient to ensure a dissociation lifetime greater than the collisional lifetime at all pressures. Kistiakowsky and Roberts¹⁹⁴ give the effective rate coefficient for the recombination of methyl radicals in the presence of acetone at 180 °C



as

$$k_{\text{eff}} = k''/(1 + 4.6/p) \quad (270a)$$

where k'' is a second order rate coefficient, and p the pressure of acetone in torr. At pressures above a few cm Hg, therefore, $k_{\text{eff}} \approx k''$, although at low pressures $k_{\text{eff}} \approx k''p/4.6$ and the process is third order.

One requirement of any third body in a recombination process is that it shall be able to remove vibrational energy efficiently on collision. The details of vibrational energy transfer are not within the scope of this chapter (see last Chapter, Volume 3), but two general points should be made. Both vibration-vibration and vibration-translation energy exchange processes can occur, and the exchange appears to become more efficient if the energy difference between exchanging states is small. Secondly, the probability that quenching will occur on collision by a vibration-vibration exchange process is increased by an increase in the number of vibrational degrees of freedom in the quenching molecule. The validity of these considerations may be seen qualitatively from Table 9 which shows the third order recombination rate coefficients for the reaction



TABLE 9

THIRD ORDER RATE COEFFICIENTS FOR THE GAS-PHASE RECOMBINATION OF O WITH O₂
(from Kaufman and Kelso¹⁹⁵)

<i>Third body</i>	<i>10³⁴ × rate coefficient (cm⁶. molec⁻². sec⁻¹)</i>
He	4
Ar	4
N ₂	5.6
O ₂	6.5
CO ₂	15
N ₂ O	15
CF ₄	16
SF ₆	34
H ₂ O	60

with a number of different third bodies¹⁹⁵. The increasing rate coefficients with increasing atomicity of M reflect the greater efficiency of the polyatomic molecules in vibrational quenching processes.

The problem now arises as to what exactly constitutes a termolecular process. The examples given so far appear to be complex reactions which display third-order kinetics. For instance, the recombination of atomic hydrogen might be written formally



It seems fairly well established that the recombination of iodine atoms in the presence of certain third bodies (*e.g.* aromatic hydrocarbons¹⁹⁶) involves the formation of a complex



and several complex schemes have been put forward for the nitric oxide reactions. One such scheme involves "sticky" collisions which may be represented for the NO + Cl₂ reaction as



In every case, the third order behaviour depends on competition between the reverse (dissociation) of the first reaction of the pair and the second reaction.

The concept of a termolecular reaction is itself untenable on a hard-sphere collision theory, since the chance of simultaneous contact of the surfaces of three spherical molecules is zero. Several authors have discussed three-body collision rates. For example, Tolman¹⁹⁷ assumed that the approach of rigid spheres to within an arbitrary distance of each other constituted a collision, and introduced a rather ill-defined distance parameter, δ . The collision rate calculated, Z_{ABC} , for three-body collision was given by

$$Z_{ABC}^2 = n_A n_B n_C 4\pi \sigma_{AB}^2 \sigma_{BC}^2 \delta \sqrt{8\pi kT} \left(\frac{1}{\sqrt{\mu_{AB}}} + \frac{1}{\sqrt{\mu_{BC}}} \right) \quad (278)$$

where the symbols have the meanings associated with equation (9). An alternative approach was adopted by Steiner¹⁹⁸, who assumed that unstable binary complexes of lifetime τ were formed, and these then collided with the third species. The result for Z_{ABC} is

$$Z_{ABC} = 54\sqrt{3} \pi \frac{kT}{m} d^4 \tau \cdot n_A n_B n_C \quad (279)$$

where d and m are mean molecular diameters and masses for A, B and C. The lifetime τ was estimated from the time spent by one particle close to another when they are moving freely (*i.e.* d/v where v is the mean relative velocity of the two particles). Smith^{199, 200} has developed new equations for the rates of three-body collisions, using a symmetric representation. The collisions are assumed to occur with attractive potentials with a sharp cut-off, and the potential is represented in a 6-dimensional normalised centre of mass system. Simplification of the basic equations leads to the result

$$Z_{ABC} = n_A n_B n_C \cdot \left(\frac{15\pi}{8} \right) \left(\frac{kT}{2\pi\mu} \right)^{\frac{1}{2}} \sigma_I \quad (280)$$

where

$$\mu = \left[\frac{m_A m_B m_C}{(m_A + m_B + m_C)} \right]^{\frac{1}{2}}$$

and σ_I is the 3-body reaction cross-section. In principle, σ_I is temperature dependent, although in this work a less realistic temperature independent cross-section was employed. Smith²⁰⁰ gives methods for the calculation of σ , and quotes results for collision rate coefficients ($Z_{ABC}/n_A n_B n_C$) for the recombination of iodine atoms in the presence of various third bodies. For M = Ne, Ar or Xe, the frequency factor is about $1.3 \times 10^{-30} \text{ cm}^6 \cdot \text{molec}^{-2} \cdot \text{sec}^{-1}$ at 293 °K. Using lighter masses of around 30–40 units, and molecular diameters of 3 Å, formulae (278) and (279)

give frequency factors of 10^{-32} to 10^{-31} $\text{cm}^6 \cdot \text{molec}^{-2} \cdot \text{sec}^{-1}$ at 300 °K with $\delta \sim 1$ Å or $\tau \sim 10^{-12}$ sec. Comparison of these values with the bimolecular collision frequency derived under the same conditions (section 3.4) of about 10^{-10} $\text{cm}^3 \cdot \text{molec}^{-1} \cdot \text{sec}^{-1}$ indicates that the ratio of termolecular to bimolecular frequency factors is about $10^{-22} \times (\text{concentration in molec. cm}^{-3})$. Thus up to about 3×10^4 torr, the termolecular frequency factor will always be less than the bimolecular factor. At ordinary pressures it seems reasonable to suppose that only those termolecular processes with a low activation energy can be of importance in relation to possible bimolecular mechanisms. Smith²⁰⁰ used his calculated collision rates to obtain an overall efficiency of the iodine atom recombination at 293 °K by comparison with the experimental results of Christie *et al.*²⁰¹. The efficiency was found to increase with increasing mass of the inert gas, being 1/720 for helium as third body and 1/44 for xenon.

Porter²⁰² has discussed a series of mechanisms for third-order recombination reactions. These are, formally, processes of the types (272–273) and (274–275) together with



The three mechanisms all explain two notable features of third-order atomic recombination processes: (1) the temperature coefficients are negative, and (2) the efficiency of different third bodies varies over a factor of at least 10^3 . The recombination of iodine has been studied in great detail, and the temperature dependence has been obtained by the variable temperature flash photolysis experiments of Porter and Smith¹⁹⁶ for a number of third bodies. Porter²⁰² shows that, at any rate for the iodine recombination, the energy transfer theory is less successful in predicting the absolute rates and dependence on the nature of M than are the radical-molecule complex mechanisms. One particularly interesting aspect of the experimental work on iodine recombination is the variation of the temperature dependence of rate with M. Porter and Smith¹⁹⁶ plot their results in the form

$$k_R = A_2 T e^{-E_2/RT} \quad (284)$$

as well as in the ordinary Arrhenius form. Table 10 shows the values obtained for k_R at 20 °C, and A_2 and E_2 . The correlation that is immediately apparent is the increase in $-E_2$ with increasing efficiency of the third body (as reflected in larger values of k_R). The very high efficiencies of aromatic hydrocarbons have been ascribed to the formation of charge-transfer IM complexes, and in the case of

iodine itself, it is believed that the species I_3 is formed, *viz.*



The formation of a complex in processes such as (274) or in (281) with $R = I$, can be defined in terms of an equilibrium constant, K , and the overall rate of recombination is given by

$$-\frac{d[I]}{dt} = PZ_2K \quad (287)$$

where P and Z_2 are the probability factor and collision rate for the last process of the sequence (275) or (283). Porter and Smith¹⁹⁶ obtained the constant K by a statistical thermodynamic calculation, using a binding energy for the complex estimated from the observed temperature coefficient of the overall reaction, and hence calculated rate coefficients for the various third bodies. The last column of Table 10 shows these rate coefficients calculated on the basis of the rate equation (284), and it is seen that at least the relative values are satisfyingly close to the experimental ones. A rather similar treatment has been given by Bunker and Davidson²⁰³, except that they were able to calculate the temperature dependence of k_R since they estimated the binding energy of IM from independent sources. They found good agreement with the experimental results, even to the extent that the calculations predicted an increased temperature dependence with increased efficiency of M. Kim²⁰⁴ has extended the calculations using the radical-molecule complex mechanism, and obtained an expression for the rate coefficient for the iodine recombination reaction as a function of temperature. The equilibrium

TABLE 10
RATE COEFFICIENTS AND MODIFIED ARRHENIUS FACTORS FOR THE REACTION
 $I + I + M \rightarrow I_2 + M$ (from Porter and Smith¹⁹⁶)

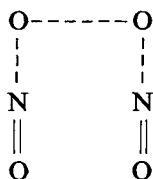
Third body	$10^{82}k_R$ (20° C) ($cm^6 \cdot molec^{-2} \cdot sec^{-1}$)	$10^{86}A_2$ ($cm^6 \cdot molec^{-2} \cdot sec^{-1}$)	E_2 ($kcal \cdot mole^{-1}$)	$10^{82}k_{calc}$ ($cm^6 \cdot molec^{-2} \cdot sec^{-1}$)
He	0.84	2.6	-1.4	3.0
Ar	1.64	1.8	-2.0	8.1
O ₂	3.72	2.9	-2.2	11
CO ₂	7.41	4.1	-2.4	17
benzene	43.9	24	-2.4	12
toluene	107	11	-3.4	86
ethyl iodide	144	24	-3.1	54
mesitylene	223	2.0	-4.8	1500
I ₂	760	2.9	-5.3	2100

constant K is calculated by a method using the Lennard-Jones (6-12) potential, and the rate coefficient for the process equivalent to (275) was estimated from a collision rate assuming a Sutherland potential between I and IM. The temperature dependence of the overall termolecular rate coefficient is expressed in terms of a transcendental function of the temperature, although it is admitted that the theoretical expression is approximated equally well by the Arrhenius equation or by an inverse power dependence of the temperature over the temperature range of most experiments. The actual calculated rate coefficients are in good agreement for argon as third body with all experimental data tested. The best fit was obtained with a Lennard-Jones potential whose depth was $700 k$ (k is here the Boltzmann constant).

The preceding discussion of third order processes has been given in terms of the collision theory of reaction. It is also possible to use the transition state theory to predict the rates of termolecular processes. Gershinowitz and Eyring²⁰⁵ applied the transition state formulation to the termolecular reactions of nitric oxide. For example, they investigated the process



and assumed a form



for the activated complex. From an analysis of the experimental results, it was shown that the energy, E_c , in the exponential of the transition state expression was small, and that most of the observed *negative* temperature dependence arose from a high negative order temperature term in the pre-exponential part of the expression. The calculated and observed rate coefficients were within a factor of about two over the temperature range 80 °K to 662 °K. Eyring *et al.*⁷⁷ subsequently gave a general formulation of the transition state theory for termolecular reactions of the type



It seems more appropriate at the present time to suggest orders of magnitude for rate coefficients predicted by the transition state theory for certain types of termolecular reaction, rather than to present detailed analyses of specific systems. The approach to be adopted is identical to that used in section 4.3.4, and the orders of magnitude used for f^{trans} , f^{vib} and f^{rot} will be the lower of those in Table 4. For a reaction of three atoms, as in (289) with a linear transition state, the rate

coefficient will be given by

$$k = \frac{kT}{h} \cdot \frac{(f'^{\text{trans}})(f'^{\text{rot}})^2(f'^{\text{vib}})^3}{(f'^{\text{trans}})^3} e^{-E_c/RT} \approx 10^{-33} e^{-E_c/RT} \text{ cm}^6 \cdot \text{molec}^{-2} \cdot \text{sec}^{-1} \quad (290)$$

In a more complex system, say the recombination reaction of two atoms with a diatomic molecule as third-body, proceeding *via* a linear complex, *viz.*



the rate coefficient will be

$$k = \frac{kT}{h} \frac{(f'^{\text{trans}})(f'^{\text{rot}})^2(f'^{\text{vib}})^6}{(f'^{\text{trans}})^3(f'^{\text{rot}})^2(f'^{\text{vib}})} e^{-E_c/RT} \approx 10^{-35} e^{-E_c/RT} \quad (292)$$

Thus it is seen that the pre-exponential factors are roughly of the same order of magnitude as those observed, although the estimate is crude. Perhaps of greater importance is the form of the cancelled expression $kT/h(f'^{\text{vib}})^5/(f'^{\text{trans}})^2$, since if vibrations are not excited the temperature dependence of the pre-exponential factor will be affected by the temperature dependence of $(f'^{\text{trans}})^2$ alone (*i.e.* the dependence will be T^{-1}). For more complex systems, even higher negative temperature exponents can be obtained. These results may well explain the observed negative temperature coefficients observed for termolecular processes, although it should be remembered that *experimentally* the results can well be fitted to an Arrhenius expression with a negative activation energy. In any case, the transition state formulation should be used with care for termolecular reactions which are thought to occur *via* a complex mechanism.

Some theoretical investigations of termolecular processes have been made which follow the general lines of the statistical trajectory studies described in other sections. Of these studies, that of Keck²⁰⁶ may be given as an example. Rate coefficients were obtained as a function of temperature for the $I + I + \text{Ar}$ recombination, although the results were not in very good accord with experiment. Considerably more attention has been paid by theorists to the reverse process of bimolecular dissociation, *viz.*



both along the lines of the trajectory studies (*e.g.* Keck²⁰⁷) and in the direction indicated by Montroll and Shuler³⁶ in their "stochastic" treatment. For the present, however, it would appear that the most satisfactory treatment of termolecular reactions lies in the collision theory, and the important object is to elucidate the exact mechanism of the processes.

BIBLIOGRAPHY

- BELL, R. P., *The Proton in Chemistry*, Methuen, London, 1959.
- BENSON, S. W., *The Foundations of Chemical Kinetics*, McGraw-Hill, New York, 1960.
- BUNKER, D. L., *Theory of Elementary Gas Reaction Rates*, Pergamon Press, Oxford, 1966.
- EYRING, H. AND EYRING, E. M., *Modern Chemical Kinetics*, Chapman and Hall, London, 1965.
- FROST, A. A. AND PEARSON, R. G., *Kinetics and Mechanism*, Wiley, New York, 1961.
- GLASSTONE, S., LAIDLER, K. J. AND EYRING, H., *The Theory of Rate Processes*, McGraw-Hill, New York, 1941.
- HINSHELWOOD, C. N., *The Kinetics of Chemical Change*, O.U.P., Oxford, 1940.
- JOHNSTON, H. S., *Gas Phase Reaction Rate Theory*, The Ronald Press, New York, 1966.
- KASSEL, L. S., *The Kinetics of Homogeneous Gas Reactions*, Chemical Catalog Co. Inc., New York, 1932.
- KONDRATIEV, V. N., *Chemical Kinetics of Gas Reactions*, Pergamon Press, Oxford, 1964.
- LAIDLER, K. J., *Chemical Kinetics*, McGraw-Hill, New York, 1965.
- LAIDLER, K. J., *The Chemical Kinetics of Excited States*, O.U.P., Oxford, 1955.
- MELANDER, L., *Isotope Effects on Reaction Rates*, The Ronald Press, New York, 1959.
- RICE, O. K., *Statistical Mechanics, Thermodynamics and Kinetics*, Freeman, San Francisco, 1967.
- SLATER, N. B., *Theory of Unimolecular Reactions*, Methuen, London, 1959.
- TROTMAN-DICKENSON, A. F., *Gas Kinetics*, Butterworths, London, 1955.

REFERENCES

- 1 S. ARRHENIUS, *Z. Physik. Chem.*, 4 (1889) 226.
- 2 C. N. HINSHELWOOD, *The Kinetics of Chemical Change*, O.U.P., Oxford, 1940, pp. 36-39.
- 3 M. BODENSTEIN, *Z. Physik. Chem.*, 13 (1894) 22; 22 (1897) 1; 29 (1899) 295.
- 4 W. C. MCC. LEWIS, *J. Chem. Soc.*, (1918) 471.
- 5 J. H. SULLIVAN, *J. Chem. Phys.*, 46 (1967) 73.
- 6 A. B. NALBANDYAN AND V. V. VOEVODSKII, *Mekhanizm Okisleniya i Goreniya Vodoroda*, Acad. Sci. USSR, Moscow, 1949.
- 7 L. I. AVRAMENKO AND V. N. KONDRATIEV, *Zh. Fiz. Khim.*, 24 (1950) 207.
- 8 P. G. ASHMORE AND J. CHANMUGAM, *Trans. Faraday Soc.*, 49 (1953) 254.
- 9 M. VAN MEERSCHE, *Bull. Soc. Chim. Belg.*, 60 (1951) 99.
- 10 C. F. GOODEVE AND A. W. TAYLOR, *Proc. Roy. Soc. (London)*, Ser. A, 152 (1935) 221; 154 (1936) 181.
- 11 H. S. JOHNSTON AND D. M. YOST, *J. Chem. Phys.*, 17 (1949) 386.
- 12 P. G. ASHMORE AND J. CHANMUGAM, *Trans. Faraday Soc.*, 49 (1953) 270.
- 13 M. R. BERLIE AND D. J. LEROY, *J. Chem. Phys.*, 20 (1952) 200.
- 14 M. A. A. CLYNE, B. A. THRUSH AND R. P. WAYNE, *Trans. Faraday Soc.*, 60 (1964) 359.
- 15 L. MANDELORN AND E. W. R. STEACIE, *Can. J. Chem.*, 30 (1952) 800.
- 16 A. F. TROTMAN-DICKENSON, J. R. BIRCHARD AND E. W. R. STEACIE, *J. Chem. Phys.*, 19 (1951) 163.
- 17 D. ROWLEY AND H. STEINER, *Discussions Faraday Soc.*, 10 (1951) 198.
- 18 C. C. SCHUBERT AND R. N. PEASE, *J. Chem. Phys.*, 24 (1956) 919.
- 19 E. P. WIGNER, *Gött. Nachr.*, 4 (1927) 375.
- 20 A. B. CALLEAR, Chapter 7 in *Photochemistry and Reaction Kinetics*, P. G. ASHMORE, F. S. DANTON AND T. M. SUGDEN (Eds.), Cambridge University Press, 1967.
- 21 J. HIRSCHFELDER, *J. Chem. Phys.*, 9 (1941) 645.
- 22 M. G. EVANS AND M. POLANYI, *Trans. Faraday Soc.*, 34 (1938) 11.
- 23 E. T. BUTLER AND M. POLANYI, *Trans. Faraday Soc.*, 39 (1943) 19.
- 24 N. N. SEMENOV, Chapter 1 in *Some Problems in Chemical Kinetics and Reactivity*, Pergamon Press, Oxford, 1958.
- 25 M. BODENSTEIN AND H. LÜTKEMEYER, *Z. Physik. Chem.*, 114 (1925) 208.
- 26 D. J. LEROY, *Discussions Faraday Soc.*, 14 (1953) 120.
- 27 H. O. PRITCHARD, J. B. PYKE AND A. F. TROTMAN-DICKENSON, *J. Am. Chem. Soc.*, 76 (1954) 1201.
- 28 G. B. KISTIAKOWSKY AND E. R. VAN ARTSDALEN, *J. Chem. Phys.*, 12 (1944) 469.
- 29 E. H. TAYLOR AND S. DATZ, *J. Chem. Phys.*, 23 (1955) 1711.
- 30 H. S. JOHNSTON, *Advan. Chem. Phys.*, 3 (1960) 131.
- 31 H. S. JOHNSTON AND D. RAPP, *J. Am. Chem. Soc.*, 83 (1961) 1.
- 32 T. E. SHARP AND H. S. JOHNSTON, *J. Chem. Phys.*, 37 (1962) 1541.
- 33 W. R. SCHULZ AND D. J. LEROY, *J. Chem. Phys.*, 42 (1965) 3869.
- 34 B. WIDOM AND S. H. BAUER, *J. Chem. Phys.*, 21 (1953) 1670.
- 35 H. A. KRAMERS, *Physica*, 7 (1940) 284.
- 36 E. W. MONTROLL AND K. E. SCHULER, *Advan. Chem. Phys.*, 1 (1958) 361.
- 37 M. A. ELIASON AND J. O. HIRSCHFELDER, *J. Chem. Phys.*, 30 (1959) 1426.
- 38 WANG-CHANG AND G. E. UHLENBECK, *Transport Phenomena in Polyatomic Molecules*, University of Michigan, 1951.
- 39 R. D. LEVINE, *J. Chem. Phys.*, 46 (1967) 331.
- 40 J. O. HIRSCHFELDER, H. EYRING AND B. TOPLEY, *J. Chem. Phys.*, 4 (1936) 170.
- 41 F. T. WALL, L. A. HILLER AND J. MAZUR, *J. Chem. Phys.*, 29 (1958) 255.
- 42 F. T. WALL, L. A. HILLER AND J. MAZUR, *J. Chem. Phys.*, 35 (1961) 1284.
- 43 M. KARPLUS, R. N. PORTER AND R. D. SHARMA, *J. Chem. Phys.*, 43 (1965) 3259.
- 44 S. GOLDEN, *J. Chem. Phys.*, 17 (1949) 620.
- 45 S. GOLDEN AND A. M. PEISER, *J. Chem. Phys.*, 17 (1949) 630.
- 46 G. C. FETTIS AND J. H. KNOX, *Progr. Reaction Kinetics*, 2 (1964) 1.
- 47 E. BAUER AND T. Y. WU, *J. Chem. Phys.*, 21 (1953) 726.

- 48 E. M. MORTENSEN AND K. S. PITZER, *Chem. Soc., Special Publ.*, 16 (1962) 57.
49 M. SZWARC, *Chem. Soc., Special Publ.*, 16 (1962) 25.
50 H. EYRING, *Chem. Soc., Special Publ.*, 16 (1962) 27.
51 C. STEEL AND K. J. LAIDLER, *J. Chem. Phys.*, 34 (1961) 1827.
52 D. R. HERSCHBACH, *Discussions Faraday Soc.*, 33 (1962) 149.
53 D. R. HERSCHBACH, *Advan. Chem. Phys.*, 10 (1966) 319.
54 S. DATZ AND E. H. TAYLOR, *J. Chem. Phys.*, 39 (1963) 1896.
55 R. D. PRESENT, *J. Chem. Phys.*, 31 (1959) 747.
56 H. EYRING, *J. Chem. Phys.*, 3 (1935) 107.
57 M. G. EVANS AND M. POLANYI, *Trans. Faraday Soc.*, 31 (1935) 875.
58 K. J. LAIDLER AND J. C. POLANYI, *Progr. Reaction Kinetics*, 3 (1964) 3.
59 D. M. BISHOP AND K. J. LAIDLER, *J. Chem. Phys.*, 42 (1965) 1688.
60 E. W. SCHLAG, *J. Chem. Phys.*, 38 (1963) 2480.
61 E. W. SCHLAG, *J. Chem. Phys.*, 42 (1965) 584.
62 J. C. LIGHT, *J. Chem. Phys.*, 40 (1964) 3221.
63 J. LIN AND J. C. LIGHT, *J. Chem. Phys.*, 45 (1966) 2545.
64 I. PRIGOGINE AND E. XHROUET, *Physica*, 15 (1949) 913.
65 I. PRIGOGINE AND M. MAHIEU, *Physica*, 16 (1950) 51.
66 B. H. MAHAN, *J. Chem. Phys.*, 32 (1960) 362.
67 J. ROSS AND P. MAZUR, *J. Chem. Phys.*, 35 (1961) 19.
68 C. W. PYNN AND J. ROSS, *J. Chem. Phys.*, 40 (1964) 2572.
69 H. C. EU AND J. ROSS, *J. Chem. Phys.*, 44 (1966) 2467.
70 H. C. EU AND J. ROSS, *J. Chem. Phys.*, 46 (1967) 411.
71 R. A. MARCUS, *J. Chem. Phys.*, 41 (1964) 2614, 2624.
72 R. A. MARCUS, *J. Chem. Phys.*, 41 (1964) 610.
73 R. A. MARCUS, *J. Chem. Phys.*, 43 (1965) 1598.
74 R. A. MARCUS, *J. Chem. Phys.*, 45 (1966) 2138.
75 R. A. MARCUS, *J. Chem. Phys.*, 45 (1966) 2630.
76 R. A. MARCUS, *J. Chem. Phys.*, 46 (1967) 959.
77 H. EYRING, H. GERSHINOWITZ AND C. E. SUN, *J. Chem. Phys.*, 3 (1935) 786.
78 F. LONDON, *Z. Elektrochem.*, 35 (1929) 552.
79 A. S. COOLIDGE AND H. M. JAMES, *J. Chem. Phys.*, 2 (1934) 811.
80 Y. SUGIURA, *Z. Physik*, 45 (1927) 484.
81 H. EYRING AND M. POLANYI, *Z. Physik. Chem.*, B12 (1931) 279.
82 J. O. HIRSCHFELDER, H. EYRING AND N. ROSEN, *J. Chem. Phys.*, 4 (1936) 121.
83 J. O. HIRSCHFELDER, H. DIAMOND AND H. EYRING, *J. Chem. Phys.*, 5 (1937) 695.
84 J. O. HIRSCHFELDER, *J. Chem. Phys.*, 6 (1938) 795.
85 G. E. KIMBALL AND J. G. TRULIO, *J. Chem. Phys.*, 28 (1958) 493.
85a S. F. BOYS AND I. SHAVITT, University of Wisconsin Naval Res. Lab. Tech. Rep. WIS-AF-13 (1959).
86 S. SATO, *J. Chem. Phys.*, 23 (1955) 592, 2465.
87 R. E. WESTON, *J. Chem. Phys.*, 31 (1959) 892.
88 P. J. KUNTZ, E. M. NEMETH, J. C. POLANYI, S. D. ROSNER AND C. E. YOUNG, *J. Chem. Phys.*, 44 (1966) 1168.
89 R. N. PORTER AND M. KARPLUS, *J. Chem. Phys.*, 40 (1964) 1105.
90 H. S. JOHNSTON AND C. PARR, *J. Am. Chem. Soc.*, 85 (1963) 2544.
91 F. T. WALL AND R. N. PORTER, *J. Chem. Phys.*, 36 (1962) 3256.
92 N. C. BLAIS AND D. L. BUNKER, *J. Chem. Phys.*, 37 (1962) 2713.
93 M. KARPLUS AND L. M. RAFF, *J. Chem. Phys.*, 41 (1964) 1267.
94 M. G. EVANS AND M. POLANYI, *Trans. Faraday Soc.*, 35 (1939) 178.
95 M. C. MOULTON AND D. R. HERSCHBACH, *J. Chem. Phys.*, 44 (1966) 3010.
96 S. DATZ AND R. E. MINTURN, *J. Chem. Phys.*, 41 (1964) 1153.
97 J. C. POLANYI AND S. D. ROSNER, *J. Chem. Phys.*, 38 (1963) 1028.
98 J. K. CASHION AND J. C. POLANYI, *Proc. Roy. Soc. (London), Ser. A*, 258 (1960) 570.
99 P. E. CHARTERS AND J. C. POLANYI, *Discussions Faraday Soc.*, 33 (1962) 107.
100 J. R. AIREY, R. R. GETTY, J. C. POLANYI AND D. R. SNELLING, *J. Chem. Phys.*, 41 (1964) 3255.

- 101 W. D. McGRATH AND R. G. W. NORRISH, *Proc. Roy. Soc. (London), Ser. A*, 242 (1957) 165.
- 102 N. BASCO AND R. G. W. NORRISH, *Proc. Roy. Soc. (London), Ser. A*, 268 (1962) 291.
- 103 D. GARVIN AND M. BOUDART, *J. Chem. Phys.*, 23 (1955) 784.
- 104 J. K. CASHION AND J. C. POLANYI, *J. Chem. Phys.*, 35 (1961) 600.
- 105 L. M. RAFF AND M. KARPLUS, *J. Chem. Phys.*, 44 (1966) 1212.
- 106 A. E. GROSSER, A. R. BLYTHE AND R. B. BERNSTEIN, *J. Chem. Phys.*, 42 (1965) 1268.
- 107 A. E. GROSSER AND R. B. BERNSTEIN, *J. Chem. Phys.*, 43 (1965) 1140.
- 108 J. H. BIRELY AND D. R. HERSCHBACH, *J. Chem. Phys.*, 44 (1966) 1690.
- 109 T. T. WARNOCK, R. B. BERNSTEIN AND A. E. GROSSER, *J. Chem. Phys.*, 46 (1967) 1685.
- 110 R. P. BELL, *Proc. Roy. Soc. (London), Ser. A*, 139 (1933) 466.
- 111 R. P. BELL, *The Proton in Chemistry*, Cornell University Press, Ithaca, N.Y., 1959, p. 205.
- 112 R. P. BELL, *Trans. Faraday Soc.*, 55 (1959) 1.
- 113 E. P. WIGNER, *Z. Physik. Chem.*, B19 (1932) 203.
- 114 C. ECKART, *Phys. Rev.*, 35 (1930) 1303.
- 115 H. S. JOHNSTON AND J. HEICKLEN, *J. Phys. Chem.*, 66 (1962) 532.
- 116 I. SHAVITT, *J. Chem. Phys.*, 31 (1959) 1359.
- 117 H. SHIN, *J. Chem. Phys.*, 39 (1963) 2934.
- 118 G. HERZBERG, *Infrared and Raman Spectra*, Van Nostrand, Princeton, New Jersey, 1945, p. 221.
- 119 J. BIGEISEN AND M. WOLFSBERG, *Advan. Chem. Phys.*, 1 (1958) 15.
- 120 H. CARMICHAEL AND H. S. JOHNSTON, *J. Chem. Phys.*, 41 (1964) 1975.
- 121 J. BIGEISEN, F. S. KLEIN, R. E. WESTON AND M. WOLFSBERG, *J. Chem. Phys.*, 30 (1959) 1340.
- 122 R. B. TIMMONS AND R. E. WESTON, *J. Chem. Phys.*, 41 (1964) 1654.
- 123 J. H. SULLIVAN, *J. Chem. Phys.*, 39 (1963) 3001.
- 124 A. PERSKY AND F. S. KLEIN, *J. Chem. Phys.*, 44 (1966) 3617.
- 125 F. S. KLEIN, A. PERSKY AND R. E. WESTON, *J. Chem. Phys.*, 41 (1964) 1799.
- 126 D. R. HERSCHBACH, H. S. JOHNSTON, K. S. PITZER AND R. E. POWELL, *J. Chem. Phys.*, 25 (1956) 736.
- 127 E. WIGNER, *Trans. Faraday Soc.*, 34 (1938) 29.
- 128 J. O. HIRSCHFELDER AND E. WIGNER, *J. Chem. Phys.*, 7 (1939) 616.
- 129 R. DE L. KRONIG, *Z. Physik*, 43 (1927) 524.
- 130 A. E. STERN AND H. EYRING, *J. Chem. Phys.*, 3 (1935) 778.
- 131 K. J. LAIDLER, *J. Chem. Phys.*, 10 (1942) 34.
- 132 J. L. MAGEE, *J. Chem. Phys.*, 8 (1940) 677.
- 133 L. LANDAU, *Physik. Z. Sowjetunion*, 1 (1932) 88.
- 134 C. ZENER, *Proc. Roy. Soc. (London), Ser. A*, 137 (1932) 696; 140 (1933) 666.
- 135 F. A. LINDEMANN, *Trans. Faraday Soc.*, 17 (1922) 598.
- 136 C. N. HINSHELWOOD, *Proc. Roy. Soc. (London), Ser. A*, 113 (1927) 230.
- 137 L. S. KASSEL, *J. Phys. Chem.*, 32 (1928) 225.
- 138 O. K. RICE AND H. C. RAMSPERGER, *J. Am. Chem. Soc.*, 49 (1927) 1617.
- 139 N. B. SLATER, *Proc. Roy. Soc. (London), Ser. A*, 194 (1948) 112.
- 140 R. A. MARCUS AND O. K. RICE, *J. Phys. and Colloid Chem.*, 55 (1951) 894.
- 141 R. A. MARCUS, *J. Chem. Phys.*, 20 (1952) 359.
- 142 F. W. SCHNEIDER AND B. S. RABINOVITCH, *J. Am. Chem. Soc.*, 85 (1963) 2365.
- 143 S. W. BENSON AND W. B. DEMORE, *Ann. Rev. Phys. Chem.*, 16 (1965) 397.
- 144 T. S. CHAMBERS AND G. B. KISTIAKOWSKY, *J. Am. Chem. Soc.*, 56 (1934) 399.
- 145 R. A. OGG, *J. Chem. Phys.*, 15 (1947) 337.
- 146 R. L. MILLS AND H. S. JOHNSTON, *J. Am. Chem. Soc.*, 73 (1951) 938.
- 147 L. A. K. STAVELEY AND C. N. HINSHELWOOD, *Nature*, 137 (1936) 29.
- 148 C. STEEL AND A. F. TROTMAN-DICKENSON, *J. Chem. Soc.*, (1959) 975.
- 149 J. C. GIDDINGS AND H. EYRING, *J. Chem. Phys.*, 22 (1954) 538.
- 150 H. M. ROSENSTOCK, M. B. WALLENSTEIN, A. L. WAHRHAFTIG AND H. EYRING, *Proc. Natl. Acad. Sci. U.S.*, 38 (1952) 667.
- 151 L. S. KASSEL, *J. Phys. Chem.*, 32 (1928) 1065.
- 152 O. K. RICE AND H. C. RAMSPERGER, *J. Am. Chem. Soc.*, 50 (1928) 617.
- 153 R. A. MARCUS, *J. Chem. Phys.*, 43 (1965) 2658.

- 154 G. M. WIEDER AND R. A. MARCUS, *J. Chem. Phys.*, 37 (1962) 1835.
155 B. S. RABINOVITCH AND R. W. DIESEN, *J. Chem. Phys.*, 30 (1959) 735.
156 G. Z. WHITTEN AND B. S. RABINOVITCH, *J. Chem. Phys.*, 38 (1963) 2466.
157 G. Z. WHITTEN AND B. S. RABINOVITCH, *J. Chem. Phys.*, 41 (1964) 1883.
158 N. B. SLATER, *Proc. Cambridge Phil. Soc.*, 35 (1939) 56.
159 N. B. SLATER, *Nature*, 159 (1947) 264.
160 N. B. SLATER, *Phil. Trans. Roy. Soc. (London)*, Ser. A, 246 (1953) 57.
161 F. W. SCHNEIDER AND B. S. RABINOVITCH, *J. Am. Chem. Soc.*, 84 (1962) 4215.
162 F. W. SCHNEIDER AND B. S. RABINOVITCH, *J. Am. Chem. Soc.*, 87 (1965) 158.
163 B. S. RABINOVITCH, D. W. SETSER AND F. W. SCHNEIDER, *Can. J. Chem.*, 39 (1961) 2609.
164 E. W. SCHLAG AND B. S. RABINOVITCH, *J. Am. Chem. Soc.*, 82 (1960) 5996.
165 N. B. SLATER, *Proc. Roy. Soc. (London)*, Ser. A, 218 (1953) 224.
166 T. S. CHAMBERS AND G. B. KISTIAKOWSKY, *J. Amer. Chem. Soc.*, 56 (1934) 399.
167 H. O. PRITCHARD, R. G. SOWDEN AND A. F. TROTMAN-DICKENSON, *Proc. Roy. Soc. (London)* Ser. A, 217 (1953) 563.
168 E. K. GILL AND K. J. LAIDLER, *Proc. Roy. Soc. (London)*, Ser. A, 250 (1959) 121.
169 E. K. GILL AND K. J. LAIDLER, *Proc. Roy. Soc. (London)*, Ser. A, 251 (1959) 66.
170 E. K. GILL AND K. J. LAIDLER, *Trans. Faraday Soc.*, 55 (1959) 753.
171 R. K. LYON, *J. Chem. Phys.*, 46 (1967) 4504.
172 B. S. RABINOVITCH AND M. C. FLOWERS, *Quart. Rev.*, 18 (1964) 122.
173 B. S. RABINOVITCH, E. TSCHNIKOW-ROUX AND E. W. SCHLAG, *J. Am. Chem. Soc.*, 81 (1959) 1081.
174 J. N. BUTLER AND G. B. KISTIAKOWSKY, *J. Am. Chem. Soc.*, 82 (1960) 759.
175 H. M. FREY, *Proc. Roy. Soc. (London)*, Ser. A, 251 (1959) 575.
176 D. W. SETSER AND B. S. RABINOVITCH, *Can. J. Chem.*, 40 (1962) 1425.
177 B. S. RABINOVITCH, R. F. KUBIN AND R. E. HARRINGTON, *J. Chem. Phys.*, 38 (1963) 405.
178 D. C. TARDY, B. S. RABINOVITCH AND C. W. LARSON, *J. Chem. Phys.*, 45 (1966) 1163.
179 J. C. HASSLER AND D. W. SETSER, *J. Chem. Phys.*, 45 (1966) 3246.
180 N. B. SLATER, *J. Chem. Phys.*, 24 (1956) 1256.
181 E. THIELE, *J. Chem. Phys.*, 36 (1962) 1466.
182 D. J. WILSON AND E. THIELE, *J. Chem. Phys.*, 40 (1964) 3425.
183 E. THIELE, *J. Chem. Phys.*, 45 (1966) 491.
184 B. H. MAHAN, *J. Phys. Chem.*, 62 (1958) 100.
185 F. P. BUFF AND D. J. WILSON, *J. Chem. Phys.*, 32 (1960) 677.
186 M. HOARE, *J. Chem. Phys.*, 38 (1963) 1630.
187 R. D. LEVINE, *J. Chem. Phys.*, 44 (1966) 1567, 2029, 2035, 2046, 3597.
188 W. G. VALENCE AND E. W. SCHLAG, *J. Chem. Phys.*, 45 (1966) 216, 4280.
189 E. THIELE AND D. J. WILSON, *J. Chem. Phys.*, 35 (1961) 1256.
190 N. C. HUNG AND D. J. WILSON, *J. Chem. Phys.*, 38 (1963) 828.
191 R. J. HARTE, E. B. ALTERMAN AND D. J. WILSON, *J. Chem. Phys.*, 40 (1964) 2137.
192 E. THIELE, *J. Chem. Phys.*, 38 (1963) 1959.
193 D. L. BUNKER, *J. Chem. Phys.*, 40 (1964) 1946.
194 G. B. KISTIAKOWSKY AND E. K. ROBERTS, *J. Chem. Phys.*, 21 (1953) 1637.
195 F. KAUFMAN AND J. R. KELSO, *J. Chem. Phys.*, 46 (1967) 4541.
196 G. PORTER AND J. A. SMITH, *Proc. Roy. Soc. (London)*, Ser. A, 261 (1961) 28.
197 R. C. TOLMAN, *Statistical Mechanics*, Chemical Catalogue Co., New York, 1921, p. 245.
198 W. STEINER, *Z. Physik. Chem.*, B15 (1932) 249.
199 F. T. SMITH, *J. Chem. Phys.*, 36 (1962) 248.
200 F. T. SMITH, *Discussions Faraday Soc.*, 33 (1962) 183.
201 M. I. CHRISTIE, A. J. HARRISON, R. G. W. NORRISH AND G. PORTER, *Proc. Roy. Soc. (London)*, Ser. A, 231 (1955) 446.
202 G. PORTER, *Discussions Faraday Soc.*, 33 (1962) 198.
203 D. L. BUNKER AND N. DAVIDSON, *J. Am. Chem. Soc.*, 80 (1958) 5090.
204 S. K. KIM, *J. Chem. Phys.*, 46 (1967) 123.
205 H. GERSHINOWITZ AND H. EYRING, *J. Am. Chem. Soc.*, 57 (1935) 985.
206 J. KECK, *J. Chem. Phys.*, 40 (1964) 1166.
207 J. KECK, *Discussions Faraday Soc.*, 33 (1962) 173.